

Finite excess in transfinite nim multiplication

Exact reductions, order bounds, and the 0/1/4 frontier

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Abstract

Conway's field On_2 makes nim-multiplication below ω^{ω^ω} depend on Lenstra's finite *excess* m_p in the Kummer carry $\alpha_p = \kappa_{f(p)} + m_p$. The observed rule is that every excess is 0, 1, or 4. We prove that this rule is equivalent to four exact statements: a primitive-support norm condition, a projective-order condition on each ordinary odd Kummer spine, Gaussian-period primitivity on the cubic spine, and a current-factor condition on the exceptional $2 \cdot 3^k$ spine. This is a complete reduction, not a proof of the rule.

The zero arm is sharpened in two directions. For arbitrary component sets we isolate the synchronized Frobenius phase that is lost by separate component norms, and for two components express it by a resultant over the intersection field. For the infinite power-of-two two-spine family $h = 2^a r^b$, where $d(\alpha_r) = 2^t$ and $a \geq t+1$, the distinguished Conway class in the primitive-support quotient is nontrivial and the zero arm is equivalent to generation of that explicit cyclic quotient. This proves the zero arm outright at $h = 12$ and $h = 24$, where the quotient orders are prime.

On an ordinary odd spine, the selected class is an explicit relative Hilbert–90 unit. A signed conjugate-ball argument gives an unconditional order lower bound $B_r^\pm$ and proves every primary factor $p > \mathcal{L}_{r,a}/B_r^\pm$. For the first source-pinned safe-prime row lacking an independent analytical certificate, $p = 359$, we reduce the desired value exactly to one trace bit in $\mathbb{F}_{2^{179}}$ and identify the selected phase among 179 resultant factors; the trace bit itself remains open. A Teichmüller-weighted Hilbert-product formula proves that base-field and pure order-(p) selectors have zero weight, while mixed Jacobi sums recover the desired phase as one explicit inverse-Fourier coordinate; its nonvanishing remains open. A projective minimal polynomial and its binary section expansion make failure equivalent to simultaneous divisibility by the actual Conway minimal polynomial; norm and complementation are automatic on that system. A tower-faithful character-sum construction shows that this complete section pattern occurs abundantly with irreducible ancestry, smooth exact degree, and full primary bookkeeping, even within a fixed bottom Kummer coset. It leaves only the unscaled Conway selector distinguished. On the singleton-even Fermat arm, a five-point generalized-Jacobian calculation proves that the full natural equivariant divisor-and-pairing package carries exactly the original Kummer scalar and supplies no independent relation. An exact root-extraction descent now shows more: all lower trace, norm, and maximal multiplicative-order ancestry is compatible with failure, which is equivalent to one explicit Dickson power-sum equation at the top step. The corresponding Dickson polynomial either splits completely in the failure case or is irreducible in the success case, with an exact resultant and absolute Capelli factorization. A reverse-Dickson transform sharpens failure to divisibility of the trace-one polynomial by an explicit absolute norm containing the selected minimal polynomial. Its degree count and long coefficient gap are compatible with every composite Fermat case, so the selected divisibility remains open. An exact semiconjugacy identifies the trace-one polynomial with the inversion quotient of the norm-one torus and the reverse-Dickson factor with one transported Kummer coset. It also gives the precise fibotomic support, proving that all additive and support constraints are realized by nonselected cosets.

The cubic simple-zero derivative regulator is $\lambda_k^{3^k}/3$; its reduction is $\gamma_k^{3^k}$, so both it and its relative norm-one quotient carry exactly the original unknown Kummer class. A cyclic group-algebra reduction sharpens cubic failure to one exact subset-sum parity pattern: every residue occurs oddly except one. After normalization inside the augmentation ideal, this is an extra proper-subproduct relation among Frobenius-conjugate units; the full product and the apparent ℓ -th-power exponent test are unconditional identities. A block compression shows that mask complementation forces the unique even residue into exactly the position required by failure. The surviving condition is equivalently the occurrence of a distinguished reciprocal pair of degree- 3^k factors in a binary Wendt gcd; all Wendt factors have that degree and are paired, so factor existence alone is insufficient. An exact Singer-difference-set factorization and its quotient moments go further: for a substantial small-index range every multiplicative coset contains such factors. The selected pair descends through the S_3 invariant to one lower trace polynomial. Its selector is exactly the reciprocal Gaussian period γ_{k-1}^{-2} , and an exact order product proves that this recursion is lossless rather than inductive. Failure is the extremal value of one prescribed-trace character sum; other trace fibres genuinely attain that value, so only the selected fibre can be excluded. The cubic and exceptional arms also admit a common norm-coherent antiunit. For the exception column, a distinct-part regulator gives order at least $B_-(h) = \exp((\pi/(6\sqrt{2}) + o(1))\sqrt{h})$ and proves every current primary factor above the corresponding threshold. Exceptional failure is also equivalent to simultaneous first-order flatness of a complete mixed Jacobi family, and to an exact complementary binary-submask parity pattern modulo $15 \cdot 3^k$. Raising that parity identity to the ℓ -th power is again unconditional. A half-block factorization makes the parity identity a quadratic norm equation. Its fixed-residue pushforward is automatic, and the remaining selected residue symbol is nontrivial exactly when one explicit relative trace is nonzero. Parallel partition bounds hold on the cubic spine. A further antiunit descent makes the selected exceptional phase a root of an explicit reciprocal sextic, and failure is exactly divisibility of one named irreducible cubic by a Dickson polynomial whose complete factorization consists of cubic squares. The ambient factorization contains many admissible cubics, and every such cubic has the same trace-constant shape as the selected one. Two further absolute traces of its coefficients are explicit but yield no conflicting ambient identity, so the remaining problem is exclusion of the recursively selected one. The remaining small factors in these three order problems, and maximality of the zero-arm primitive quotient, are genuine distinguished-value nonvanishing statements not forced by degree, norm, reciprocity, or generic character-sum bounds. Thus the full 0/1/4 rule and Lenstra's absolute boundedness question remain open, but the exact proof frontier and several infinite subfamilies are now rigorous.

Evidence taxonomy. Statements in theorem, proposition, lemma, and corollary environments are proved in this note or deduced from a cited theorem. **[certified I]** denotes an exact finite computation for the displayed instances I , with no claim about untested levels. **[source-pinned]** denotes a value imported from an external table. **[consistent]** denotes agreement with incomplete computational evidence. **[conjectured]** and **[open]** mark statements for which no proof or counterexample is known.

Contents

1	Introduction and main results	4
1.1	What is proved and what remains open	5
1.2	Organization	6
2	Mathematical setup	6
2.1	Tower notation	6

3	The order criterion and structural norm reduction	6
3.1	The exact criterion	6
3.2	Norm reduction	7
4	The zero arm	7
4.1	Primitive-support norm criterion	7
5	Ordinary odd Kummer spines	14
5.1	Transverse and projective reductions	14
5.2	The selected relative cyclotomic unit	17
5.3	Unconditional projective order bounds	25
5.4	Limits of generic criteria	27
5.5	The selected phase at $p = 359$	32
5.6	Open problem O	33
6	The cubic 3^k spine	34
6.1	Structure	34
6.2	Equivalent algebraic and cyclotomic forms	35
6.3	The selected Kummer class and reciprocity boundary	37
6.4	Kernel tower and quantitative order bounds	43
6.5	The exact remaining character test	46
6.6	Open problem C	59
7	The singleton-even Conway–Fermat arm	59
7.1	Maximal projective order	59
7.2	The selected factor and the limits of one-step induction	61
7.3	Ancestry refinements and exact no-go results	64
7.4	Elliptic and generalized-Jacobian reformulations	75
7.5	Comparison with the cubic arm	86
7.6	Open problem Z	86
8	The exceptional $2 \cdot 3^k$ arm	86
8.1	Why component data do not decide the excess	86
8.2	The corrected norm and current-factor target	87
8.3	Projection, fibotomic, and translated-binomial forms	88
8.4	Cyclotomic antiunit and regulator bounds	99
8.5	Exact bad-locus exhaustions	107
8.6	The limits of norm induction	113
8.7	Open problem D	115
9	Consequences and exact frontier	115
9.1	Boundedness	115
9.2	The four remaining theorem targets	116
A	Computational and implementation evidence	118
A.1	Source-pinned tower data	118
A.2	Independent small-row oracle	118
A.3	Cubic finite evidence	119
A.4	Exceptional finite evidence	119

1 Introduction and main results

Conway's nim-multiplication turns the ordinals into a field of characteristic two. Below $\omega^{\omega^{\omega}}$, the multiplication law is organized by a Kummer tower. At an odd prime p , its new carry has the form

$$\alpha_p = \kappa_{f(p)} + m_p, \quad f(p) = \text{ord}_p(2),$$

where the transfinite term $\kappa_{f(p)}$ is structural and the finite integer m_p is Lenstra's *excess*. Lenstra proved general lower bounds but left absolute boundedness open. All presently tabulated values suggest the following much sharper rule.

Conjecture 1.1 (the 0/1/4 rule). *For every odd prime p ,*

$$m_p = \begin{cases} 0 & \text{if } \mathcal{Q}(f(p)) \text{ is not a singleton odd prime-power,} \\ 4 & \text{if } f(p) = 2 \cdot 3^k \text{ for some } k \geq 1, \\ 1 & \text{otherwise.} \end{cases}$$

The principal result of this note is a case-exhaustive reduction of the conjecture to four explicit order problems. The notation $\Pi_h, T_h, \mathcal{L}_{r,a}, \gamma_k, \Psi_k, M_k$ is developed in the corresponding arm sections below.

Theorem 1.2 (four-arm reduction). *Conjecture 1.1 is equivalent to the conjunction of the following four universal statements.*

(Z) *For every $h > 1$ for which $\mathcal{Q}(h)$ is not a singleton odd prime-power,*

$$\Pi_h \mid \text{ord}\left(N_{\mathbb{F}_{2^{d(\kappa_h)}/\mathbb{F}_{2^h}}(\kappa_h)}\right).$$

(O) *For every odd prime $r \neq 3$ and every $a \geq 1$, the class of $y_a = \kappa_{r^a} + 1$ in the projective quotient $G_{r,a} = F_a^\times / F_{a-1}^\times$ has maximal prime-to- r order: for every prime $p \neq r$ dividing $\mathcal{L}_{r,a}$,*

$$v_p(\text{ord}(y_a F_{a-1}^\times)) = v_p(\mathcal{L}_{r,a}).$$

(C) *The Gaussian period $\gamma_k \in \mathbb{F}_{2^{3k}}^\times$ is primitive for every $k \geq 1$.*

(D) *The current exceptional factor satisfies $\Psi_k \mid \text{ord}(M_k)$ for every $k \geq 1$.*

Proof. If $\mathcal{Q}(h)$ is not a singleton odd prime-power, the conjecture asks for $m_p = 0$ at every prime with $f(p) = h$. Proposition 4.1 identifies this exactly with (Z).

If $\mathcal{Q}(h) = \{r^a\}$ with $r \neq 3$, then Proposition 5.3 partitions the eligible row primes among the primary divisors of $\mathcal{L}_{r,a}$ and identifies $m_p = 1$ with the equality in (O).

For $r = 3$, one has $d(\alpha_3) = 2$, so the possible residue degrees are 3^a and $2 \cdot 3^a$. Proposition 6.39 identifies the first family with (C). The unconditional lower bound $m_p \geq 4$, together with Proposition 8.3, identifies the second family with (D). These cases are disjoint and exhaust the definition of Conjecture 1.1. $\square$

1.1 What is proved and what remains open

The theorem is a reduction, not a proof of any universal arm. The present frontier is summarized below; finite certifications are listed separately in Appendix A.

Arm	Exact target	Proved here	Remaining universal assertion
(Z)	full primitive-support order	synchronized-phase reduction; exact two-component resultant; nontrivial support in the power-of-two two-spine family; complete $h = 12, 24$; exact root-ancestry, Dickson dichotomy, and reverse-Dickson trace-divisor and Kummer-coset criteria	generation for composite two-spine quotients, the general synchronized phase, and the singleton-even Conway–Fermat quotient
(O)	maximal primary projective order	transverse norm and relative-unit formulas; nontrivial order; signed conjugate-ball factor range; weighted-selector and mixed-Jacobi formulas; projective minimal polynomial and binary-section ancestry; tower-faithful common-section no-go	nondivisibility at the selected Conway polynomial for the remaining smaller primary factors
(C)	γ_k primitive	proof for $k \leq 3$; partition order bounds; exact derivative regulator, reciprocity boundary, cyclic-parity obstruction, and augmentation-ideal and block normalization; Singer–Wendt factorization and selected S_3 -trace quotient; reciprocal order seam and prescribed-trace character boundary	exclusion of the selected Wendt/trace fibre at the remaining current factors
(D)	$\Psi_k \mid \text{ord}(M_k)$	exact current-factor criterion; projection and antiunit partition bounds; equivalent polynomial, character, Jacobi-flatness, and binary cyclic-parity formulations; half-block norm, selected-trace form, selected reciprocal-sextic/Dickson factorization, universal cubic-shape no-go, and selected absolute-trace fingerprint	exclusion of the explicit selected cubic from the Dickson factor list at the remaining current factors

The extensive non-determination results below serve one purpose: they separate these four selected-value problems from statements that follow only from degree, trace, norm, reciprocity, or generic character-sum bounds. They are obstructions to standard proof strategies, not additional conjectural arms.

1.2 Organization

Section 2 fixes the tower notation, and Section 3 gives the exact power and norm criteria. Sections 4, 5, 6, 7, and 8 treat the zero, ordinary, cubic, singleton-even, and exceptional structures. The singleton-even section is part of arm (Z) , but is separated because its quadratic tower and generalized-Jacobian reformulations are substantial. Section 9 records the boundedness consequence and the four exact remaining theorem targets. Computational and implementation evidence is isolated in Appendix A.

2 Mathematical setup

2.1 Tower notation

For an odd prime p :

- $f(p) = \text{ord}_p(2)$, the multiplicative order of 2 modulo p ;
- $d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2]$, the least finite-field degree containing the algebraic number x ;
- κ_h is the Lenstra/DiMuro tower element indexed by h ;
- $\mathcal{Q}(h)$ is Lenstra's set of prime-power components appearing in κ_h ;
- the *excess* m_p is the least finite m such that $\kappa_{f(p)} + m$ has no p -th root in $\mathbb{F}_{2^{d(\kappa_{f(p)} + m)}}$;
- the Kummer carry is $\alpha_p = \kappa_{f(p)} + m_p$.

For the ordinal tower, a row with $\mathcal{Q}(f(p)) = \{q\}$ and finite excess m gives the ordinal sum corresponding to $\kappa_q + m$. Addition of numbers is XOR; $\otimes$ is the nim product.

The implementation boundary, external table provenance, and finite certifications are collected in Appendix A; none is used as evidence for an unbounded theorem.

3 The order criterion and structural norm reduction

3.1 The exact criterion

Throughout, E denotes the degree of the component field containing $\beta = \kappa_{f(p)} + m$, so $f = f(p) \mid E$ and hence $p \mid 2^E - 1$.

Proposition 3.1 (power criterion). $\beta \in \mathbb{F}_{2^E}^\times$ has no p -th root in $\mathbb{F}_{2^E}$ iff $\beta^{(2^E-1)/p} \neq 1$. Hence

$$m_p = \min\{m \geq 0 : (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

Proof. $\mathbb{F}_{2^E}^\times$ is cyclic of order $2^E - 1$ and $p \mid 2^E - 1$; in a cyclic group of order N , β is a p -th power iff $\beta^{N/p} = 1$. $\square$

Corollary 3.2 (order form). If $v_p(2^E - 1) = 1$, then β has no p -th root in $\mathbb{F}_{2^E}$ iff $p \mid \text{ord}(\beta)$, so m_p is the least m with $p \mid \text{ord}(\kappa_{f(p)} + m)$.

Remark 3.3 (Wieferich caveat). In general “no p -th root” is equivalent to $v_p(\text{ord}(\beta)) = v_p(2^E - 1)$, which collapses to $p \mid \text{ord}(\beta)$ only when $v_p(2^E - 1) = 1$. By lifting the exponent,

$$v_p(2^E - 1) = v_p(2^{f(p)} - 1) + v_p(E/f(p)).$$

Thus the hypothesis can fail either at base-2 Wieferich-type primes with $p^2 \mid 2^{f(p)} - 1$ or when $p \mid E/f(p)$. The only known base-2 Wieferich primes are $p = 1093$ and $p = 3511$, both of which lie inside the contributor's extended A380496 auxiliary range. Its calculator uses the exact test of Proposition 3.1, so the reported zeroes assert full p^2 -primary order; Appendix A records their thinner certificate boundary. For those rows, and for any row with $p \mid E/f(p)$, the exact power test (or the full- p -part order condition) must be used. The local certifications of Section 6 compute orders exactly and include squarefreeness checks. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

3.2 Norm reduction

Proposition 3.4 (norm reduction). *For $\beta \in \mathbb{F}_{2^E}^\times$ with $f = f(p) \mid E$,*

$$\beta^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

Proof. $N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$, and $((2^E-1)/(2^f-1)) \cdot ((2^f-1)/p) = (2^E-1)/p$. $\square$

Frobenius conjugacy preserves multiplicative order, and relative norms are Frobenius invariant. Consequently each later selected-value test depends only on the Conway conjugacy class fixed by the irreducible Kummer steps, not on a choice of root or field presentation.

4 The zero arm

4.1 Primitive-support norm criterion

For $h > 1$, write

$$\Pi_h := \prod_{\substack{\ell \text{ odd prime} \\ \text{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}.$$

This is the full prime-power part of $2^h - 1$ supported at primes whose order of 2 is exactly h ; using the full valuations, rather than only the radical, incorporates the Wieferich cases.

Proposition 4.1 (exact-order support). *Let $x = \kappa_h$, let $E = d(x)$, and put*

$$T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x).$$

Then $m_p = 0$ for every odd prime p with $f(p) = h$ if and only if

$$\Pi_h \mid \text{ord}(T_h).$$

For an individual such p , $m_p = 0$ iff $v_p(\text{ord } T_h) = v_p(2^h - 1)$.

Proof. The definition of κ_h gives $h \mid E$. Proposition 3.4 with $\beta = x$ identifies the p -power test in $\mathbb{F}_{2^E}$ with the same test on T_h in $\mathbb{F}_{2^h}$. In the cyclic group $\mathbb{F}_{2^h}^\times$, failure to be a p -th power is exactly the presence of the full p -primary part of the group order in $\text{ord}(T_h)$. Taking the product over the distinct eligible primes proves the claim. No hypothesis on $p \nmid E/h$ is hidden here: if $c = v_p((2^E-1)/(2^h-1))$, the norm exponent gives $v_p(\text{ord } T_h) = \max\{0, v_p(\text{ord } x) - c\}$, so equality with $v_p(2^h-1)$ is equivalent to $v_p(\text{ord } x) = v_p(2^E-1)$. $\square$

Proposition 4.2 (multicomponent Kummer no-collapse). *Suppose $\mathcal{Q}(h) = \{q_1, \dots, q_t\}$ with $t \geq 2$. Put $x_i = \kappa_{q_i}$, $x = \kappa_h = \sum_i x_i$, $E_i = d(x_i)$, $E = d(x)$, and $D = \text{lcm}(E_1, \dots, E_t)$. For a prime p with $\text{ord}_p(2) = h$, let*

$$T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x).$$

Then $p > h$, $p \nmid D$, and, in additive notation for $\mathbb{F}_{2^E}^\times/(\mathbb{F}_{2^E}^\times)^p$,

$$[T_h] = \frac{E}{h}[x]. \quad (\text{MC}_1)$$

The scalar E/h is nonzero modulo p , so norm reduction is an isomorphism on the relevant Kummer line. On the other hand every x_i is a p -th power in the common compositum $\mathbb{F}_{2^D}$. Thus no product of the individual component Kummer classes, or of their one-component norms, can produce the obstruction carried by the additive sum x .

If $P_h(Y) \in \mathbb{F}_2[Y]$ is the minimal polynomial for κ_h , then

$$m_p = 0 \iff P_h(Y^p) \text{ is irreducible over } \mathbb{F}_2. \quad (\text{MC}_2)$$

In contrast, every individual component has trivial p -Kummer class. Hence the multicomponent zero arm is a genuinely new family of additive-resultant/Capelli irreducibility assertions, not a multiplicative consequence of the singleton components.

Proof. Since $h = \text{ord}_p(2)$ divides $p - 1$, one has $p > h$; this is conditional on such a p and uses no Zsigmondy assertion. Write $q_i = r_i^{a_i}$. Lenstra gives $E_i = q_i b_i$ with every prime divisor of b_i smaller than r_i . Since $r_i \mid h$, every prime divisor of D is at most h , so $p \nmid D$. Also $x \in \mathbb{F}_{2^D}$, hence $E \mid D$ and $p \nmid E/h$.

Writing $n = E/h$, the norm exponent satisfies

$$\frac{2^E - 1}{2^h - 1} = 1 + 2^h + \dots + 2^{(n-1)h} \equiv n = \frac{E}{h} \pmod{p},$$

because $2^h \equiv 1 \pmod{p}$. This proves (MC_1) ; the natural map from the p -Kummer quotient of $\mathbb{F}_{2^h}$ to that of $\mathbb{F}_{2^E}$ is an isomorphism because the same scalar n is nonzero modulo p .

For each i , $q_i \mid h$ gives $\kappa_{q_i} \leq \kappa_h$. If $h \mid E_i$, the defining minimality of κ_h gives the reverse inequality, hence $\kappa_h = \kappa_{q_i}$, contradicting uniqueness of the multicomponent expansion. Thus $h \nmid E_i$, so $p \nmid 2^{E_i} - 1$. The p -power map is consequently an automorphism of $\mathbb{F}_{2^{E_i}}^\times$, and x_i is a p -th power there and hence in $\mathbb{F}_{2^D}$. Because $p \nmid D/E$, extension from $\mathbb{F}_{2^E}$ to $\mathbb{F}_{2^D}$ is injective on p -Kummer quotients, so this comparison loses no class of x .

Finally Capelli's lemma says $P_h(Y^p)$ is irreducible iff $Y^p - x$ is irreducible over $\mathbb{F}_{2^E}$. Since $\mu_p \subset \mathbb{F}_{2^E}$, the latter holds iff $x \notin (\mathbb{F}_{2^E}^\times)^p$, which is exactly $m_p = 0$. This proves (MC_2) . $\square$

Proposition 4.3 (synchronized additive resultants and the missing phase). *Put $Q = 2^h$, $K = \mathbb{F}_Q$, and suppose $p \mid Q - 1$. For $1 \leq i \leq t$, let $L_i = \mathbb{F}_{Q^{r_i}}$ and choose $u_i \in L_i$ of full degree r_i over K . Set*

$$R = \text{lcm}(r_1, \dots, r_t), \quad M = \mathbb{F}_{Q^R}, \quad u = \sum_i u_i \neq 0,$$

and assume $p \nmid R$. Exponents in the following formula are read modulo the corresponding r_i . The diagonal translation action on

$$\prod_i \mathbb{Z}/r_i \mathbb{Z}$$

has orbit set

$$\Omega = \left(\prod_i \mathbb{Z}/r_i\mathbb{Z} \right) / \langle (1, \dots, 1) \rangle.$$

For a representative $\mathbf{d} = (d_i)$ of an orbit, define

$$F_{\mathbf{d}}(Z) = \prod_{j=0}^{R-1} \left(Z + \sum_i u_i^{Q^{j+d_i}} \right). \quad (\text{MC}_3)$$

Then $F_{\mathbf{d}} \in K[Z]$, it depends only on the class of $\mathbf{d}$, and the ordinary componentwise additive composed product factors as

$$\prod_{0 \leq j_i < r_i} \left(Z + \sum_i u_i^{Q^{j_i}} \right) = \prod_{\mathbf{d} \in \Omega} F_{\mathbf{d}}(Z). \quad (\text{MC}_4)$$

The distinguished common-embedding sum is the zero phase, and

$$F_{\mathbf{0}}(0) = N_{M/K}(u), \quad [F_{\mathbf{0}}(0)] = R[u] \quad \text{in } M^\times/M^{\times p}. \quad (\text{MC}_5)$$

Consequently

$$u \notin M^{\times p} \iff F_{\mathbf{0}}(0) \notin K^{\times p}. \quad (\text{MC}_6)$$

If the r_i are pairwise coprime, Ω is a singleton and the ordinary additive resultant is the exact obstruction. Otherwise it gives only the product of several synchronized Frobenius phases. Independently replacing u_i by $u_i^{Q^{e_i}}$ preserves every separate minimal polynomial and merely permutes these phases. Thus separate component polynomials do not identify the distinguished factor $F_{\mathbf{0}}$.

For the Conway components of Proposition 4.2, take

$$r_i = \frac{E_i}{\gcd(E_i, h)}, \quad R = \frac{D}{h}, \quad L_i = K(x_i) = \mathbb{F}_{2^{\text{lcm}(h, E_i)}}, \quad M = \mathbb{F}_{2^D}.$$

Then $p \nmid R$, and

$$F_{\mathbf{0}}(0) = N_{\mathbb{F}_{2^D}/\mathbb{F}_{2^h}}(x) = T_h^{D/E}.$$

Since $p \nmid D/E$, this is exactly the same Kummer obstruction as (MC_1) , now expressed as one synchronized additive-resultant factor rather than the product over all relative embeddings.

Proof. The diagonal element $(1, \dots, 1)$ has order R , so its action is free and its orbits partition the tuples (j_i) into sets of size R . The Q -Frobenius cycles the factors in (MC_3) , proving that $F_{\mathbf{d}}$ has coefficients in K ; changing the representative translates the index j . Partitioning the factors on the left of (MC_4) by these diagonal orbits proves the factorization.

For the zero phase, the factors are precisely the M/K -conjugates of u , so the constant term is its norm (signs disappear in characteristic two). Under the natural inclusion of $K^\times/K^{\times p}$ into $M^\times/M^{\times p}$, the norm exponent is

$$1 + Q + \dots + Q^{R-1} \equiv R \pmod{p}.$$

The assumption $p \nmid R$ proves (MC_5) – (MC_6) . The description of Ω proves the conditional pairwise-coprime case, and independent Frobenius shifts translate the corresponding phase class, proving the information-loss statement.

In the Conway application, $K(x_i) = \mathbb{F}_{2^{\text{lcm}(h, E_i)}}$, whose degree over $K = \mathbb{F}_{2^h}$ is the displayed r_i . Taking least common multiples gives $R = D/h$. Proposition 4.2 proves $p \nmid D$, hence $p \nmid R$, and norm transitivity gives the last displayed equality. It also gives $p \nmid D/E$, completing the comparison. No pairwise-coprimality assertion for the actual r_i is used. $\square$

Proposition 4.4 (strict phase no-go). *Separate component degrees, minimal polynomials, and trivial p -Kummer classes do not determine the Kummer class of an additively composed sum, even when both possible sums have full compositum degree.*

Proof. Here is an exact example. In this proof, hexadecimal bit j denotes the coefficient of T^j . Put

$$k = \mathbb{F}_2[T]/(m), \quad m = 11\mathbf{b} = T^8 + T^4 + T^3 + T + 1.$$

Rabin's criterion is certified by

$$T^{256} = T \pmod{m}, \quad T^{16} + T = 5\mathbf{c} \pmod{m},$$

and the exact Bézout identity

$$13m + 515\mathbf{c} = 1,$$

so $k = \mathbb{F}_{256}$. Let $K = \mathbb{F}_4 \subset k$, put $\gamma = \mathbf{bc}$, so $\gamma^2 + \gamma + 1 = 0$, and take

$$a = 0\mathbf{c} = T^3 + T^2, \quad b = 2\mathbf{e} = T^5 + T^3 + T^2 + T.$$

Their 4-Frobenius data are

$$\begin{aligned} a^4 &= \mathbf{b}0, & a^{16} &= a, \\ b^4 &= 37, & b^{16} &= \mathbf{c}3, & b^{64} &= 67, \end{aligned}$$

so a and b have degrees two and four over K , respectively. Moreover

$$a^{85} = b^{85} = 1,$$

and hence both are cubes in $k^\times$. The elements b and b^4 are K -conjugate and have the same minimal polynomial; explicitly

$$\begin{aligned} \min_K(a) &= U^2 + \gamma U + 1, \\ \min_K(b) &= \min_K(b^4) = U^4 + (\gamma + 1)U^3 + \gamma U^2 + 1. \end{aligned}$$

Nevertheless

$$a + b = 22, \quad (a + b)^{85} = \gamma \neq 1,$$

whereas

$$a + b^4 = 3\mathbf{b}, \quad (a + b^4)^{85} = 1. \tag{MC_7}$$

Both sums have degree four over K : their 4-Frobenius orbits are

$$(22, 87, \mathbf{cf}, \mathbf{d7}), \quad (3\mathbf{b}, 73, 6\mathbf{b}, 9\mathbf{e}).$$

Since $k^\times$ has order 255, a nonzero element is a cube exactly when its 85-th power is one. Thus the first sum is a noncube and the second is a cube, although the separate component data are identical. As $\mu_3 \subset K$, Capelli's lemma says that composing their irreducible degree-four minimal polynomials with X^3 gives, respectively, an irreducible and a reducible polynomial. This proves the claimed strict no-go. It is a counterexample to a componentwise inference, not to the distinguished Conway statement. $\square$

Corollary 4.5 (two-component intersection resultant). *In Proposition 4.3, suppose $t = 2$. Put*

$$A = K(u_1) = \mathbb{F}_{Q^{r_1}}, \quad B = K(u_2) = \mathbb{F}_{Q^{r_2}}, \quad g = \gcd(r_1, r_2), \quad J = A \cap B = \mathbb{F}_{Q^g},$$

and let $A_i(U)$ be the minimal polynomial of u_i over J . Then

$$\text{Res}_U(A_1(U), A_2(U)) = N_{AB/J}(u_1 + u_2), \quad (\text{MC}_8)$$

and

$$u_1 + u_2 \notin (AB)^{\times p} \iff \text{Res}_U(A_1, A_2) \notin J^{\times p}. \quad (\text{MC}_9)$$

Thus the exact positive additive invariant is a single resultant over the intersection field, whose coefficients retain the relative Frobenius alignment omitted by the separate K -polynomials.

Suppose in addition that $p \nmid \text{ord}(u_1)\text{ord}(u_2)$, as for the Conway components $u_i = x_i$, which lie in the original fields $\mathbb{F}_{2^{E_i}}$ with $p \nmid 2^{E_i} - 1$. Then each u_i is a p -th power and

$$[u_1 + u_2] = \left[1 + \frac{u_1}{u_2} \right]. \quad (\text{MC}_{10})$$

Writing $N = \text{ord}(u_1/u_2)$, one has $p \nmid N$. After taking the Teichmüller lift $u_1/u_2 \leftarrow \zeta_N$, the remaining assertion is exactly nontriviality of the fixed power-residue symbol

$$\left(\frac{1 + \zeta_N}{\mathfrak{P}} \right)_p$$

at the prime above two selected by the common Conway embedding. Hence the two-component zero arm is a cyclotomic-unit Artin-symbol problem after the intersection resultant has removed all generic additive data.

Proof. The extensions A/J and B/J have coprime degrees r_1/g and r_2/g , so they are linearly disjoint and their pairs of relative conjugates are exactly the AB/J -conjugates of $u_1 + u_2$. Taking their product gives (MC₈). Here $\mu_p \subset K \subset J$, and

$$[AB : J] = \frac{R}{g}$$

is prime to p , because $p \nmid R$. After inclusion in $(AB)^{\times}/(AB)^{\times p}$, the resultant class is therefore $[AB : J][u_1 + u_2]$, proving (MC₉).

Under the additional order hypothesis, division by the p -th power u_2 gives (MC₁₀), and the ratio has order prime to p . The standard definition of the power-residue symbol identifies nontriviality of the residue class $1 + u_1/u_2$ with the displayed symbol of its Teichmüller lift. This proves the final formulation. $\square$

Proposition 4.6 (a power-of-two two-spine family has primitive support). *Let r be an odd prime for which*

$$b_r := d(\alpha_r) = 2^t.$$

Take $a \geq t + 1$, $b \geq 1$, and $h = 2^a r^b$. Then

$$\mathcal{Q}(h) = \{2^a, r^b\}, \quad d(\kappa_h) = h, \quad \Pi_h = \Phi_h(2). \quad (\text{MC}_{11})$$

Put $x = \kappa_{r^b}$, $c = \kappa_{2^a}$, and $z = \kappa_h = x + c$, and define the two maximal proper subfields

$$A = \mathbb{F}_{2^{h/2}}, \quad B = \mathbb{F}_{2^{h/r}}, \quad J = A \cap B = \mathbb{F}_{2^{h/(2r)}}.$$

The primitive-support quotient is cyclic of order Π_h :

$$\mathcal{T}_h := \mathbb{F}_{2^h}^\times / (A^\times B^\times), \quad |\mathcal{T}_h| = \Phi_h(2) = \Pi_h. \quad (\text{MC}_{12})$$

The class of z in $\mathcal{T}_h$ is nontrivial. Consequently $\text{ord}(z)$ is divisible by at least one prime p with $\text{ord}_p(2) = h$. More precisely, the full zero-support assertion at this h is equivalent to

$$\langle z(A^\times B^\times) \rangle = \mathcal{T}_h. \quad (\text{MC}_{13})$$

The full zero-support assertion in particular holds whenever $\Phi_h(2)$ is prime. This gives, for example, a proof without an order computation of the complete multicomponent zero arm at $h = 12$ and $h = 24$, whose quotients have orders 13 and 241, respectively.

Proof. By the odd Kummer-spine degree formula,

$$d(\kappa_{r^b}) = b_r r^b = 2^t r^b.$$

Lenstra's component recursion first constructs $\mathcal{Q}(r^b) = \{r^b\}$, and then adjoins the 2^a -component exactly when $2^a \nmid d(\kappa_{r^b})$. The hypothesis $a \geq t + 1$ therefore gives the first assertion in (MC₁₁). Moreover

$$d(\kappa_{2^a}) = 2^a, \quad \text{lcm}(2^a, 2^t r^b) = h.$$

Thus κ_h lies in $\mathbb{F}_{2^h}$; its defining degree is divisible by h , so $d(\kappa_h) = h$.

Write $q = 2^{h/(2^r)}$. The standard prime-power substitution identity for cyclotomic polynomials gives

$$\Phi_h(2) = \Phi_{2^r}(q) = \frac{q^r + 1}{q + 1}.$$

If a prime ℓ divides this value but $\text{ord}_\ell(2) < h$, the cyclotomic-divisor criterion forces $\ell \mid h$. The only possible odd exception is therefore r . The Kummer construction contains μ_r in $\mathbb{F}_{2^{b_r}}$, so $f(r) \mid b_r = 2^t$; write $f(r) = 2^u$ with $u \leq t$. The exceptional case would require

$$\frac{h}{\text{ord}_r(2)} = 2^{a-u} r^b$$

to be a power of r , which it is not because $a \geq t + 1 > u$. Hence every prime divisor of $\Phi_h(2)$ has exact order h , and all of its valuation in $2^h - 1$ occurs for the first time in $\Phi_h(2)$. This proves $\Pi_h = \Phi_h(2)$.

The intersection and compositum rules for finite fields give $A \cap B = J$ and $AB = \mathbb{F}_{2^h}$. Since the multiplicative group of a finite field is cyclic,

$$|A^\times B^\times| = \text{lcm}(q^r - 1, q^2 - 1) = \frac{(q^r - 1)(q^2 - 1)}{q - 1}.$$

Its index in $q^{2^r} - 1$ is

$$\frac{(q^{2^r} - 1)(q - 1)}{(q^r - 1)(q^2 - 1)} = \frac{q^r + 1}{q + 1} = \Phi_h(2),$$

which proves (MC₁₂).

It remains to show that $z \notin A^\times B^\times$. The degree formulas and $a \geq t + 1$ give

$$x \in A \setminus J, \quad c \in B \setminus J.$$

Because $[A : J] = r$ and $[B : J] = 2$ are coprime, multiplication induces an isomorphism

$$A \otimes_J B \xrightarrow{\sim} \mathbb{F}_{2^h}.$$

Under it, z is the tensor

$$x \otimes 1 + 1 \otimes c.$$

The pairs $\{x, 1\} \subset A$ and $\{1, c\} \subset B$ are both J -linearly independent, so this tensor has rank two. Every element of $A^\times B^\times$, on the other hand, is the image of a nonzero rank-one tensor $u \otimes v$. Thus $z \notin A^\times B^\times$, and its quotient class is nontrivial.

Every prime dividing the order of the quotient class divides Π_h , hence has exact order h . Conversely no such prime divides $|A^\times B^\times|$, so the full Π_h -part of $\text{ord}(z)$ is exactly the order of its class in $\mathcal{T}_h$. This proves (MC₁₃). If $\Phi_h(2)$ is prime, nontriviality of the class already means generation. Finally $\Phi_{12}(2) = 13$ and $\Phi_{24}(2) = 241$, so the complete zero arm holds at those two levels. $\square$

Remark 4.7 (the two-spine cross ratio and its present limit). Put $q = 2^{h/(2r)}$, $L_0 = (q^r - 1)(q + 1)$, $P = z^{L_0}$, and $c' = c^q$. The same Hilbert–90 calculation gives the exact representative

$$P = \left(\frac{x + c'}{x + c} \right)^{q+1} = \frac{(x + c')(\zeta x + c)}{(x + c)(\zeta x + c')}, \quad (\text{MC}_{14})$$

where $x^q = \zeta x$ and ζ is a primitive r -th root in J . This formula does not presently imply a nontrivial lower bound for $\text{ord}(P)$. Indeed, in the natural subset-product comparison of its r conjugates, clearing a relation with cyclic differences δ_j produces polynomials of degree $\sum_j |\delta_j|$, which may be as large as $2r - 2$; it is not bounded by $r - 1$. Since x has degree only r over B , evaluation at x cannot then be promoted to a polynomial identity. Thus tensor rank proves nontriviality, but maximality of the composite quotient remains open. We record (MC₁₄) as an exact starting point, not as an order estimate.

Remark 4.8 (sharp boundary of the tensor argument). Write more generally

$$d(\alpha_r) = 2^t c, \quad c > 1 \text{ odd},$$

and take $a > t$, $b \geq 1$. Over $J = \mathbb{F}_{2^{2t}}$, the fields generated by κ_{r^b} and κ_{2^a} have coprime relative degrees

$$m = cr^b, \quad n = 2^{a-t}.$$

Tensor rank therefore still proves that their sum is nontrivial in the product quotient of their compositum. If $q = |J|$, however, that quotient has order

$$\frac{(q^{mn} - 1)(q - 1)}{(q^m - 1)(q^n - 1)} = \prod_{\substack{d_1 | m, d_1 > 1 \\ d_2 | n, d_2 > 1}} \Phi_{d_1 d_2}(q). \quad (\text{MC}_{18})$$

The desired $h = 2^{a_r b}$ support occupies only the factor $\Phi_{nr^b}(q)$. Tensor nontriviality may instead be supported on a factor whose d_1 uses a nontrivial divisor of c ; norming the degree- hc compositum down to $\mathbb{F}_{2^h}$ kills every such prime-order component that does not divide $2^h - 1$. Thus the rank-two proof does not project to the required zero phase when $c > 1$. The hypothesis $d(\alpha_r) = 2^t$ is exactly where the present tensor/divisor argument isolates a single primitive-support factor; the general case still needs a theorem controlling the synchronized norm.

5 Ordinary odd Kummer spines

5.1 Transverse and projective reductions

The singleton-odd case admits a sharper field square. Let r be an odd prime, $q = r^a$, and set $b_r = d(\alpha_r)$, where $\kappa_q^q = \alpha_r$. Since $\alpha_r < \kappa_r$, Lenstra's field description gives $r \nmid b_r$, while the successive Kummer equations have degree r [13, Proposition 1.8]; hence

$$d(\kappa_q) = qb_r, \quad \gcd(q, b_r) = 1.$$

Proposition 5.1 (singleton transverse-norm criterion). *Suppose $\mathcal{Q}(h) = \{q\}$ with $q = r^a$ odd. Then $h = qs$ for a divisor $s \mid b_r$. In the field square*

$$F = \mathbb{F}_{2^{qb_r}}, \quad B = \mathbb{F}_{2^{b_r}}, \quad K = \mathbb{F}_{2^{qs}}, \quad C = \mathbb{F}_{2^s},$$

one has $B \cap K = C$ and $F = BK$. Put $x = \kappa_q$ and

$$\Theta_{q,s} := N_{F/K}(x+1) \in K^\times.$$

For every prime p with $f(p) = h$:

(i) x is a p -th power in F , so $m_p \geq 1$;

(ii) $m_p = 1$ iff

$$v_p(\text{ord } \Theta_{q,s}) = v_p(2^{qs} - 1);$$

(iii) the easy Kummer norm is

$$N_{F/B}(x+1) = 1 + \alpha_r,$$

but $p \nmid |B^\times|$, so this norm cannot see the required p -primary obstruction. The norm to K is genuinely transverse.

Proof. Lenstra's component theorem gives $\kappa_h = \sum_{t \in \mathcal{Q}(h)} \kappa_t$ and, for each component t , $t \mid h$ and $\gcd(t, h/t) = 1$ [13, Corollary 2.2]. Thus the singleton assumption identifies $\kappa_h = \kappa_q$. Since $h \mid d(\kappa_h) = d(\kappa_q) = qb_r$, one has $h = qs$ with $s \mid b_r$. The intersection and compositum statements follow from the gcd/lcm rules for finite subfields and $\gcd(q, b_r) = 1$.

Since $p > h \geq r$, one has $p \nmid q$. Also $f(p) = qs \nmid b_r$, so $p \nmid 2^{b_r} - 1$. The relation $x^q = \alpha_r \in B$ therefore gives $p \nmid \text{ord}(x)$, proving (i), including the full p -primary statement. Part (ii) is Proposition 3.4 applied to $x+1$. Finally, $X^q + \alpha_r$ is the minimal polynomial of x over B , and evaluating it at 1 gives (iii); the invisibility follows again from $p \nmid 2^{b_r} - 1$. $\square$

Corollary 5.2 (Hilbert–90 cross-ratio for the transverse class). *Retain Proposition 5.1, put $y = x+1$, and define*

$$z = x^{2^{b_r}-1}, \quad U = y^{2^{b_r}-1} = \frac{zx+1}{x+1}.$$

Then z has exact order q , and

$$U \in \ker(N_{F/B} : F^\times \longrightarrow B^\times).$$

For every prime p with $f(p) = h = qs$,

$$v_p(\text{ord } U) = v_p(\text{ord}(x+1)) = v_p(\text{ord } \Theta_{q,s}). \quad (\text{T})$$

Thus the singleton-one arm is equivalently the assertion that the explicit Hilbert–90 cross-ratio $(1 + zx)/(1 + x)$ contains the full p -primary part. In particular, the large transverse norm may be replaced, for the unresolved primary component, by one norm-one cyclotomic cross-ratio. On the one-dimensional quotient

$$\mathcal{V} = F^\times / (F^\times)^p$$

the automorphism $\sigma = \text{Frob}^{b_r}$ acts by $\lambda = 2^{b_r} \pmod{p}$, an element of exact order q , and

$$[U] = (\lambda - 1)[y], \quad [\Theta_{q,s}] = (b_r/s)[y]. \quad (\text{T}_1)$$

Both displayed scalars are nonzero, whereas

$$(1 + \sigma + \cdots + \sigma^{q-1})[y] = 0. \quad (\text{T}_2)$$

Finally the same target has the exact translated-composition form

$$m_p = 1 \iff ((T^p + 1)^q + \alpha_r) \text{ is irreducible over } B. \quad (\text{T}_3)$$

Proof. The 2^{b_r} -Frobenius generates $\text{Gal}(F/B)$ and sends x to zx . Since $x^q = \alpha_r \in B$, the class $xB^\times$ has exact order q : if $x^d \in B$ for $0 < d < q$, then x would satisfy a polynomial of degree d over B , contradicting $[B(x) : B] = q$. In the cyclic group $F^\times$, the map

$$F^\times / B^\times \longrightarrow F^\times, \quad gB^\times \longmapsto g^{|B^\times|}$$

is injective, because $B^\times$ is the unique subgroup of order $|B^\times| = 2^{b_r} - 1$. Its value at $xB^\times$ is z , so z has exact order q . The displayed formula for U is immediate, and its form $y^{\text{Frob}^{b_r} - 1}$ gives norm one by Hilbert 90.

Now $f(p) = qs \nmid b_r$, so $p \nmid 2^{b_r} - 1$. Exponentiation by $2^{b_r} - 1$ therefore preserves the p -primary part of the order of y , proving the first equality in (T). On the other hand

$$\Theta_{q,s} = y^A, \quad A = \frac{2^{qb_r} - 1}{2^{qs} - 1} \equiv \frac{b_r}{s} \pmod{p}.$$

Every prime divisor of $b_r = d(\alpha_r)$ is smaller than r , by Lenstra’s field description and $\alpha_r < \kappa_r$ [13, Proposition 1.8]; since $p > h \geq r$, one has $p \nmid b_r/s$. Thus $p \nmid A$, so this exponentiation also preserves the p -primary part, proving the second equality.

The quotient $\mathcal{V}$ has order p . Since $\text{ord}_p(2) = qs$, $b_r = s(b_r/s)$, and $\gcd(q, b_r) = 1$, the order of 2^{b_r} modulo p is exactly q . The definitions give $[U] = (\sigma - 1)[y] = (\lambda - 1)[y]$, while the preceding congruence for A gives the second equality in (T₁). The geometric sum in (T₂) vanishes because λ has exact order q . This is the representation-theoretic reason that the easy B -valued norm sees none of the target class.

The minimal polynomial of $y = x + 1$ over B is $(Y + 1)^q + \alpha_r$. Since p is odd, Capelli’s lemma says that its composition with T^p is irreducible exactly when y is not a p -th power in F . Proposition 5.1 identifies this with $m_p = 1$, proving (T₃). $\square$

Proposition 5.3 (one projective-order target for each odd Kummer spine). *Fix an odd prime r , put $b = d(\alpha_r)$, and define*

$$x_0 = \alpha_r, \quad x_a = \kappa_{r^a} \ (a \geq 1), \quad y_a = x_a + 1, \quad F_a = \mathbb{F}_{2^{r^a b}}.$$

Then, for every $a \geq 1$,

$$N_{F_a/F_{a-1}}(y_a) = y_{a-1}. \quad (\text{P}_1)$$

The projective quotient

$$G_{r,a} = F_a^\times / F_{a-1}^\times$$

is cyclic of order

$$\mathcal{L}_{r,a} = \frac{2^{r^a b} - 1}{2^{r^{a-1}b} - 1} = \Phi_{r^a}(2^b). \quad (\text{P}_2)$$

If $p \neq r$ is a prime divisor of $\mathcal{L}_{r,a}$, then

$$\text{ord}_p(2^b) = r^a, \quad f(p) = r^a s \text{ for some } s \mid b, \quad \mathcal{Q}(f(p)) = \{r^a\}. \quad (\text{P}_3)$$

Conversely every prime p in a singleton row $\mathcal{Q}(f(p)) = \{r^a\}$ occurs in this way. For every such p ,

$$m_p = 1 \iff v_p(\text{ord}(y_a F_{a-1}^\times)) = v_p(\mathcal{L}_{r,a}). \quad (\text{P}_4)$$

Consequently every singleton-one test for a fixed r is a primary-order test on this one projective class, simultaneously for all residue-degree divisors $s \mid b$. For $r \neq 3$, the 0/1/4 rule asks that its prime-to- r part be maximal. For $r = 3$, the two cyclotomic halves instead split as in the following remark.

Proof. The Kummer tower equations are $x_1^r = \alpha_r$ and $x_a^r = x_{a-1}$ for $a > 1$. The conjugates over F_{a-1} are ξx_a , $\xi \in \mu_r$; evaluating the corresponding binomial at one gives

$$N_{F_a/F_{a-1}}(x_a + 1) = x_{a-1} + 1,$$

which proves (P₁). The field degrees give (P₂).

For $p \neq r$, the standard cyclotomic-divisor criterion applied to $p \mid \Phi_{r^a}(2^b)$ gives $\text{ord}_p(2^b) = r^a$. If $h = f(p)$, then

$$\frac{h}{\gcd(h, b)} = r^a.$$

Since $h \mid r^a b$, this writes $h = r^a s$ with $s \mid b$. Now $h \mid d(\kappa_{r^a}) = r^a b$, so the defining minimality gives $\kappa_h \leq \kappa_{r^a}$. But $r^a \mid h$, and κ_{r^a} is the least element whose degree is divisible by r^a , so the reverse inequality holds. Hence $\kappa_h = \kappa_{r^a}$, and uniqueness in Lenstra's component theorem gives $\mathcal{Q}(h) = \{r^a\}$. Conversely Proposition 5.1 writes every singleton order as $h = r^a s$, $s \mid b$, and then $\text{ord}_p(2^b) = r^a$, so $p \mid \mathcal{L}_{r,a}$.

Such a p does not divide $|F_{a-1}^\times|$, because the order of 2^b modulo p is r^a . Its entire primary contribution to $|F_a^\times|$ therefore lies in the projective quotient. The exact power criterion now identifies non- p -power of y_a , equivalently $m_p = 1$, with the full p -primary order of its projective class. This proves (P₄). $\square$

Proposition 5.4 (exact two-half decomposition of the cubic projective quotient). *Let $a \geq 1$, put $h = 3^a$, $\zeta = \kappa_h$, $y = \zeta + 1$, and $\gamma_a = \zeta + \zeta^{-1} \in \mathbb{F}_{2h}^\times$. Then*

$$\Phi_h(4) = A_a B_a, \quad A_a = \Phi_h(2), \quad B_a = \Phi_{2h}(2),$$

with $\gcd(A_a, B_a) = 1$ and $v_3(B_a) = 1$. The projective order of y in $\mathbb{F}_{2h}^\times / \mathbb{F}_{2^{2h/3}}^\times$ is

$$3 \frac{\text{ord}(\gamma_a)}{\gcd(\text{ord}(\gamma_a), 2^{h/3} - 1)}. \quad (\text{CPS})$$

Consequently every prime $\ell \neq 3$ dividing B_a is absent from the translate-one projective class, unconditionally; it satisfies $\text{ord}_\ell(2) = 2h$. The A_a -part is full exactly when the new projective part of γ_a is full, and the all-level assertion is equivalent to Gaussian-period primitivity C_a . There is no overlap between the ordinary and exceptional halves. The missing $B_a/3 = \Psi_a$ part is precisely the translate-four residue-symbol problem D'_a .

Proof. The identity $\Phi_{3^a}(X^2) = \Phi_{3^a}(X)\Phi_{2 \cdot 3^a}(X)$ gives the factorization. With $X = 2^{3^{a-1}}$, the two factors are $X^2 + X + 1$ and $X^2 - X + 1$; a common odd divisor divides their difference $2X$, so the gcd is one. Since the exponent in X is odd, LTE applied to $(X^3 + 1)/(X + 1)$ gives $v_3(B_a) = 1$.

The half-angle identity gives

$$\text{ord}(y) = 3^{a+1} \text{ord}(\gamma_a), \quad \text{ord}(\gamma_a) \mid 2^h - 1.$$

The base group has 3-valuation a , and

$$\gcd(2^h - 1, 2^{2h/3} - 1) = 2^{h/3} - 1.$$

In a cyclic group, the order modulo its unique base subgroup is the element order divided by its gcd with the base-group order. This proves (CPS). If $\ell \neq 3$ divides B_a , the cyclotomic-divisor criterion gives $\text{ord}_\ell(2) = 2h$, so ℓ divides neither 3^{a+1} nor $2^h - 1$. The remaining assertions follow from the two displayed factors and the definitions of C_a and D'_a . $\square$

Remark 5.5 (the first nontrivial split). At $a = 2$, one has $\text{ord}(\gamma_2) = 2^9 - 1 = 7 \cdot 73$, and therefore

$$\text{ord}((\kappa_9 + 1)\mathbb{F}_{26}^\times) = 3 \cdot 73, \quad \Phi_9(4) = 3 \cdot 19 \cdot 73.$$

Thus the missing prime 19 is exactly the exceptional $\Phi_{18}(2)/3$ half, not a failure of the projective reduction.

5.2 The selected relative cyclotomic unit

Proposition 5.6 (selected-prime cyclotomic Kummer form). *Fix an odd prime r , put $b = d(\alpha_r)$ and $M_r = \text{ord}(\alpha_r)$, and retain $x_a = \kappa_{r^a}$, $y_a = x_a + 1$, $F_a = \mathbb{F}_{2^{r^a b}}$. Set $N_a = r^a M_r$. Then*

$$\text{ord}(x_a) = N_a, \quad \text{ord}_{N_a}(2) = r^a b. \quad (\text{CK}_1)$$

Let $p \neq r$ divide $\mathcal{L}_{r,a} = \Phi_{r^a}(2^b)$, choose a primitive complex N_a -th root ζ_{N_a} , and choose a prime $\mathfrak{P} \mid 2$ of

$$K_{a,p} = \mathbb{Q}(\mu_{N_a}, \mu_p)$$

whose residue map sends ζ_{N_a} to x_a . Its residue field is F_a . For $\varepsilon_a = 1 - \zeta_{N_a}$, the following are equivalent:

- (i) the class $y_a F_{a-1}^\times$ has full p -primary order $p^{v_p(\mathcal{L}_{r,a})}$;
- (ii) $y_a \notin (F_a^\times)^p$;
- (iii) $\left(\frac{\varepsilon_a}{\mathfrak{P}}\right)_p \neq 1$;
- (iv) $\text{Frob}_{\mathfrak{P}}$ is nontrivial in $K_{a,p}(\sqrt[p]{\varepsilon_a})/K_{a,p}$.

Thus, for $r \neq 3$, the predicted maximal prime-to- r projective order is exactly a simultaneous non-splitting assertion at the Conway-selected primes over two for these cyclotomic S -unit Kummer extensions. If N_a is not a prime power, ε_a is a global cyclotomic unit; in every case it is a unit at $\mathfrak{P}$.

Proof. The binomial $X^r - \alpha_r$ is irreducible over $B = \mathbb{F}_{2^b}$. Hence $\alpha_r \notin (B^\times)^r$. In particular $r \mid 2^b - 1$, and cyclicity of $B^\times$ gives

$$v_r(M_r) = v_r(2^b - 1).$$

If an element u has order $r^c m$, with $c \geq 1$ and $r \nmid m$, and $v^r = u$, then $\text{ord}(v) = r^{c+1}m$. Applying this first to $x_1^r = \alpha_r$ and then to $x_a^r = x_{a-1}$ proves $\text{ord}(x_a) = r^a M_r$. Since $\mathbb{F}_2(x_a) = F_a$, the degree of a root of unity of order N_a gives $\text{ord}_{N_a}(2) = r^a b$.

For $p \neq r$ dividing $\Phi_{r^a}(2^b)$, the cyclotomic-divisor criterion gives $\text{ord}_p(2^b) = r^a$. Hence p divides $|F_a^\times|$ but not $|F_{a-1}^\times|$. Also $p \nmid N_a$: if $p \mid M_r$, then $\text{ord}_p(2) \mid b$, contradicting $\text{ord}_p(2^b) = r^a > 1$. Therefore the entire p -Sylow subgroup of the projective quotient comes from $F_a^\times$, and in a cyclic group an element has full p -primary order iff it is not a p -th power. This proves (i) $\Leftrightarrow$ (ii).

The prime above two exists because x_a is a primitive N_a -th root with degree $r^a b$. The completion of $\mathbb{Q}(\mu_{N_a})$ at this prime is the unramified extension of $\mathbb{Q}_2$ of degree $\text{ord}_{N_a}(2) = r^a b$. Since $\text{ord}_p(2) \mid r^a b$, its residue field already contains μ_p , so adjoining μ_p does not enlarge that residue field. Its reduction sends ε_a to $1 + x_a = y_a$. The usual power-residue formula gives

$$\left(\frac{\varepsilon_a}{\mathfrak{P}} \right)_p = \overline{\varepsilon_a}^{(|F_a| - 1)/p},$$

which is nontrivial exactly when the residue is not a p -th power, proving (ii) $\Leftrightarrow$ (iii). Finally, p is odd and the residue characteristic is two, so the p -power map is an automorphism on the principal-unit group at $\mathfrak{P}$. The local Kummer extension is therefore unramified of degree p when the residue is not a p -th power, and split when it is; its Frobenius acts on $\sqrt[p]{\overline{\varepsilon_a}}$ by the displayed residue symbol. This proves (iii) $\Leftrightarrow$ (iv). $\square$

Proposition 5.7 (relative cyclotomic unit at the selected prime). *Retain the notation of Proposition 5.6, and put*

$$q_0 = 2^{r^{a-1}b}, \quad Q = q_0^r = 2^{r^a b}, \quad N = N_a.$$

The automorphism

$$\sigma(\zeta_N) = \zeta_N^{q_0}$$

generates $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{N/r}))$. For $\varepsilon = 1 - \zeta_N$, the relative cyclotomic unit

$$\rho = \frac{\sigma(\varepsilon)}{\varepsilon} = \frac{1 - \zeta_N^{q_0}}{1 - \zeta_N} \tag{RU_1}$$

is a global unit and has relative norm one. If $p \neq r$ divides $\mathcal{L}_{r,a}$, extend σ to $K_{a,p} = \mathbb{Q}(\mu_N, \mu_p)$ by fixing μ_p , and let $\mathfrak{P}$ be the Conway-selected prime of the preceding proposition. Then

$$\bar{\rho} = y_a^{q_0-1}, \quad \left(\frac{\rho}{\mathfrak{P}} \right)_p = \left(\frac{\varepsilon}{\mathfrak{P}} \right)_p^{q_0-1}. \tag{RU_2}$$

The residue class $\lambda = q_0 \pmod{p}$ has exact order r . Consequently $\lambda - 1 \neq 0$, and the four equivalent conditions of Proposition 5.6 are also equivalent to

$$\left(\frac{\rho}{\mathfrak{P}} \right)_p \neq 1. \tag{RU_3}$$

Thus the ordinary target is already a residue-symbol problem for an explicit norm-one global unit; the inherited lower-layer factor has been removed.

Proof. Since $x_a^r = x_{a-1} \in F_{a-1}$, one has $q_0 \equiv 1 \pmod{N/r}$. The q_0 -Frobenius has exact order r on F_a/F_{a-1} , so q_0 has exact order r modulo N . Moreover $r \mid M_r$ by the preceding proof, so $[\mathbb{Q}(\mu_N) : \mathbb{Q}(\mu_{N/r})] = r$. This proves the assertion about σ . The quotient in (RU₁) is a global unit: if N is not a prime power, both numerator and denominator are cyclotomic units, while if N is a prime power their principal ideals are the same unique prime above the conductor prime. Telescoping gives

$$N_{\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{N/r})}(\rho) = \prod_{j=0}^{r-1} \frac{\sigma^{j+1}(\varepsilon)}{\sigma^j(\varepsilon)} = 1.$$

Reduction at $\mathfrak{P}$, in characteristic two, gives

$$\bar{\rho} = \frac{1 + x_a^{q_0}}{1 + x_a} = \frac{(1 + x_a)^{q_0}}{1 + x_a} = y_a^{q_0-1},$$

and the power-residue formula gives (RU₂). Write $h = \text{ord}_p(2) = r^a s$, where $s \mid b$, as in (P₃). Every prime divisor of $b = d(\alpha_r)$ is smaller than r , so $r \nmid b$. Therefore

$$\text{ord}_p(q_0) = \frac{r^a s}{\gcd(r^a s, r^{a-1}b)} = r.$$

Raising an element of μ_p to $q_0 - 1$, equivalently to $\lambda - 1$, is therefore an automorphism. This proves (RU₃). $\square$

Proposition 5.8 (ordinary projective polynomial and binary sections). *Retain Proposition 5.7, and write*

$$F_0 = F_{a-1} = \mathbb{F}_{q_0}, \quad F = F_a, \quad x = x_a, \quad A = x_{a-1} = x^r, \quad y = 1 + x.$$

Put

$$\zeta = x^{q_0-1} \in F_0, \quad \rho = y^{q_0-1} = \frac{1 + \zeta x}{1 + x}, \quad L = \frac{q_0^r - 1}{q_0 - 1}.$$

Then ζ has exact order r , and the monic irreducible polynomial of ρ over F_0 is

$$H_{r,a}(T) = \frac{(T+1)^r + A(T+\zeta)^r}{1+A}. \quad (\text{OB}_1)$$

In particular,

$$\begin{aligned} H_{r,a}(0) &= 1, & \text{Tr}_{F/F_0}(\rho) &= \frac{1 + A\zeta}{1 + A}, \\ H_{r,a}(1) &= \frac{A(1 + \zeta)^r}{1 + A}, & H_{r,a}(\zeta) &= \frac{(1 + \zeta)^r}{1 + A}. \end{aligned} \quad (\text{OB}_2)$$

Also $N_{F/F_0}(\rho) = 1$, so $H_{r,a} \mid T^L - 1$.

Let $p \neq r$ be a prime divisor of L , and put $e = L/p$. The ordinary arm succeeds at p , with its full p -primary valuation, exactly when

$$H_{r,a}(T) \nmid T^e - 1. \quad (\text{OB}_3)$$

For $0 \leq k < r$, define the binary section polynomial

$$S_{e,k}(Z) = \sum_{\substack{t \geq 0 \\ k+tr \leq e}} \left(\binom{e}{k+tr} \bmod 2 \right) Z^t.$$

Then failure at p is exactly the simultaneous selected-value system

$$S_{e,k}(A) = 0 \quad (1 \leq k < r). \quad (\text{OB}_4)$$

This system descends all the way to the original Conway carry. Put $n = r^{a-1}$, let P_{α_r} be the minimal polynomial of α_r over $\mathbb{F}_2$, and, for $1 \leq c < r^a$ with $r \nmid c$, define

$$R_{e,c}(Z) = \sum_{\substack{v \geq 0 \\ c+r^a v \leq e}} \left(\binom{e}{c+r^a v} \bmod 2 \right) Z^v.$$

Then

$$\text{failure at } p \iff P_{\alpha_r}(Z) \mid R_{e,c}(Z) \quad \text{for every } c \pmod{r^a} \text{ with } r \nmid c. \quad (\text{OB}_5)$$

Thus the unresolved inverse-Fourier phase has an equivalent purely binary form, but it is still a divisibility assertion at the selected minimal polynomial P_{α_r} , not a generic Kummer-step identity.

Proof. The relative Frobenius sends x to ζx , and its order is r ; hence $\zeta \in F_0$ has exact order r . Solving

$$T = \frac{1 + \zeta x}{1 + x}$$

for x gives $x = (T + 1)/(T + \zeta)$. Substitution in $x^r = A$ gives (OB_1) . The inverse Möbius formula shows that $F_0(\rho) = F_0(x) = F$, so this degree- r polynomial is irreducible. Its constant term, next-to-leading coefficient, and the two displayed values give (OB_2) . Telescoping the Frobenius conjugates of y^{q_0-1} proves its norm is one.

The prime p does not divide $q_0 - 1$, because $\text{ord}_p(q_0) = r$. Therefore the full p -primary order of ρ equals that of the projective class of y . In a cyclic group of order L , an element contains the full p -primary part exactly when its L/p -th power is nonidentity, with no squarefreeness hypothesis. This proves (OB_3) .

In the F_0 -basis $1, x, \dots, x^{r-1}$, reduction by $x^r = A$ gives

$$(1 + x)^e = \sum_{k=0}^{r-1} x^k S_{e,k}(A). \quad (\text{OB}_6)$$

Now $\rho^e = 1$ exactly when $y^e \in F_0^\times$. The nonconstant coordinates in (OB_6) therefore prove both directions of (OB_4) .

Finally $A^n = \alpha_r$, and $1, A, \dots, A^{n-1}$ is a basis over $\mathbb{F}_{2^b}$. Writing $t = u + nv$ in each section gives

$$S_{e,k}(A) = \sum_{u=0}^{n-1} A^u R_{e,k+ru}(\alpha_r).$$

As k and u vary, the residues $k+ru$ are exactly the nonmultiples of r modulo r^a . Basis independence, followed by the minimal-polynomial criterion over $\mathbb{F}_2$, proves (OB_5) . $\square$

Corollary 5.9 (norm and complement are automatic in the ordinary sections). *Retain Proposition 5.8. Write $e = rE$. Then*

$$\frac{L}{r} = pE \equiv 1 \pmod{q_0 - 1}. \quad (\text{ON}_1)$$

Under failure, the sole scalar coordinate in (OB_6) is forced to be

$$S_{e,0}(A) = (1 + A)^E, \quad (\text{ON}_2)$$

and the sections satisfy the reciprocal symmetry

$$S_{e,r-k}(Z) = Z^{E-1} S_{e,k}(Z^{-1}) \quad (1 \leq k < r). \quad (\text{ON}_3)$$

Thus relative norm and binary complementation are exactly compatible with the simultaneous vanishings in (OB_4) .

Proof. LTE gives $v_r(L) = 1$, since $r \mid q_0 - 1$, while $p \neq r$; hence $e = rE$. Write

$$L - r = (q_0 - 1)D.$$

Modulo r , the usual geometric-sum quotient gives

$$D \equiv \sum_{j=1}^{r-1} j = 0,$$

so division by r proves (ON_1) . If failure holds, $(1+x)^e = S_{e,0}(A) \in F_0^\times$. Raising to p and using

$$(1+x)^L = N_{F/F_0}(1+x) = 1+A$$

together with (ON_1) proves (ON_2) . Finally e is odd, and complementing a binary submask $j = k+tr$ sends it to

$$e - j = (r - k) + r(E - t - 1).$$

Pairing the coefficients gives (ON_3) . □

Proposition 5.10 (four exact blind spots for the relative unit). *With the preceding notation, let*

$$\nu(T) = 1 + T + \cdots + T^{r-1}, \quad \mathcal{L} = \nu(q_0) = \Phi_{r^a}(2^b), \quad \lambda = q_0 \pmod{p}.$$

The following standard structural inputs do not determine $\left(\frac{\rho}{\mathfrak{P}}\right)_p$.

(i) *Relative norm and distribution give only*

$$\prod_{j=0}^{r-1} \left(\frac{\sigma^j(\varepsilon)}{\mathfrak{P}} \right)_p = A^{\nu(\lambda)} = 1, \quad A = \left(\frac{\varepsilon}{\mathfrak{P}} \right)_p, \quad (\text{RUB}_1)$$

and $\nu(\lambda) = 0$ has a simple zero.

(ii) *For the finite-field norm $N : F_a^\times \rightarrow F_{a-1}^\times$, every nonempty norm fibre maps surjectively to*

$$F_a^\times / (F_a^\times)^p. \quad (\text{RUB}_2)$$

In particular, fixing the exact lower norm—even norm one—permits both the trivial and every nontrivial p -Kummer class.

(iii) *Put $H = \langle \sigma \rangle$, $R = \mathbb{Z}_p[H]$, and $\mathfrak{m} = (p, \sigma - q_0)$. In $R_{\mathfrak{m}}$, if $u = \mathcal{L}/p$, then the complete first-order group-ring relation is*

$$0 = pu + \nu'(q_0)(\sigma - q_0) \pmod{\mathfrak{m}^2}, \quad (\text{RUB}_3)$$

where $\nu'(q_0)$ is a unit modulo $\mathfrak{m}$. This Bockstein/cotangent relation depends only on the cyclotomic layer and contains no coordinate of ρ .

(iv) *Global p -Hilbert reciprocity for the pair $(\rho, 2)$ is an identity independently of the selected local symbol. More precisely, if $\mathfrak{q} \mid 2$ is the selected prime of $\mathbb{Q}(\mu_N)$, its $p-1$ lifts $\mathfrak{P}_c$ to $K_{a,p}$, indexed by $c \in \mathbb{F}_p^\times$, satisfy*

$$(\rho, 2)_{\mathfrak{P}_{c,p}} = R_2^c, \quad \prod_{c \in \mathbb{F}_p^\times} (\rho, 2)_{\mathfrak{P}_{c,p}} = R_2^{p(p-1)/2} = 1 \quad (\text{RUB}_4)$$

for an arbitrary $R_2 \in \mu_p$.

Hence all four displayed consequences are compatible with every value of the selected symbol. Without an additional identity peculiar to ρ , they do not determine (RU₃). An argument must evaluate the distinguished additive unit $1 - \zeta_N$, or an equivalent explicit p -adic or Jacobi-sum congruence, beyond ideal factorization.

Proof. The conjugates of ε in the current degree- r layer have reductions

$$1 + x_a^{q_0^j} = (1 + x_a)^{q_0^j} = y_a^{q_0^j}.$$

Their residue symbols are therefore A^{λ^j} , proving (RUB₁). Equivalently, their product is the reduction of $1 - \zeta_N^r$, which lies in $F_{a-1}^\times$; it is automatically a p -th power because $p \nmid |F_{a-1}^\times|$. Differentiating $(T-1)\nu(T) = T^r - 1$ at $T = \lambda$ gives

$$(\lambda - 1)\nu'(\lambda) = r\lambda^{r-1} \neq 0,$$

so the vanishing is simple but supplies no value of A .

The kernel of the finite-field norm has order $\mathcal{L} = (Q-1)/(q_0-1)$. Since $p \mid \mathcal{L}$ but $p \nmid q_0-1$, the full p -Sylow subgroup of $F_a^\times$ lies in this kernel. Its image in the cyclic order- p quotient $F_a^\times / (F_a^\times)^p$ is nontrivial, hence surjective. Every norm fibre is a coset of the kernel, which proves (RUB₂).

In the group ring, $(\sigma - 1)\nu(\sigma) = 0$. The element $\sigma - 1$ is a unit after localization at $\mathfrak{m}$, because its residue is $\lambda - 1 \neq 0$; hence $\nu(\sigma) = 0$ there. Taylor expansion about the integer q_0 , together with $\nu(q_0) = \mathcal{L} = pu$, gives (RUB₃). The derivative is a unit by the preceding simple-root calculation. In particular this is a universal relation in $\mathbb{Z}_p[H]$, not an evaluation of a Kummer class.

Finally set $K = \mathbb{Q}(\mu_N)$ and $E = K(\mu_p)$. Since $p \nmid N$, one has $K \cap \mathbb{Q}(\mu_p) = \mathbb{Q}$. The local field $K_{\mathfrak{q}}$ already contains μ_p , because its residue degree is $r^a b$ and $\text{ord}_p(2) \mid r^a b$. Thus $\mathfrak{q}$ splits completely in E/K . Galois equivariance of the Hilbert symbol under the automorphisms $\zeta_p \mapsto \zeta_p^c$ proves (RUB₄); at a prime above two, R_2 is the residue symbol of ρ , up to the harmless choice of inversion convention.

For completeness, all remaining local factors are one. Away from $2p$, both entries are units and the tame Hilbert symbol is trivial. At a prime above p , $K/\mathbb{Q}$ is unramified locally and both entries come from that unramified subfield. Their Teichmüller parts are p -th powers; their principal parts in E lie in $U_{p-1} = 1 + p\mathcal{O}_E$, and the standard unit-filtration orthogonality gives $(U_{p-1}, U_{p-1})_p = 1$, since $2p-2 > p$. The archimedean places are complex. Global reciprocity therefore reduces to the already trivial products (RUB₄), one for each prime of K above two. Moreover $(\rho) = 1$, so ideal-theoretic Stickelberger or Jacobi-sum factorizations alone retain no datum that can distinguish the selected value. $\square$

Remark 5.11 (every unweighted base-field selector cancels). The cancellation in (RUB₄) is not peculiar to the second argument 2. Retain $K = \mathbb{Q}(\mu_N)$, $E = K(\mu_p)$, and the selected prime $\mathfrak{q} \mid 2$. Label its $p-1$ lifts $\mathfrak{P}_c$, $c \in \mathbb{F}_p^\times$, so that the automorphism carrying $\mathfrak{P}_1$ to $\mathfrak{P}_c$ sends ζ_p to ζ_p^c . For every $w \in K^\times$, if

$$R_w = (\rho, w)_{\mathfrak{P}_{1,p}},$$

then Galois equivariance gives

$$(\rho, w)_{\mathfrak{P}_{c,p}} = R_w^c, \quad \prod_{c \in \mathbb{F}_p^\times} (\rho, w)_{\mathfrak{P}_{c,p}} = R_w^{p(p-1)/2} = 1. \quad (\text{RUB}_5)$$

Thus no unweighted selector from the base cyclotomic field can make global reciprocity isolate one chosen lift above two. A weighted eigenspace identity or a Jacobi-sum selector outside K may evade (RUB₅), but ideal factorization alone then leaves local unit symbols to evaluate. The same cancellation applies to the exceptional arm with the current prime ℓ in place of p : since $\ell \nmid 15h$ and $\text{ord}_\ell(2) = 2h \mid 4h$, each of the two base primes above two splits into $\ell - 1$ lifts and its unweighted orbit product is one. Thus merely distinguishing the two base primes does not distinguish their individual lifts.

Proposition 5.12 (Teichmüller-weighted selector reciprocity). *Let M be odd, let $p \nmid M$ be an odd prime, and put*

$$K = \mathbb{Q}(\mu_M), \quad E = K(\mu_p), \quad f = \text{ord}_p(2), \quad d = \text{ord}_M(2).$$

Assume $f \mid d$. For each prime $\mathfrak{q} \mid 2$ of K , choose $\mathfrak{P}_{\mathfrak{q},1} \mid \mathfrak{q}$ in E , and write

$$\mathfrak{P}_{\mathfrak{q},c} = \tau_c \mathfrak{P}_{\mathfrak{q},1}, \quad \tau_c(\zeta_p) = \zeta_p^c \quad (c \in \mathbb{F}_p^\times).$$

Normalize every valuation above two by $v_{\mathfrak{P}}(2) = 1$. For a global unit $u \in \mathcal{O}_K^\times$, set

$$A_{\mathfrak{q}} = (u, 2)_{\mathfrak{P}_{\mathfrak{q},1,p}}, \quad W_{\mathfrak{q}}(w) = \sum_{c \in \mathbb{F}_p^\times} c v_{\mathfrak{P}_{\mathfrak{q},c}}(w) \in \mathbb{F}_p.$$

Then, for every $w \in E^\times$,

$$\prod_{c \in \mathbb{F}_p^\times} (u, w)_{\mathfrak{P}_{\mathfrak{q},c,p}} = A_{\mathfrak{q}}^{W_{\mathfrak{q}}(w)}. \quad (\text{WS}_1)$$

If the divisor of w is supported above two, global reciprocity becomes the exact weighted identity

$$\prod_{\mathfrak{q} \mid 2} A_{\mathfrak{q}}^{W_{\mathfrak{q}}(w)} = \prod_{\lambda \mid p} (u, w)_{\lambda,p}^{-1}. \quad (\text{WS}_2)$$

Moreover

$$W_{\mathfrak{q}}(\tau_e w) = e W_{\mathfrak{q}}(w), \quad (\text{WS}_3)$$

and

$$W_{\mathfrak{q}}(w) = 0 \quad \text{if } w \in K^\times \quad \text{or } w \in \mathbb{Q}(\mu_p)^\times. \quad (\text{WS}_4)$$

Thus a selector which isolates a chosen lift above two must be genuinely mixed between the M - and p -cyclotomic directions. Even then, (WS₂) retains the local factors above p : they are not determined by the ideal-theoretic weights.

Proof. The completion $K_{\mathfrak{q}}$ has residue degree d , so $f \mid d$ implies $\mu_p \subset K_{\mathfrak{q}}$. Hence $\mathfrak{q}$ splits completely in E/K . At a place above two, the tame p -Hilbert symbol of two local units is one. Since u is a unit, bilinearity therefore gives

$$(u, w)_{\mathfrak{P},p} = (u, 2)_{\mathfrak{P},p}^{v_{\mathfrak{P}}(w)}.$$

Galois equivariance gives

$$(u, 2)_{\mathfrak{P}_{\mathfrak{q},c,p}} = \tau_c(A_{\mathfrak{q}}) = A_{\mathfrak{q}}^c,$$

which proves (WS₁). If (w) is supported above two, then away from $2p$ both entries are units and their tame symbol is one; the archimedean factors are also one. The global product formula therefore gives (WS₂). Notice that no triviality is asserted at the wild places above p .

Reindexing the valuations proves (WS₃). If $w \in K^\times$, its valuations are constant in c , and $\sum_{c \in \mathbb{F}_p^\times} c = 0$. If $w \in \mathbb{Q}(\mu_p)^\times$, the restrictions of $\mathfrak{P}_{q,c}$ and $\mathfrak{P}_{q,c2^j}$ to $\mathbb{Q}(\mu_p)$ agree for $0 \leq j < f$. The valuations are therefore constant on cosets of $\langle 2 \rangle \subset \mathbb{F}_p^\times$, while each weighted coset sum is

$$c \sum_{j=0}^{f-1} 2^j = c(2^f - 1) = 0 \pmod{p}.$$

This proves (WS₄). □

Proposition 5.13 (the first mixed Jacobi phase). *Let $Q = 2^D$, let M be odd, and let $p \nmid M$ be an odd prime such that $Mp \mid Q - 1$. Choose a generator $g \in \mathbb{F}_Q^\times$, put*

$$z = g^{(Q-1)/M}, \quad \text{ind}_p(g^v) = v \pmod{p},$$

and choose characters, extended by zero at zero, by

$$\chi_M(g) = \zeta_M, \quad \chi_p(g) = \zeta_p.$$

For $1 \leq j < M$, define

$$J_j = \sum_{x \in \mathbb{F}_Q} \chi_M(x)^j \chi_p(1 - x).$$

Fix a place above p of $\mathbb{Q}(\mu_M, \mu_p)$, put $\pi = \zeta_p - 1$, and let an overline denote reduction at that place—not complex conjugation. Then $J_j \equiv -1 \pmod{\pi}$, and

$$C_j := \frac{\overline{J_j + 1}}{\pi} = \sum_{a=1}^{M-1} \overline{\zeta_M}^{ja} \text{ind}_p(1 - z^a). \quad (\text{MJ}_1)$$

With $J_0 = -1$ and $C_0 = 0$, Fourier inversion gives

$$\text{ind}_p(1 - z^a) = M^{-1} \sum_{j=0}^{M-1} \overline{\zeta_M}^{-ja} C_j \quad (1 \leq a < M). \quad (\text{MJ}_2)$$

Consequently the following are equivalent:

- (i) *every $1 - z^a$, $1 \leq a < M$, is a p -th power in $\mathbb{F}_Q^\times$;*
- (ii) *$C_j = 0$ for every j ;*
- (iii) *$J_j \equiv -1 \pmod{\pi^2}$ for every $1 \leq j < M$.*

For prime M , the first-order equivalence is the $k = 2$ layer of Arul's Jacobi congruence [1, Theorem 1.5]; the elementary argument here also permits composite M .

Proof. Modulo π^2 , one has $\zeta_p^v = (1 + \pi)^v \equiv 1 + v\pi$. The constant term of J_j for $j \neq 0$ is

$$\sum_{x \in \mathbb{F}_Q^\times \setminus \{1\}} \chi_M(x)^j = -1,$$

while the trivial-character identity gives $J_0 = -1$ exactly. The coefficient of π is

$$\sum_{v=1}^{Q-2} \zeta_M^{jv} \text{ind}_p(1 - g^v).$$

Partition $v = a + tM$, and put $N = (Q - 1)/M$. For $a \neq 0$,

$$\prod_{t=0}^{N-1} (1 - g^{a+tM}) = 1 - g^{aN} = 1 - z^a,$$

so the sum of the indices on that block is $\text{ind}_p(1 - z^a)$. In the $a = 0$ block the omitted zero factor leaves

$$\prod_{t=1}^{N-1} (1 - g^{tM}) = N = 1 \quad \text{in characteristic two,}$$

because N is odd; its index is zero. This proves (MJ₁). The Fourier matrix is invertible because $p \nmid M$, giving (MJ₂). Finally, $\text{ind}_p(y) = 0$ exactly when y is a p -th power in the cyclic group $\mathbb{F}_Q^\times$, which proves the three equivalences. $\square$

Corollary 5.14 (ordinary target as one inverse-Fourier coordinate). *Retain Proposition 5.6. Put*

$$M = N_a, \quad Q = 2^{r^a b}, \quad z = x_a,$$

and let $p \neq r$ divide $\Phi_{r^a}(2^b)$. Choose g in the preceding proposition so that $g^{(Q-1)/M} = z$, and form its mixed Jacobi sums J_j and coefficients C_j . Then the ordinary condition at p is exactly

$$\text{ind}_p(y_a) = M^{-1} \sum_{j=0}^{M-1} \overline{\zeta_M}^{-j} C_j \neq 0. \quad (\text{OMJ})$$

Indeed $y_a = 1 - z$ in characteristic two, and

$$\text{ind}_p(\bar{\rho}) = (q_0 - 1) \text{ind}_p(y_a),$$

where $q_0 - 1 \not\equiv 0 \pmod{p}$. Thus mixed Jacobi sums evade the zero weight in (WS₄), but their required first local phase is exactly the original selected Kummer coordinate. Formula (WS₂) relates its ideal weights to the still-essential local symbols above p ; it does not itself prove nonvanishing.

5.3 Unconditional projective order bounds

Proposition 5.15 (unconditional nontrivial projective order). *Let*

$$d_{r,a} = \text{ord}(y_a F_{a-1}^\times) \mid \mathcal{L}_{r,a}.$$

Then

$$d_{r,a} \notin \{1, r\}, \quad v_r(\mathcal{L}_{r,a}) = 1. \quad (\text{PO}_1)$$

Consequently $d_{r,a}$ has a prime divisor $p \neq r$ at every level. This asserts only $p \mid d_{r,a}$: when $v_p(\mathcal{L}_{r,a}) > 1$, it does not by itself give the full p -primary equality in (O). If the chosen divisor satisfies $v_p(\mathcal{L}_{r,a}) = 1$, however, its instance of (O) follows. In particular, if $r \neq 3$ and $\mathcal{L}_{r,a}/r$ is prime, then the entire ordinary target (O) at this layer holds.

Proof. If $d_{r,a} = 1$, then y_a , and hence $x_a = y_a + 1$, lies in F_{a-1} , contrary to $[F_a : F_{a-1}] = r$. If $d_{r,a} = r$, then $y_a^r = c \in F_{a-1}^\times$. Since y_a has degree r over F_{a-1} , its minimal polynomial would be $Y^r - c$. On the other hand $x_a^r = x_{a-1}$ shows that its monic degree- r minimal polynomial is

$$(Y + 1)^r + x_{a-1}.$$

The coefficient of Y^{r-1} in the latter is $r \equiv 1 \pmod{2}$, whereas it is zero in $Y^r - c$, a contradiction. Because $r \mid 2^b - 1$, also $q_0 \equiv 1 \pmod{r}$. LTE gives

$$v_r(\mathcal{L}_{r,a}) = v_r(q_0^r - 1) - v_r(q_0 - 1) = v_r(r) = 1.$$

Thus a divisor of $\mathcal{L}_{r,a}$ other than 1 and r has a prime divisor different from r . The final assertions follow from (P_4) , with the stated full-valuation caveat. $\square$

Proposition 5.16 (signed conjugate-ball order sieve). *Retain the notation of Proposition 5.3. Put*

$$R = \frac{r-1}{2}, \quad B_r^\pm = \sum_{u=0}^R \sum_{v=0}^R [X^u Y^v] \left(1 + \frac{X}{1-X} + \frac{Y}{1-Y} \right)^r. \quad (\text{CS}_0)$$

Then

$$d_{r,a} = \text{ord}(y_a F_{a-1}^\times) \geq B_r^\pm. \quad (\text{CS}_1)$$

Consequently, if $p \neq r$ divides $\mathcal{L}_{r,a}$ and

$$p > \frac{\mathcal{L}_{r,a}}{B_r^\pm}, \quad (\text{CS}_2)$$

then the Conway-selected p -th power-residue symbol is nontrivial and

$$v_p(d_{r,a}) = v_p(\mathcal{L}_{r,a}). \quad (\text{CS}_3)$$

Thus every prime factor above the explicit complementary threshold is proved in the ordinary arm, including its full primary valuation.

Proof. Write $q_0 = |F_{a-1}|$, let $\sigma(z) = z^{q_0}$ generate $\text{Gal}(F_a/F_{a-1})$, and set

$$\zeta = \frac{\sigma(x_a)}{x_a}.$$

The Kummer equation $x_a^r = x_{a-1}$ and the degree- r irreducibility show that ζ is a primitive r -th root of unity. Since $r \mid q_0 - 1$, it belongs to F_{a-1} , and

$$\sigma^j(y_a) = 1 + \zeta^j x_a \quad (0 \leq j < r). \quad (\text{CS}_4)$$

For $\mathbf{e} = (e_0, \dots, e_{r-1}) \in \mathbb{Z}^r$, write $e_j^+ = \max(e_j, 0)$ and $e_j^- = \max(-e_j, 0)$, and consider

$$\mathcal{S}_R = \{ \mathbf{e} \in \mathbb{Z}^r : |\mathbf{e}^+| \leq R, |\mathbf{e}^-| \leq R \}.$$

The single-coordinate generating series

$$1 + X + X^2 + \dots + Y + Y^2 + \dots = 1 + \frac{X}{1-X} + \frac{Y}{1-Y}$$

shows that $|\mathcal{S}_R| = B_r^\pm$.

For $\mathbf{e} \in \mathcal{S}_R$, form the nonzero signed product

$$Z_{\mathbf{e}} = \prod_{j=0}^{r-1} \sigma^j(y_a)^{e_j} \in F_a^\times.$$

Suppose $Z_{\mathbf{e}}$ and $Z_{\mathbf{f}}$ represent the same class modulo $F_{a-1}^\times$, and put $\mathbf{g} = \mathbf{e} - \mathbf{f}$. Then

$$\prod_j (1 + \zeta^j x_a)^{g_j^+} = c \prod_j (1 + \zeta^j x_a)^{g_j^-} \quad \text{for some } c \in F_{a-1}^\times. \quad (\text{CS}_5)$$

The two polynomial products in (CS_5) have degrees at most

$$|\mathbf{g}^+| \leq |\mathbf{e}^+| + |\mathbf{f}^-| \leq 2R = r - 1,$$

and the same bound holds for $|\mathbf{g}^-|$. Their difference therefore has degree less than r , whereas x_a has degree r over F_{a-1} ; it must be the zero polynomial. Both constant terms are one, so $c = 1$. The linear factors $1 + \zeta^j X$ are pairwise distinct and normalized. Unique factorization gives $\mathbf{g}^+ = \mathbf{g}^-$, hence $\mathbf{e} = \mathbf{f}$.

Negative powers cause no problem in the projective quotient:

$$[Z_{\mathbf{e}}] = \prod_{j=0}^{r-1} [\sigma^j(y_a)]^{e_j} = [y_a]^{\sum_j e_j q_0^j}.$$

Thus the $B_r^\pm$ elements of $\mathcal{S}_R$ give distinct powers of the single class $[y_a]$, proving (CS_1) .

If the selected p -th residue symbol were trivial, then y_a would be a p -th power in F_a . Since the full p -part of $|F_a^\times|$ lies in the projective quotient, this would give $d_{r,a} \mid \mathcal{L}_{r,a}/p$. Under (CS_2) that contradicts (CS_1) . Nontriviality is equivalent to the full primary equality by Proposition 5.6, which proves (CS_3) . $\square$

5.4 Limits of generic criteria

Thus the singleton-one arm is not an unspecified root search. It is the uniform assertion that the explicitly defined transverse norm $\Theta_{q,s}$ contains the full p -primary part for every prime p of exact order qs . The C_k norms below are the special case $q = 3^k$, $b_r = 2$, $s = 1$; the m_{359} obstruction is the same square at $q = 179$, $b_r = 19580$, $s = 1$. Lenstra's use of Carlitz's primitive-root distribution theorem gives an effective bound at which *some* finite translate works [4, 13]; it does not evaluate this fixed first transverse norm.

There is a sharp reason that this effective argument cannot be replaced, for arbitrary finite-field elements, by any fixed initial translate window.

Proposition 5.17 (finite-window Kummer no-go). *Let p be an odd prime and let*

$$A = \{b_1, \dots, b_r\} \subset \overline{\mathbb{F}_2}$$

be a finite set of distinct elements. Choose $k_0 = \mathbb{F}_{2^d}$ containing both A and μ_p . There are arbitrarily large integers n and elements $x_n \in \mathbb{F}_{2^{dn}}$ of exact degree dn over $\mathbb{F}_2$ such that

$$x_n + b_i \in (\mathbb{F}_{2^{dn}}^\times)^p \quad (1 \leq i \leq r).$$

Consequently no absolute truncation of Lenstra's translate-generation lemma, nor any argument using only full degree, an order- p character, and a fixed affine translate set, can prove an absolute bound for arbitrary elements. This does not decide the distinguished elements $\kappa_{f(p)}$: any uniform excess theorem must use their special Conway-tower identities.

Proof. Over $k_0(X)$ form the Kummer extension

$$\mathcal{K} = k_0(X)(Y_1, \dots, Y_r), \quad Y_i^p = X + b_i.$$

The classes of the $X + b_i$ are independent in $\bar{k}_0(X)^\times / \bar{k}_0(X)^{\times p}$: in a relation $\prod_i (X + b_i)^{e_i} = G^p$, $0 \leq e_i < p$, the valuation at $X = -b_i$ forces $e_i = 0$. Thus the smooth projective curve C with function field $\mathcal{K}$ is geometrically integral and

$$[\mathcal{K} : k_0(X)] = p^r, \quad \text{Gal}(\mathcal{K}/k_0(X)) \cong (\mathbb{Z}/p\mathbb{Z})^r.$$

The cover $C \rightarrow \mathbb{P}_X^1$ is ramified at the r points $-b_i$ and at infinity, each with inertia order p . Hence Riemann–Hurwitz gives

$$2g(C) - 2 = -2p^r + (r+1)p^r \left(1 - \frac{1}{p}\right).$$

Put $Q = 2^d$. Hasse–Weil, followed by deletion of the finitely many branch points, supplies

$$Q^n + O_{p,r}(Q^{n/2})$$

unramified $\mathbb{F}_{Q^n}$ -points. Since the map to the X -line has degree p^r , the number of resulting X -coordinates is at least

$$\frac{2^{dn} - O_{p,r}(2^{dn/2})}{p^r}.$$

The union of all proper subfields of $\mathbb{F}_{2^{dn}}$ has at most

$$\sum_{\substack{e|dn \\ e < dn}} 2^e \leq dn 2^{dn/2}$$

elements. For all sufficiently large n the former lower bound is larger. Choose an unramified X -coordinate x_n outside that union. It has exact degree dn , and its lifted coordinates satisfy $x_n + b_i = Y_i^p \neq 0$ for every i , as required. $\square$

Proposition 5.18 (Kummer-eigenline translate no-go). *Let r and p be distinct odd primes, let q be a prime power with*

$$r \mid q - 1, \quad \text{ord}_p(q) = r,$$

and put $K = \mathbb{F}_{q^r}$. Fix $\zeta \in \mathbb{F}_q^\times$ of order r , and let

$$L_\zeta = \{x \in K : x^q = \zeta x\}.$$

Every nonzero $x \in L_\zeta$ has degree r over $\mathbb{F}_q$, the element x^r lies in $\mathbb{F}_q^\times$, and

$$X^r - x^r$$

is irreducible over $\mathbb{F}_q$. Nevertheless, if

$$q - (p-1)(r-1)\sqrt{q} > p, \tag{KE}$$

there is a nonzero $x \in L_\zeta$ such that

$$x + 1 \in (K^\times)^p.$$

Thus irreducibility of the Kummer binomial, exact degree, and the Frobenius eigenrelation still do not force the ordinary singleton translate to have a nonzero p -Kummer class.

For every odd prime r , such characteristic-two countermodels occur over arbitrarily large base fields. Indeed, choose a prime p with $\text{ord}_p(2) = r$, put $t = \text{ord}_r(2)$, and take

$$q = 2^{t(1+kr)} \quad (k \geq 0).$$

Then the two displayed congruence conditions hold, and (KE) holds for all sufficiently large k .

Proof. By the normal-basis theorem, the q -Frobenius on K , viewed as an $\mathbb{F}_q$ -linear operator, is the regular representation of the cyclic group of order r . Since $\mu_r \subset \mathbb{F}_q$ and the characteristic does not divide r , its r eigenspaces are one-dimensional. Hence L_ζ is a line. A nonzero $v \in L_\zeta$ does not lie in $\mathbb{F}_q$, and, as r is prime, it has degree r . Moreover

$$(v^r)^q = v^r.$$

If $v^r = a^r$ for some $a \in \mathbb{F}_q^\times$, then $v/a \in \mu_r \subset \mathbb{F}_q$, a contradiction. Kummer's prime-degree criterion therefore makes $X^r - v^r$ irreducible. The same argument applies to $x = cv$ for every $c \in \mathbb{F}_q^\times$.

Let $\chi : K^\times \rightarrow \mu_p$ be a character of exact order p ; its kernel is $(K^\times)^p$. No element $1 + cv$, with $c \in \mathbb{F}_q$, vanishes. Character orthogonality gives

$$\begin{aligned} N &:= \#\{c \in \mathbb{F}_q : 1 + cv \in (K^\times)^p\} \\ &= \frac{1}{p} \left(q + \sum_{j=1}^{p-1} \sum_{c \in \mathbb{F}_q} \chi^j(1 + cv) \right). \end{aligned}$$

The element v^{-1} has degree r , so Katz's affine-line estimate [11] gives, for $1 \leq j < p$,

$$\left| \sum_{c \in \mathbb{F}_q} \chi^j(1 + cv) \right| = \left| \chi^j(v) \sum_{c \in \mathbb{F}_q} \chi^j(c + v^{-1}) \right| \leq (r-1)\sqrt{q}.$$

Consequently

$$N \geq \frac{q - (p-1)(r-1)\sqrt{q}}{p} > 1.$$

The value $c = 0$ accounts for only one solution, so another has $c \neq 0$; the corresponding $x = cv$ proves the first assertion.

For the final assertion, Zsigmondy's theorem (whose exponent-six exception is irrelevant for the odd prime r) supplies a prime divisor p of $2^r - 1$ with $\text{ord}_p(2) = r$. Since $t = \text{ord}_r(2) \mid r - 1$, the exponent $t(1 + kr)$ is divisible by t and coprime to r . Hence $r \mid q - 1$ and $\text{ord}_p(q) = r$. Letting k grow proves (KE). $\square$

Proposition 5.19 (tower-faithful common-section countermodels). *Let r and p be distinct odd primes, let $\nu \geq 1$, put $R = r^\nu$, and let q be even with*

$$r \mid q - 1, \quad \text{ord}_p(q) = R.$$

Choose any

$$A_* \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^r, \quad \theta^R = A_*,$$

in $K = \mathbb{F}_{q^R}$. Then $X^R - A_*$ is irreducible over $\mathbb{F}_q$. If

$$\mathcal{N} = \#\{c \in \mathbb{F}_q : 1 + c\theta \in (K^\times)^p\},$$

then

$$\mathcal{N} \geq \frac{q - (p-1)(R-1)\sqrt{q}}{p}. \quad (\text{TN}_1)$$

For every nonzero scalar counted by $\mathcal{N}$, put

$$x_\nu = c\theta, \quad x_i = x_\nu^{r^{\nu-i}} \quad (0 \leq i < \nu), \quad \alpha_c = x_0 = c^R A_*.$$

Then

$$x_i^r = x_{i-1}, \quad [\mathbb{F}_q(x_i) : \mathbb{F}_q] = r^i, \quad (\text{TN}_2)$$

and every binomial $X^{r^i} - \alpha_c$ is irreducible over $\mathbb{F}_q$. If $M = \text{ord}(\alpha_c)$, then

$$v_r(M) = v_r(q-1), \quad \text{ord}(x_i) = r^i M. \quad (\text{TN}_3)$$

Put

$$q_0 = q^{R/r}, \quad L = \frac{q^R - 1}{q_0 - 1}, \quad e = \frac{L}{p}.$$

The projective class of $1 + x_\nu$ in $K^\times / \mathbb{F}_{q_0}^\times$ misses the full p -primary part of its order, with no assumption on $v_p(L)$. More precisely, for the section polynomials of Proposition 5.8,

$$P_{\alpha_c}(Z) \mid R_{e,d}(Z) \quad (1 \leq d < R, \ r \nmid d), \quad (\text{TN}_4)$$

where P_{α_c} is the minimal polynomial over $\mathbb{F}_2$.

These countermodels occur with every Conway-like degree constraint. Fix r, ν , let p be a primitive prime divisor of $2^R - 1$, put $\tau = \text{ord}_r(2)$, and set

$$b = \tau 2^j, \quad q = 2^b.$$

For all sufficiently large j , there is a choice above for which α_c has exact absolute degree b . Then

$$[\mathbb{F}_2(x_i) : \mathbb{F}_2] = br^i, \quad \text{ord}_{r^i M}(2) = br^i, \quad (\text{TN}_5)$$

$r \nmid b$, and every prime divisor of b is smaller than r . Thus the full Kummer ancestry, smooth degree, binomial irreducibility, primary order, Lucas sections, and exact p -primary bookkeeping all admit failure. The remaining distinction is the unscaled Conway selector $c = 1$.

Proof. Because A_* is an r -nonresidue, its order contains the full r -primary part of $q - 1$. The finite-field binomial criterion therefore makes $X^R - A_*$ irreducible. Let $\chi : K^\times \rightarrow \mu_p$ have exact order p . Character orthogonality and Katz's affine-line estimate, applied to the degree- R element θ^{-1} , give

$$\mathcal{N} = \frac{1}{p} \left(q + \sum_{h=1}^{p-1} \sum_{c \in \mathbb{F}_q} \chi^h(1 + c\theta) \right),$$

$$\left| \sum_{c \in \mathbb{F}_q} \chi^h(1 + c\theta) \right| \leq (R-1)\sqrt{q}.$$

This proves (TN₁).

Multiplication by c^R does not change the r -power coset of A_* , so every α_c is again an r -nonresidue with the full r -part in its order. Applying the same binomial criterion to $X^{r^i} - \alpha_c$ proves (TN₂); the usual order induction proves (TN₃).

Since $\text{ord}_p(q_0) = r$, one has $p \mid L$ and $p \nmid q_0 - 1$. If $1 + x_\nu = z^p$, then in the cyclic group $K^\times / \mathbb{F}_{q_0}^\times$, of order L ,

$$[1 + x_\nu]^e = [z]^L = 1.$$

Thus the full p -primary valuation of the projective order is strictly deficient even when $p^2 \mid L$.

Let $B = x_{\nu-1}$ and $n = R/r$. Expanding in the $\mathbb{F}_{q_0}$ -basis $1, x_\nu, \dots, x_\nu^{r-1}$ gives

$$S_{e,k}(B) = 0 \quad (1 \leq k < r).$$

Now $B^n = \alpha_c$, and $1, B, \dots, B^{n-1}$ is an $\mathbb{F}_q$ -basis. The ancestry expansion

$$S_{e,k}(B) = \sum_{u=0}^{n-1} B^u R_{e,k+ru}(\alpha_c)$$

therefore proves every divisibility in (TN₄).

For the final family, Zsigmondy's theorem supplies p because R is odd. The choice $b = \tau 2^j$ gives $\gcd(b, R) = 1$, $r \mid q - 1$, and $\text{ord}_p(q) = R$. Take A_* primitive. The union of the proper subfields of $\mathbb{F}_q$ has at most $b\sqrt{q}$ elements, while the map $c \mapsto c^R A_*$ has fibres of size at most R . Hence a counted nonzero c with α_c of exact degree b exists whenever

$$\frac{q - (p-1)(R-1)\sqrt{q}}{p} > 1 + Rb\sqrt{q}, \quad (\text{TN}_6)$$

which holds for all sufficiently large j . Since $\mathbb{F}_2(x_i)$ contains $\mathbb{F}_2(\alpha_c) = \mathbb{F}_q$, (TN₂) gives the degree in (TN₅); (TN₃) gives its order statement. Finally $\tau \mid r - 1$, so $r \nmid b$ and every prime divisor of b is smaller than r . $\square$

Remark 5.20 (the common zero locus is large). In Proposition 5.19, define

$$D_e(Z) = \gcd_{\substack{1 \leq d < R \\ r \nmid d}} R_{e,d}(Z), \quad g_R = \gcd(R, q-1).$$

Every nonzero scalar counted by $\mathcal{N}$ gives the common root $\alpha_c = c^R A_*$, and each such value has at most g_R preimages. Thus

$$\deg D_e \geq \left\lceil \frac{q - (p-1)(R-1)\sqrt{q} - p}{pR} \right\rceil. \quad (\text{TZ}_1)$$

In the binary family of the proposition, let $M_b(D_e)$ denote the number of distinct degree- b irreducible factors. Excluding the proper-subfield values in the same count gives

$$M_b(D_e) \geq \left\lceil \max \left\{ 0, \frac{q - ((p-1)(R-1) + pRb)\sqrt{q} - p}{pRb} \right\} \right\rceil. \quad (\text{TZ}_2)$$

This tends to infinity with j . Even the exact simultaneous section pattern is therefore abundant among tower-faithful selectors; only membership of the actual Conway value in this common zero locus remains unresolved.

5.5 The selected phase at $p = 359$

Set

$$d = 19,580, \quad E = 179d = 3,504,820, \quad B = \mathbb{F}_{2^d}, \quad F = \mathbb{F}_{2^E}, \quad L = \mathbb{F}_{2^{179}}.$$

For an independent derivation of $m_{359} = 1$, $f(359) = 179$, $\mathcal{Q}(f(359)) = \mathcal{Q}(179) = \{179\}$, and the candidate element is $\beta = \kappa_{179} + 1$. To represent κ_{179} , the tower uses the lower prime row $f(179) = 178$, $\mathcal{Q}(178) = \{89\}$, $m_{179} = 1$. Thus in the tower model $F = B(x)$ with $x = \kappa_{179}$ and

$$x^{179} = \kappa_{89} + 1 \in B,$$

because the lower row $m_{179} = 1$ has already been certified.

The cheap Kummer norm down the top tower step is not the 359-test norm. Indeed,

$$N_{F/B}(x + 1) = 1 + (\kappa_{89} + 1) = \kappa_{89}$$

(evaluate the monic polynomial $T^{179} + (\kappa_{89} + 1)$ at $T = 1$, in characteristic 2). But $359 \nmid |B^\times| = 2^{19580} - 1$, since $\text{ord}_{359}(2) = 179$ and $179 \nmid 19580$. Thus the structurally cheap norm lands in exactly the field where the 359-primary obstruction is invisible.

The norm forced by Proposition 3.4 is the transverse one

$$N_{F/L}(\beta) = \prod_{i=0}^{19579} \text{Frob}^{179i}(\beta) \in L.$$

Here $\gcd(179, 19580) = 1$, so $B \cap L = \mathbb{F}_2$ and $F = BL$. The Frobenius Frob^{179} preserves B as a set but acts nontrivially on it; it is not the relative Kummer action $x \mapsto \zeta x$ over B . Consequently the known lower certificates for m_{89} and m_{179} do not propagate to m_{359} by taking the easy top-step norm.

The Wieferich caveat is absent at the next two pressure points: direct modular checks give $2^{179} \not\equiv 1 \pmod{359^2}$ and $2^{359} \not\equiv 1 \pmod{719^2}$. The order form of Corollary 3.2 is therefore equivalent to the full power criterion at $p = 359$ and at the next unknown table row $p = 719$.

Proposition 5.21 (the selected phase at $p = 359$). *Retain the notation above and put*

$$\Theta = N_{F/L}(x + 1), \quad c = \frac{2^{179} - 1}{359}, \quad A = \Theta^c \in \mu_{359}(L).$$

Then

$$2^{19580} \equiv 3 \pmod{359}, \quad v_{359}(2^{179} - 1) = 1,$$

and the independent order condition has the exact binary form

$$m_{359} = 1 \iff A \neq 1 \iff \text{Tr}_{L/\mathbb{F}_2}(A + A^{-1}) = 1. \quad (\text{SPH}_1)$$

There is also an intrinsic selected-phase resultant description. Let $a = x^{179} = \kappa_{89} + 1$, and let $P_a(T) \in \mathbb{F}_2[T]$ be its degree-19580 minimal polynomial. Over L one has

$$P_a(X^{179}) = \prod_{j=0}^{178} G_j(X), \quad G_j(X) = \prod_{i=0}^{19579} (X - x^{2^{j+179i}}), \quad (\text{SPH}_2)$$

where the G_j are the 179 Frobenius-conjugate irreducible factors of degree 19580. The common-embedding factor is G_0 , and

$$G_0(1) = \Theta, \quad \prod_{j=0}^{178} G_j(1) = P_a(1) = 1. \quad (\text{SPH}_3)$$

Thus the full resultant symmetrizes the required class away; the only remaining datum is the selected value in (SPH_1) .

Proof. Since $19580 \equiv 69 \pmod{179}$, repeated squaring gives $2^{69} \equiv 3 \pmod{359}$, proving the first congruence. The same calculation one power higher gives

$$2^{179} \equiv 10412 = 1 + 29 \cdot 359 \pmod{359^2},$$

so the displayed valuation is one. Proposition 3.4 now identifies the 359-power test on $x + 1$ with the corresponding test on $\Theta \in L^\times$. Since the 359-part of $|L^\times|$ has order exactly 359, that test is nontrivial exactly when $A \neq 1$.

The subgroup $H = \langle 2 \rangle \subset \mathbb{F}_{359}^\times$ has order 179, hence is the subgroup of quadratic residues. As $359 \equiv 3 \pmod{4}$, one has $-1 \notin H$. If $A \neq 1$, then A is a primitive 359th root, and the two absolute-Frobenius orbits of A and A^{-1} have exponent sets uH and $-uH$ for some $u \in \mathbb{F}_{359}^\times$. These sets partition $\mathbb{F}_{359}^\times$, so in characteristic two

$$\text{Tr}_{L/\mathbb{F}_2}(A + A^{-1}) = \sum_{t=1}^{358} A^t = 1.$$

For $A = 1$ the traced element is zero. This proves (SPH_1) .

The lower-row condition says that $a = \kappa_{89} + 1$ is not a 179-th power in B , while $\mu_{179} \subset B$. The standard prime Kummer criterion therefore makes $X^{179} - a$ irreducible over B ; equivalently $P_a(X^{179})$ is irreducible of degree $179 \cdot 19580$ over $\mathbb{F}_2$. Over $L = \mathbb{F}_{2^{179}}$, its roots split into the 179 orbits in (SPH_2) , each of length 19580; this proves the asserted factorization and irreducibility. Evaluating the common-embedding orbit at one gives

$$G_0(1) = \prod_{i=0}^{19579} (1 + x^{2^{179i}}) = N_{F/L}(1 + x) = \Theta.$$

Finally, evaluation of the full factorization at one gives $P_a(1)$. This is $N_{B/\mathbb{F}_2}(1 + a)$, up to the irrelevant characteristic-two sign, and $1 + a = \kappa_{89} \neq 0$; its norm lies in $\mathbb{F}_2^\times = \{1\}$. Hence (SPH_3) follows. $\square$

[source-pinned] The value $m_{359} = 1$ is present in A380496, but the trace in (SPH_1) is not independently evaluated here. Quadratic half-orbit norms do not decide it: $\sum_{h \in H} h = 0$ in $\mathbb{F}_{359}$, so their symbol product is identically one. Proposition 5.18 likewise shows why the safe-prime eigenline data alone cannot select the Conway phase.

5.6 Open problem $\mathcal{O}$

For each odd component prime $r \neq 3$, prove that the selected relative unit of Proposition 5.7 has full primary order at every divisor of $\mathcal{L}_{r,a}$. The signed conjugate-ball theorem proves the larger primary factors; the remaining smaller factors, including the independent $p = 359$ trace evaluation, are the exact unresolved core.

6 The cubic 3^k spine

Write $\zeta := \kappa_{3^k}$ and $h := 3^k$ throughout this section.

Conjecture 6.1 (C_k : Gaussian-period primitivity). *The period*

$$\gamma_k = \zeta + \zeta^{-1} \in \mathbb{F}_{2^{3^k}}^\times$$

is primitive for every $k \geq 1$.

6.1 Structure

Proposition 6.2 (cyclotomic recognition). $\zeta^{3^k} = \kappa_2$, which has order 3; hence ζ is a primitive 3^{k+1} -st root of unity. Since 2 is a primitive root modulo 3^{k+1} , the cyclotomic polynomial $\Phi_{3^{k+1}}(x) = x^{2^h} + x^h + 1$ is irreducible over $\mathbb{F}_2$, the component field is $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$, and it carries the index-2 subfield $L = \mathbb{F}_{2^h}$.

Proposition 6.3 (half-angle splitting). With $s = (3^{k+1} + 1)/2$ (the inverse of 2 modulo 3^{k+1}),

$$\kappa_{3^k} + 1 = \zeta^s (\zeta^s + \zeta^{-s}),$$

where ζ^s lies in the norm-one circle U (of order exactly 3^{k+1} within the subgroup of order $2^h + 1$) and $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders,

$$\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \text{ord}(\gamma_k), \quad \text{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

Machine-checked corollaries: $(\kappa + 1)^{2^h - 1} = \zeta^{-1}$ and $N_{F/L}(\kappa + 1) = \gamma_k$.

Proposition 6.4 (translates $m = 2, 3$). $\kappa_{3^k} + 2$ and $\kappa_{3^k} + 3$ split as in Proposition 6.3 with γ_k replaced by a Galois conjugate. Therefore $\text{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \text{ord}(\gamma_k)$ for $m = 1, 2, 3$, and $\text{ord}(\kappa_{3^k}) = 3^{k+1}$.

Proposition 6.5 (the $2 \cdot 3^k$ exception, unconditionally). Any prime p with $f(p) = 2 \cdot 3^k$ satisfies $p \mid 2^{3^k} + 1$, hence p divides neither 3^{k+1} nor $2^{3^k} - 1$, hence p never divides $\text{ord}(\kappa_{3^k} + m)$ for $m \in \{0, 1, 2, 3\}$. By the order criterion, $m_p \geq 4$. The full primitivity conjecture C_k below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g. $p = 87211$, $f = 54$.

Proposition 6.6 (the 3-power column). For a prime r with $f(r) = 3^k$, the values $m \in \{0, 2, 3\}$ are impossible: $m_r = 0$ would require $r \mid \text{ord}(\kappa_{3^k}) = 3^{k+1}$, a power of 3, while $r \equiv 1 \pmod{3^k}$ is a prime distinct from 3; and $m = 2, 3$ share $m = 1$'s order class by Proposition 6.4. Any failure of C_k therefore gives $m_r \geq 4$ directly.

Proposition 6.7 (norm tower). $\gamma_{k-1} = \gamma_k^3 + \gamma_k = N_{L_k/L_{k-1}}(\gamma_k)$ (minimal polynomial $X^3 + X + \gamma_{k-1}$), so $\text{ord}(\gamma_{k-1}) \mid \text{ord}(\gamma_k)$. With lifting-the-exponent ($v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1)$ for old primes $r \neq 3$), full-order parts propagate upward, and conjecture C_k reduces to C_{k-1} plus primitivity at the new primes $r \mid \Phi_{3^k}(2)$.

Proposition 6.8 (the norm-kernel seam). Put $q = 2^{3^{k-1}}$, so $L_k = \mathbb{F}_{q^3}$ and $L_{k-1} = \mathbb{F}_q$. There is a canonical direct product

$$L_k^\times = L_{k-1}^\times \times U_k, \quad |U_k| = q^2 + q + 1 = \Phi_{3^k}(2),$$

where $U_k = \ker N_{L_k/L_{k-1}}$. If $\gamma_k = a_k b_k$ in this product, then

$$\gamma_{k-1} = N(\gamma_k) = a_k^3,$$

and cubing is an automorphism of $L_{k-1}^\times$. Consequently

$$C_k \iff C_{k-1} \text{ and } b_k \text{ generates } U_k.$$

In particular, the norm recursion determines the old component a_k exactly but supplies no order information about the new component b_k .

Proof. The two displayed factor orders are coprime:

$$\gcd(q-1, q^2+q+1) = \gcd(q-1, 3) = 1,$$

because $q = 2^{\text{odd}} \equiv 2 \pmod{3}$. The cyclic group $L_k^\times$ therefore splits as asserted. The relative norm is $z \mapsto z^{1+q+q^2}$; it kills U_k and restricts on $L_{k-1}^\times$ to $a \mapsto a^3$, which is bijective because $3 \nmid q-1$. The final equivalence follows from coprimality of the two factor orders. $\square$

6.2 Equivalent algebraic and cyclotomic forms

Proposition 6.9 (primitive Dickson iterate). *Let $D_n(X, 1)$ be the Dickson polynomial characterized by $D_n(u + u^{-1}, 1) = u^n + u^{-n}$, and put $f(X) = X^3 + X$. Then*

$$P_k(X) := D_{3^k}(X, 1) + 1 = f^{\circ k}(X) + 1$$

is the minimal polynomial of γ_k over $\mathbb{F}_2$. In particular it is irreducible of degree 3^k , and

$$C_k \iff P_k(X) \text{ is a primitive polynomial over } \mathbb{F}_2.$$

The polynomials also obey the coefficient recursion

$$P_{k+1} = P_k^3 + P_k^2 + 1, \quad P_1 = X^3 + X + 1.$$

Proof. The Dickson composition law gives $D_{3^k} = D_3^{\circ k} = f^{\circ k}$ in characteristic two. Since ζ^{3^k} is a primitive cube root of unity,

$$D_{3^k}(\gamma_k, 1) = \zeta^{3^k} + \zeta^{-3^k} = 1.$$

On the other hand, the Frobenius orbit of $\zeta + \zeta^{-1}$ has length 3^k : the least $j > 0$ with $2^j \equiv \pm 1 \pmod{3^{k+1}}$ is 3^k . Thus the displayed monic polynomial, which has degree 3^k , is the minimal polynomial. Its root is primitive precisely when the polynomial is primitive. Finally, writing $D_{3^k} = P_k + 1$ and using $D_3(Y) = Y^3 + Y$ gives $P_{k+1} = (P_k + 1)^3 + (P_k + 1) + 1 = P_k^3 + P_k^2 + 1$. $\square$

Proposition 6.10 (the coefficient recursion does not preserve primitivity). *Over $\mathbb{F}_2$, put*

$$F = X^{10} + X^3 + 1, \quad H = F^3 + F^2 + 1 = X^{30} + X^{23} + X^{16} + X^{10} + X^9 + X^3 + 1.$$

Then F is primitive and H is irreducible, but a root of H has order

$$7161 = 3 \cdot 7 \cdot 11 \cdot 31 < 2^{30} - 1.$$

Thus neither irreducibility of the next polynomial nor primitivity of the preceding polynomial turns the recursion in Proposition 6.9 into a formal induction. Additional structure of the actual P_k sequence—for example its power-of-three degrees and cyclotomic ancestry—is required.

Proof. Rabin's criterion is certified by

$$X^{2^{10}} \equiv X \pmod{F}, \quad \gcd(F, X^{2^5} + X) = \gcd(F, X^{2^2} + X) = 1.$$

Moreover $2^{10} - 1 = 3 \cdot 11 \cdot 31$, and direct reduction gives

$$X^{1023} \equiv 1 \pmod{F},$$

while

$$\begin{aligned} X^{341} &\equiv X^7 + X^6 + X^5 + X^3 + X^2 + 1, \\ X^{93} &\equiv X^9 + X^7 + X^6 + X^3 + 1, \\ X^{33} &\equiv X^9 + X^6 + X^5 + X^3 + X^2 \end{aligned} \pmod{F}.$$

The last three residues are not one, so F is primitive.

For H , the exact Rabin reductions are

$$X^{2^{30}} \equiv X \pmod{H},$$

$$\gcd(H, X^{2^{15}} + X) = \gcd(H, X^{2^{10}} + X) = \gcd(H, X^{2^6} + X) = 1,$$

which prove irreducibility. Finally

$$X^{7161} \equiv 1 \pmod{H},$$

whereas

$$\begin{aligned} X^{2^{387}} &\equiv X^{26} + X^{23} + X^{22} + X^{16} + X^{15} + X^7 + X^6 + X^3 + 1, \\ X^{1023} &\equiv X^{20} + X^{10} + X^6 + X^3 + 1, \\ X^{651} &\equiv X^{23} + X^{21} + X^9 + X^3, \\ X^{231} &\equiv X^{23} + X^{22} + X^{21} + X^{12} + X^9 + X^8 + X^7 + X^5 + X^3 + 1 \end{aligned} \pmod{H}.$$

These residues are not one and correspond to division of 7161 by its four prime factors, so the displayed order is exact. $\square$

Proposition 6.11 (shifted cyclotomic-unit form). *Put $x_k = 1 + \zeta \in \mathbb{F}_{2^{2h}}^\times$, where $h = 3^k$ and $N = 3^{k+1}$. Then*

$$x_k^{2^h-1} = \zeta^{-1}, \quad x_k^{2^h+1} = \gamma_k, \quad \text{ord}(x_k) = N \text{ ord}(\gamma_k).$$

Consequently

$$C_k \iff \text{ord}(1 + \zeta) = 3^{k+1}(2^{3^k} - 1).$$

Thus the cubic arm is equivalently maximal order for the reduction at 2 of the cyclotomic unit $1 + \zeta_{3^{k+1}}$.

Proof. Because $2^h \equiv -1 \pmod{N}$, $x_k^{2^h} = 1 + \zeta^{-1} = \zeta^{-1}x_k$. Dividing and multiplying by x_k gives the first two identities. The factors $2^h - 1$ and $2^h + 1$ are coprime. In the corresponding direct-product decomposition of $\mathbb{F}_{2^{2h}}^\times$, the first displayed power has exact order N and the second has order $\text{ord}(\gamma_k)$, so the order of x_k is their product. $\square$

Proposition 6.12 (norm-coherent cyclotomic lift). *Let $\xi_k = \zeta_{3^{k+1}}$ in characteristic zero and $u_k = 1 + \xi_k$. Then 2 is inert in $\mathbb{Q}(\xi_k)$ and*

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_{k-1})}(u_k) = u_{k-1}.$$

The norm to the maximal real subfield is

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_k + \xi_k^{-1})}(u_k) = 2 + \xi_k + \xi_k^{-1},$$

whose reduction modulo the unique prime above 2 is γ_k . Thus Proposition 6.11 is the residue-field shadow of a norm-coherent cyclotomic-unit system in the cyclotomic $\mathbb{Z}_3$ -tower. Conditional on C_{k-1} , C_k is exactly the assertion that, for every prime $\ell \mid \Phi_{3^k}(2)$, the reduction of u_k is not an ℓ -th power.

Proof. The residue degree of 2 in $\mathbb{Q}(\xi_k)$ is $\text{ord}_{3^{k+1}}(2) = 2 \cdot 3^k = \varphi(3^{k+1})$, proving inertness. The three relative conjugates of ξ_k are $\xi_k, \xi_k\omega$, and $\xi_k\omega^2$, so

$$\prod_{j=0}^2 (1 + \xi_k\omega^j) = 1 + \xi_k^3 = 1 + \xi_{k-1}.$$

Multiplication by the complex-conjugate factor gives the real norm. The final assertion follows from Proposition 6.11: in a cyclic group an element has the full ℓ -primary part of the group order precisely when it is not an ℓ -th power. $\square$

6.3 The selected Kummer class and reciprocity boundary

Proposition 6.13 (circular-unit and ray-class form). *Let*

$$F_k = \mathbb{Q}(\xi_k + \xi_k^{-1}), \quad \lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}),$$

let $\mathfrak{p}_k = (2)$ be the unique prime of F_k above 2, and let $\mathcal{C}_k \subset \mathcal{O}_{F_k}^\times$ be the real circular-unit group. If $\sigma(\xi_k) = \xi_k^2$, then

$$c_k := \frac{\sigma(\lambda_k)}{\lambda_k} = (1 + \xi_k)(1 + \xi_k^{-1}) = 2 + \xi_k + \xi_k^{-1}$$

is a circular unit, its Galois conjugates generate a subgroup of 2-power index in $\mathcal{C}_k$, and its reduction modulo $\mathfrak{p}_k$ is γ_k . Since the residue-field unit group has odd order, that 2-power index disappears on reduction. Consequently

$$\overline{\mathcal{C}_k} = \langle \gamma_k \rangle \subset (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \simeq \mathbb{F}_{2^{3^k}}^\times, \quad (\clubsuit)$$

and hence

$$[\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle] = [(\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times : \overline{\mathcal{C}_k}]. \quad (\spadesuit)$$

In particular, $\mathcal{C}_k$ is exactly surjectivity modulo 2 of the real circular units of conductor 3^{k+1} .

More precisely, put $E_k = \mathcal{O}_{F_k}^\times$ and let $\text{Cl}_{\mathfrak{p}_k}(F_k)$ be the ray class group of finite modulus $\mathfrak{p}_k$. The ray class exact sequence gives

$$[\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle] = \frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|} [\overline{E_k} : \overline{\mathcal{C}_k}], \quad (\heartsuit)$$

where $[\overline{E_k} : \overline{\mathcal{C}_k}] \mid [E_k : \mathcal{C}_k]$. The circular-unit index theorem identifies the odd part of the latter index with the odd part of $|\text{Cl}(F_k)|$. Thus, for an odd prime $\ell \nmid |\text{Cl}(F_k)|$,

$$v_\ell([\mathbb{F}_{2^{3^k}}^\times : \langle \gamma_k \rangle]) = v_\ell\left(\frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|}\right). \quad (\diamondsuit)$$

Equation ($\heartsuit$) names the obstruction but does not reduce its difficulty: the ray class group is defined through this residue-unit quotient, and the class-number exclusion in ($\diamond$) is itself not known uniformly in k . Finally, with $h = 3^k$, $q = 2^{h/3}$, and $\tau = \sigma^{h/3}$, the relative circular unit

$$\varepsilon_k = \frac{\tau(c_k)}{c_k}, \quad N_{F_k/F_{k-1}}(\varepsilon_k) = 1,$$

reduces to η_k . Thus the exact character obstruction of Proposition 6.23 is the new relative part of the same ray-class obstruction.

Proof. The class of 2 generates $(\mathbb{Z}/3^{k+1}\mathbb{Z})^\times / \{\pm 1\}$, so σ generates $\text{Gal}(F_k/\mathbb{Q})$. The squares of the standard real circular units are generated, up to -1 , by the quotients $\sigma^j(\lambda_k)/\lambda_k$. Telescoping gives

$$\frac{\sigma^j(\lambda_k)}{\lambda_k} = \prod_{i=0}^{j-1} \sigma^i(c_k),$$

so the conjugates of c_k generate this square subgroup. Squaring is an automorphism of the odd-order residue-field unit group, so the full circular-unit group and this subgroup have the same reduction image. Reduction commutes with σ , which becomes Frobenius at the inert prime 2, and

$$\overline{c_k} = (1 + \zeta)(1 + \zeta^{-1}) = \zeta + \zeta^{-1} = \gamma_k.$$

The reductions of the conjugates are therefore $\gamma_k, \gamma_k^2, \gamma_k^{2^2}, \dots$, and they generate the single multiplicative subgroup $\langle \gamma_k \rangle$. This proves ($\clubsuit$) and ($\spadesuit$).

The exact sequence

$$E_k \longrightarrow (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \longrightarrow \text{Cl}_{\mathfrak{p}_k}(F_k) \longrightarrow \text{Cl}(F_k) \longrightarrow 1$$

proves ($\heartsuit$). The divisibility of reduction indices is immediate from $\mathcal{C}_k \subset E_k$, and the circular-unit index theorem gives ($\diamond$); see Washington [24, Ch. 8]. Finally τ generates $\text{Gal}(F_k/F_{k-1})$, so the norm of its coboundary $\tau(c_k)/c_k$ telescopes to 1. Modulo 2 it is $\gamma_k^q/\gamma_k = \eta_k$. $\square$

Proposition 6.14 (exceptional eigenspace and Hilbert-reciprocity no-go). *Put $n = 3^k$, $m = 3n$, let*

$$F = \mathbb{Q}(\xi_m + \xi_m^{-1}), \quad \mathfrak{p} = (2),$$

and write $\Delta = \text{Gal}(F/\mathbb{Q}) = \langle \sigma \rangle$, where $\sigma(\xi_m) = \xi_m^2$. Let $\ell \mid \Phi_n(2)$, and set

$$V_\ell = (\mathcal{O}_F/\mathfrak{p})^\times / ((\mathcal{O}_F/\mathfrak{p})^\times)^\ell.$$

Then V_ℓ is one-dimensional over $\mathbb{F}_\ell$, and σ acts on it as the scalar 2. Let

$$\text{ev}_2 : \mathbb{F}_\ell[\Delta] \longrightarrow \mathbb{F}_\ell, \quad \sum a_j \sigma^j \longmapsto \sum a_j 2^j.$$

The following reciprocity operators vanish on V_ℓ :

$$\text{ev}_2(1 + \sigma + \dots + \sigma^{n-1}) = 0, \quad \text{ev}_2(2 - \sigma) = 0. \quad (\text{ER}_1)$$

If $q = 2^{n/3}$ and $\tau = \sigma^{n/3}$, then

$$\text{ev}_2(1 + \tau + \tau^2) = 1 + q + q^2 = 0, \quad \text{ev}_2(\tau - 1) = q - 1 \neq 0. \quad (\text{ER}_2)$$

Moreover the plus projection of the classical Stickelberger element of $\mathbb{Q}(\xi_m)$ acts as zero on V_ℓ . Its zero is simple: for

$$\nu_n(X) = 1 + X + \cdots + X^{n-1}$$

one has

$$\nu_n(2) = 0, \quad \nu'_n(2) = n2^{n-1} \not\equiv 0 \pmod{\ell}. \quad (\text{ER}_3)$$

Let

$$c = (1 + \xi_m)(1 + \xi_m^{-1}), \quad \gamma = \bar{c} \in \mathbb{F}_{2^n}^\times, \quad \mathcal{A}_{k,\ell} = \gamma^{(2^n-1)/\ell} \in \mu_\ell. \quad (\text{ER}_4)$$

Then cubic failure at ℓ is exactly $\mathcal{A}_{k,\ell} = 1$. Although $\mathcal{A}_{k,\ell}$ is a local Kummer/Artin symbol, global Hilbert reciprocity imposes no condition on this one distinguished value. More precisely, put $E = F(\mu_\ell)$. The prime $\mathfrak{p}$ splits into $\ell - 1$ primes in E/F . If $\mathfrak{P}_0 \mid \mathfrak{p}$ is chosen so that

$$(c, 2)_{\mathfrak{P}_0, \ell} = \mathcal{A}_{k,\ell}$$

for the reciprocity convention sending the uniformizer 2 to arithmetic Frobenius, then the $\ell - 1$ symbols above two are

$$(c, 2)_{a\mathfrak{P}_0, \ell} = \mathcal{A}_{k,\ell}^a, \quad a \in \mathbb{F}_\ell^\times. \quad (\text{ER}_5)$$

Every local symbol away from two is one, and Hilbert reciprocity reduces to

$$\prod_{a \in \mathbb{F}_\ell^\times} \mathcal{A}_{k,\ell}^a = \mathcal{A}_{k,\ell}^{\ell(\ell-1)/2} = 1, \quad (\text{ER}_6)$$

which holds for every possible value of $\mathcal{A}_{k,\ell}$. Thus norm coherence, the classical Stickelberger ideal, and global Kummer reciprocity cannot distinguish failure from nonfailure. A completion must compute a first-order unit regulator in the simple exceptional eigenspace, rather than another ideal factorization or product formula.

Proof. The standard cyclotomic-divisor criterion gives $\text{ord}_\ell(2) = n$. Since $(\mathcal{O}_F/\mathfrak{p})^\times \simeq \mathbb{F}_{2^n}^\times$, its quotient by ℓ -th powers has order ℓ , and Frobenius acts by $[x] \mapsto [x^2] = 2[x]$. This proves the module assertion. Equations (ER₁) follow from $2^n = 1 \pmod{\ell}$. The element $q = 2^{n/3}$ has exact order three modulo ℓ , proving (ER₂).

Let $G = \text{Gal}(\mathbb{Q}(\xi_m)/\mathbb{Q})$, let j be complex conjugation, and write

$$\theta_m = \frac{1}{m} \sum_{\substack{1 \leq a < m \\ (a,m)=1}} a \sigma_a^{-1}$$

for the Stickelberger element. Pairing a with $m - a$ gives

$$(1 + j)\theta_m = \sum_{g \in G} g.$$

On an even representation, the right side is twice the norm element of Δ , whose evaluation at two is zero by (ER₁). Since $\ell \neq 2, 3$, division by two and by m is valid modulo ℓ . Hence θ_m , and therefore every classical Stickelberger-ideal multiple controlling Gauss–Jacobi-sum ideals, acts trivially on V_ℓ . The particularly relevant integrality factor $2 - \sigma$ vanishes separately. Differentiating

$$(X - 1)\nu_n(X) = X^n - 1$$

at $X = 2$ gives (ER₃), since $\ell \nmid 2n$.

The reduction of c is γ . In the local field $F_{\mathfrak{p}}$, which contains μ_ℓ , adjoining an ℓ -th root z of c is the unramified Kummer extension selected by the residue of c . Arithmetic Frobenius satisfies

$$\frac{\text{Frob}(z)}{z} = \bar{z}^{2^n-1} = \gamma^{(2^n-1)/\ell},$$

proving (ER₄) and the failure criterion. Also, for the relative circular unit $\varepsilon = \tau(c)/c$,

$$(\varepsilon, 2)_{\mathfrak{p}, \ell} = \mathcal{A}_{k, \ell}^{q-1};$$

because $q-1 \not\equiv 0 \pmod{\ell}$, the relative coboundary carries exactly the same unknown symbol and supplies no new relation.

It remains to audit global reciprocity. The completion $F_{\mathfrak{p}}$ is the unramified degree- n extension of $\mathbb{Q}_2$. Since $\mathbb{Q}_2(\mu_\ell)/\mathbb{Q}_2$ is also unramified of degree $\text{ord}_\ell(2) = n$, it is already contained in $F_{\mathfrak{p}}$. The conductors m and ℓ are coprime, so $F \cap \mathbb{Q}(\mu_\ell) = \mathbb{Q}$; consequently $[E : F] = \ell - 1$, and $\mathfrak{p}$ splits completely in E/F . Galois functoriality of the Hilbert symbol gives (ER₅).

We now check every place above ℓ , the only point at which wild local reciprocity might have supplied extra information. Let K_0 be the maximal unramified subfield of an ℓ -adic completion of E . That completion is $K = K_0(\mu_\ell)$, with absolute ramification index $e = \ell - 1$, and both c and 2 belong to $K_0^\times$. For any unit $a \in K_0^\times$, write

$$a = [\bar{a}] \langle a \rangle, \quad \langle a \rangle \in 1 + \ell \mathcal{O}_{K_0}.$$

The Teichmüller factor has order prime to ℓ , hence is an ℓ -th power. If $U_i = 1 + \mathfrak{m}_K^i$, then

$$1 + \ell \mathcal{O}_{K_0} \subset U_{\ell-1}.$$

The standard Hilbert-filtration orthogonality for a field containing μ_ℓ is [10]

$$(U_i, U_j)_\ell = 1 \quad \text{when} \quad i + j > \frac{\ell e}{\ell - 1} = \ell. \quad (\text{ER}_7)$$

Taking $i = j = \ell - 1$ proves $(c, 2)_{\lambda, \ell} = 1$ at every $\lambda \mid \ell$. At every finite place away from 2ℓ , both entries are units and the tame Hilbert-symbol formula is one; archimedean places contribute nothing because ℓ is odd. The global product formula is therefore precisely (ER₆). Since $\sum_{a=1}^{\ell-1} a \equiv 0 \pmod{\ell}$, this relation is tautological and leaves the single symbol at $\mathfrak{P}_0$ arbitrary. $\square$

Proposition 6.15 (the simple exceptional zero does not evaluate its Bockstein). *Retain Proposition 6.14. Put*

$$\mathcal{A} = \mathbb{Z}_\ell[\Delta], \quad \mathfrak{m} = (\ell, \sigma - 2), \quad \nu = 1 + \sigma + \cdots + \sigma^{n-1},$$

and let $u = (2^n - 1)/\ell \in \mathbb{Z}_\ell$. Then

$$\nu \equiv \ell u + n2^{n-1}(\sigma - 2) \pmod{\mathfrak{m}^2}. \quad (\text{BZ}_1)$$

The coefficient $n2^{n-1}$ is a unit modulo ℓ , but this says only that the first-order exceptional coordinate is one-dimensional; it does not determine the cyclotomic unit's value in that coordinate. Indeed $(\sigma - 1)\nu = \sigma^n - 1 = 0$, while $\sigma - 1$ is a unit modulo $\mathfrak{m}^2$. Thus (BZ₁) is the cotangent relation

$$0 = \ell u + n2^{n-1}(\sigma - 2) \quad \text{in } \mathcal{A}/\mathfrak{m}^2, \quad (\text{BZ}'_1)$$

not a nonzero value of the sought unit class.

More sharply, let $K/\mathbb{Q}_2$ be the unramified extension of degree n , and let

$$\lambda = (1 - \xi_m)(1 - \xi_m^{-1}) = tv, \quad t \in \mu_{2^n-1}, \quad v \in 1 + 2\mathcal{O}_K$$

be its Teichmüller/principal-unit decomposition at two. One has $N_{K/\mathbb{Q}_2}(t) = 1$ and $N_{K/\mathbb{Q}_2}(v) = 3$. Conversely, for every $t_0 \in \mu_{2^n-1}$ there is $v_0 \in 1 + 2\mathcal{O}_K$ such that

$$\lambda_0 = t_0 v_0, \quad N_{K/\mathbb{Q}_2}(\lambda_0) = 3, \quad \overline{\lambda_0} = \overline{t_0}. \quad (\text{BZ}_2)$$

The residue on the right may have any ℓ -power class. Hence the norm-three identity, Frobenius action, and the nonzero derivative in (BZ_1) cannot by themselves prove cubic nonfailure. The missing datum is the distinguished cyclotomic unit's actual coordinate — for example through a derived local-unit regulator — not the formal existence of the Bockstein coordinate. This is a local countermodel to those data only; it is not a global cyclotomic counterexample.

Proof. Write $\epsilon = \sigma - 2$. Modulo $(\ell^2, \ell\epsilon, \epsilon^2) = \mathfrak{m}^2$, binomial expansion gives

$$\sum_{j=0}^{n-1} (2 + \epsilon)^j \equiv (2^n - 1) + \nu'_n(2)\epsilon \equiv \ell u + n2^{n-1}\epsilon.$$

Here the discarded part of $\nu'_n(2) = n2^{n-1} - (2^n - 1)$ lies in $\ell\epsilon$. Since $\text{ord}_\ell(2) = n$, one has $n \mid \ell - 1$, so $n2^{n-1} \not\equiv 0 \pmod{\ell}$. This proves (BZ_1) . The identity $(\sigma - 1)\nu = 0$ and $\sigma - 1 = 1 + \epsilon \in (\mathcal{A}/\mathfrak{m}^2)^\times$ prove (BZ'_1) ; its one nonzero linear coefficient leaves a one-dimensional cotangent space.

The real cyclotomic norm of λ is three. The norm of every Teichmüller unit is its residue-field norm, so $N(t) = t^{2^n-1} = 1$, leaving $N(v) = 3$. Conversely the norm map from the principal units of an unramified local extension onto the base principal units is surjective. Given t_0 , choose v_0 with norm three; then (BZ_2) follows. In addition $\sigma(t_0 v_0)/(t_0 v_0)$ reduces to $t_0^2/t_0 = t_0$, so even the relative-coboundary form carries the same arbitrary residue class. Choosing t_0 inside or outside the subgroup of ℓ -th powers proves the asserted neutrality. $\square$

Proposition 6.16 (the derivative regulator is the original Kummer class). *Retain the notation of Proposition 6.13, put $n = 3^k$, and write*

$$\nu_n(X) = 1 + X + \cdots + X^{n-1}, \quad \mathcal{R}_k^\partial := \prod_{j=0}^{n-1} \sigma^j(c_k)^j = c_k^{\sigma \nu'_n(\sigma)}.$$

Then the first-order circular-unit regulator singled out by the simple exceptional zero is explicit:

$$\mathcal{R}_k^\partial = \frac{\lambda_k^n}{3}, \quad \overline{\mathcal{R}_k^\partial} = \gamma_k^n. \quad (\text{DR}_1)$$

Consequently, for every prime $\ell \mid \Phi_n(2)$, its reduction is an ℓ -th power if and only if γ_k is. Thus the derivative regulator carries exactly the unknown Kummer class in C_k : the nonzero derivative identifies the coordinate but does not evaluate it.

For $k \geq 2$ this identification is norm coherent in the sharp form

$$N_{F_k/F_{k-1}}(\mathcal{R}_k^\partial) = (\mathcal{R}_{k-1}^\partial)^3. \quad (\text{DR}_2)$$

Hence

$$\mathcal{E}_k^\partial := \frac{\mathcal{R}_k^\partial}{\mathcal{R}_{k-1}^\partial} = \left(\frac{\lambda_k^3}{\lambda_{k-1}} \right)^{n/3}$$

has relative norm one. At every current $\ell \mid \Phi_n(2)$, the lower-field factor is automatically an ℓ -th power, so $\overline{\mathcal{E}_k^\partial}$ again carries precisely the Kummer class of γ_k . In particular, passing to a derived relative unit does not create an inductive relation capable of proving C_k .

Proof. Since $c_k = \sigma(\lambda_k)/\lambda_k$, the weighted product telescopes cyclically:

$$\prod_{j=0}^{n-1} \left(\frac{\sigma^{j+1}(\lambda_k)}{\sigma^j(\lambda_k)} \right)^j = \frac{\lambda_k^{n-1}}{\prod_{i=1}^{n-1} \sigma^i(\lambda_k)} = \frac{\lambda_k^n}{N_{F_k/\mathbb{Q}}(\lambda_k)}.$$

The real cyclotomic norm of λ_k is 3, proving the first identity in (DR₁). Modulo the prime above two, both λ_k and c_k reduce to γ_k , while 3 reduces to one; hence the second identity follows. Because $\text{ord}_\ell(2) = n$, one has $\ell \nmid n$, so raising the one-dimensional ℓ -power quotient to the n -th power is an automorphism. This proves the equivalence of Kummer classes.

Finally, $N_{F_k/F_{k-1}}(\lambda_k) = \lambda_{k-1}$. Taking the relative norm of $\lambda_k^n/3$, and remembering that the relative degree is three, gives

$$N_{F_k/F_{k-1}}(\mathcal{R}_k^\partial) = \frac{\lambda_{k-1}^n}{3^3} = \left(\frac{\lambda_{k-1}^{n/3}}{3} \right)^3,$$

which is (DR₂). The displayed formula and norm-one property of $\mathcal{E}_k^\partial$ follow immediately. Since $\ell \nmid 2^{n/3} - 1$, exponentiation by ℓ is an automorphism of $\mathbb{F}_{2^{n/3}}^\times$; every lower-field factor is therefore an ℓ -th power. The relative unit has the same current Kummer class, as claimed. $\square$

Proposition 6.17 (semidirect ray extension forced by cubic failure). *Retain Proposition 6.13, put $n = 3^k$, and let $\ell \mid \Phi_{3^k}(2)$. Assume $\ell \nmid |\text{Cl}(F_k)|$. Then*

$$\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell \tag{RF}$$

if and only if there exists a cyclic extension M/F_k of degree ℓ , conductor exactly $\mathfrak{p}_k = (2)$, totally tamely ramified there and unramified at every other finite prime. This extension is Galois over $\mathbb{Q}$, with

$$\text{Gal}(M/\mathbb{Q}) \cong C_\ell \rtimes C_n, \quad \sigma\tau\sigma^{-1} = \tau^2. \tag{RF_1}$$

Moreover, the unique prime of F_k above three splits completely in M/F_k . Both prescribed local behaviors are admissible: at two this is the standard tame extension with inertia C_ℓ , residue degree n , and Frobenius action $\tau \mapsto \tau^2$; at three the C_ℓ -layer is split. Thus secant saturation and circular-unit reciprocity give no local contradiction. Without the class-number hypothesis, cubic failure yields the sharp alternative

$$\ell \mid |\text{Cl}(F_k)| \quad \text{or the extension in (RF}_1\text{) exists.} \tag{RF_2}$$

Proof. Write $E_k = \mathcal{O}_{F_k}^\times$ and $\mathcal{C}_k$ for the circular units. Proposition 6.13 identifies the reduction of $\mathcal{C}_k$ with $\langle \gamma_k \rangle$. If (RF) holds, then $\mathcal{C}_k$ maps trivially to

$$V := \mathbb{F}_{2^n}^\times / (\mathbb{F}_{2^n}^\times)^\ell \cong C_\ell.$$

The odd part of $[E_k : \mathcal{C}_k]$ is the odd part of the class number. Under the stated hypothesis its ℓ -part vanishes, so E_k also maps trivially to V . The ray sequence then supplies an order- ℓ quotient of $\text{Cl}_{\mathfrak{p}_k}(F_k)$. Because the ordinary class group has no ℓ -part, the quotient lies entirely in tame inertia at $\mathfrak{p}_k$. Class field theory gives the asserted M/F_k .

The quotient is Galois-stable. The generator σ of $\text{Gal}(F_k/\mathbb{Q})$ acts on the residue field as $x \mapsto x^2$, hence on V by exponent two. Since $\text{ord}_\ell(2) = n$, this action is faithful; Schur–Zassenhaus, or the coprimality of n and ℓ , gives (RF₁). Conversely, tame inertia in any such extension gives the unique quotient $\mathbb{F}_{2^n}^\times \twoheadrightarrow C_\ell$. Global units lie in its kernel, so in particular γ_k is an ℓ -th power.

Finally let

$$\lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}), \quad c_k = (1 + \xi_k)(1 + \xi_k^{-1}).$$

The element λ_k generates the unique prime $\mathfrak{r}_k$ above three, and $\lambda_k/c_k \equiv 1 \pmod{\mathfrak{p}_k}$, because both reduce to $(1 + \zeta)(1 + \zeta^{-1}) = \gamma_k$. Hence $\mathfrak{r}_k = (\lambda_k/c_k)$ is trivial in the full ray class group and splits in M/F_k . The local statements now follow from $\ell \mid 2^n - 1$ and the displayed semidirect action. If the class-number hypothesis fails, the first alternative in (RF₂) holds; otherwise the preceding construction proves the second. $\square$

6.4 Kernel tower and quantitative order bounds

Proposition 6.18 (explicit cubic kernel tower). *Define $\eta_0 = 0$. For $k \geq 1$ put*

$$q_k = 2^{3^{k-1}}, \quad \eta_k = \gamma_k^{q_k-1}, \quad U_k = \ker \left(N_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}} \right).$$

Then $\eta_k \in U_k$, $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$, and, with $\omega = \zeta^{3^k} \in \mathbb{F}_4$,

$$q_k \equiv 3^k - 1 \pmod{3^{k+1}}, \quad \eta_k = \frac{\omega + \omega^2 \zeta^2}{1 + \zeta^2}, \quad \zeta^2 = \frac{\eta_k + \omega}{\eta_k + \omega^2}.$$

Moreover η_k has the irreducible cubic equation over $\mathbb{F}_{q_k} = \mathbb{F}_{2^{3^{k-1}}}$

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1 = 0. \quad (*)$$

Equivalently,

$$N(\eta_k) = 1, \quad \text{Tr}(\eta_k) = \eta_{k-1}, \quad e_2(\eta_k) = \eta_{k-1} + 1.$$

It also satisfies

$$\eta_k^{q_k+1} + \eta_k + 1 = 0, \quad \eta_k + 1 = \eta_k^{q_k+1}, \quad \text{ord}(\eta_k + 1) = \text{ord}(\eta_k). \quad (\dagger)$$

Thus η_k lies in the planar Singer difference set

$$\mathcal{H}_k = \{y \in U_k : y^{q_k+1} + y + 1 = 0\}$$

of size $q_k + 1$ inside U_k [16]. Finally,

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\eta_k) = q_k^2 + q_k + 1. \quad (**)$$

(Here C_0 is the tautological assertion that 1 generates $\mathbb{F}_2^\times$.) Thus the new-prime obstruction is the single assertion that the distinguished root of $(*)$ is a generator of the cubic norm-one torus (equivalently, multiplication by it is a Singer cycle).

Proof. The congruence starts with $2 \equiv 3 - 1 \pmod{9}$ and lifts by cubing: if $a \equiv -1 + 3^k \pmod{3^{k+1}}$, then $a^3 \equiv -1 + 3^{k+1} \pmod{3^{k+2}}$. Hence $\zeta^{q_k} = \omega \zeta^{-1}$. Dividing $\gamma_k^{q_k}$ by γ_k now gives the stated Möbius formula; solving it gives the inverse formula.

Put $t = \zeta^2$ and $M(T) = (\omega + \omega^2 T)/(1 + T)$. The same formula one level down says $\eta_{k-1} = M(t^3)$ (also for $k = 1$, when it gives $\eta_0 = 0$). If $a = \eta_{k-1}$ and $X = M(t)$, then

$$\left(\frac{X + \omega}{X + \omega^2} \right)^3 = \frac{a + \omega}{a + \omega^2}.$$

Cross-multiplication and $\omega^2 + \omega + 1 = 0$ reduce exactly to (*). Also $\eta_k^{1+q_k+q_k^2} = \gamma_k^{q_k^3-1} = 1$, so $\eta_k \in U_k$. It cannot lie in $\mathbb{F}_{q_k}$: otherwise its norm-one condition would give $\eta_k^3 = 1$, hence $\eta_k = 1$ because $3 \nmid q_k - 1$; the inverse Möbius formula would then give $t = \omega$, contradicting $\text{ord}(t) = 3^{k+1} > 3$. Thus (*) is irreducible, and its coefficients give the three symmetric identities.

The earlier minimal polynomial $X^3 + X + \gamma_{k-1}$ gives $\text{Tr}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}(\gamma_k) = 0$. Dividing $\gamma_k + \gamma_k^{q_k} + \gamma_k^{q_k^2} = 0$ by γ_k yields $1 + \eta_k + \eta_k^{q_k+1} = 0$, proving the first two parts of (†). Since $\gcd(q_k + 1, q_k^2 + q_k + 1) = 1$, exponentiation by $q_k + 1$ is an automorphism of U_k , proving the equality of orders. The cited difference-set identification is the norm-one realization of the planar Singer set.

In the direct product of Proposition 6.8, raising to $q_k - 1$ kills the old-field component and is an automorphism on U_k , since $\gcd(q_k - 1, q_k^2 + q_k + 1) = 1$. Therefore the kernel component of γ_k generates U_k exactly when η_k does; Proposition 6.8 gives (**). $\square$

Proposition 6.19 (Singer-orbit sieve). *Let $k \geq 1$. Put $n = 3^k$, $q = 2^{n/3}$, and $N = q^2 + q + 1 = |U_k|$. If a prime $\ell \mid N$ satisfies*

$$\ell > \frac{N}{n(n-1)+1},$$

then $\eta_k \notin U_k^\ell$. Consequently, if $\ell^a \parallel N$, then

$$v_\ell(\text{ord } \eta_k) = a.$$

Unconditionally,

$$\text{ord}(\eta_k) \geq n(n-1)+1, \quad [U_k : \langle \eta_k \rangle] \leq \frac{N}{n(n-1)+1}.$$

Proof. Let

$$D = \{y \in U_k : y^{q+1} + y + 1 = 0\}.$$

This is the cyclic $(N, q+1, 1)$ Singer difference set: every nonidentity element of U_k has a unique expression xy^{-1} with distinct $x, y \in D$. Since the minimal polynomial of η_k over $\mathbb{F}_2$ has degree n , its Frobenius orbit

$$A = \{\eta_k^{2^i} : 0 \leq i < n\}$$

has n elements. The equation defining D is Frobenius-stable, so $A \subset D$.

Suppose $\eta_k \in H := U_k^\ell$. The subgroup H is characteristic in the cyclic group U_k , hence Frobenius-stable, and $A \subset H$. The $n(n-1)$ quotients xy^{-1} for ordered distinct pairs $(x, y) \in A^2$ are nonidentity elements of H and are all distinct by the Singer difference-set property. Therefore

$$|H| - 1 \geq n(n-1).$$

Since $|H| = N/\ell$, this contradicts the displayed strict inequality. In a cyclic group of order $\ell^a m$, $\ell \nmid m$, lying outside the subgroup of ℓ -th powers is equivalent to having the full ℓ^a part in one's order, proving the first assertion.

Finally the same distinct quotients all lie in $\langle \eta_k \rangle$, whether or not an ℓ is fixed. Thus $\text{ord}(\eta_k) - 1 \geq n(n-1)$, which gives the last two inequalities. $\square$

Remark 6.20 (why the translated orbit does not double the sieve). The identity $y + 1 = y^{q+1}$ puts the translated Frobenius orbit $A + 1 = A^{q+1}$ in the same subgroup as A . It supplies no second quotient family: modulo N one has $(q+1)^2 \equiv q$ and $q(q+1) \equiv -1$, while the quotient set of A is stable under both q -th powers and inversion. Hence the internal quotient sets of A and $A + 1$ coincide. Cross quotients already collide, for example $(y+1)/y = y^q$. Improving Proposition 6.19 therefore requires cross-correlation or character information beyond the $(N, q+1, 1)$ difference-set axiom.

Proposition 6.21 (partition-order bound). *Retain Proposition 6.19. Let $B(n)$ be the number of partitions into distinct positive parts not divisible by 3 whose total is at most $n = 3^k$. Then*

$$\text{ord}(\eta_k) \geq B(n).$$

In particular, if $p_3^{\text{dist}}(n)$ counts such partitions of n itself, then

$$\text{ord}(\eta_k) \geq p_3^{\text{dist}}(n) = \exp\left(\left(\frac{\sqrt{2}\pi}{3} + o(1)\right)\sqrt{n}\right). \quad (\text{B})$$

If a prime $\ell \mid N$ satisfies

$$\ell > \frac{N}{B(n)},$$

then $\eta_k \notin U_k^\ell$, so the order of η_k contains the full ℓ -primary part of N .

Proof. Put $r = 3n$, $t = \zeta^2$, and retain $\omega = \zeta^n$, so $t^n = \omega^2$ and

$$\eta_k = M(t), \quad M(X) = \frac{\omega + \omega^2 X}{1 + X}.$$

The element 2 generates $(\mathbb{Z}/r\mathbb{Z})^\times$, of order $2n$. For every positive j not divisible by 3, choose z_j with $2^{z_j} \equiv j \pmod{r}$. Frobenius gives

$$\eta_k^{2^{z_j}} = \omega^j \frac{1 + \omega^j t^j}{1 + t^j}. \quad (6)$$

For a finite set I of such j , put

$$s_I = \sum_{j \in I} j, \quad E_I = \sum_{j \in I} 2^{z_j}, \quad A_I(X) = \prod_{j \in I} (1 + \omega^j X^j), \quad B_I(X) = \prod_{j \in I} (1 + X^j).$$

Thus

$$\eta_k^{E_I} = \omega^{s_I} \frac{A_I(t)}{B_I(t)}. \quad (7)$$

All denominators are nonzero when $s_I \leq n < r$.

Suppose $s_I, s_J \leq n$ and $\eta_k^{E_I} = \eta_k^{E_J}$. Cross-multiplication gives

$$F(X) := \omega^{s_I} A_I(X) B_J(X) + \omega^{s_J} A_J(X) B_I(X), \quad F(t) = 0. \quad (8)$$

Now $M(t^{-1}) = M(t) + 1 = \eta_k + 1$, and the Singer equation $\eta_k + 1 = \eta_k^{q+1}$ shows that the assumed equality also holds after replacing η_k by $M(t^{-1})$. Hence $F(t^{-1}) = 0$. Over $\mathbb{F}_4$, the two elements t, t^{-1} have the distinct degree- n minimal polynomials

$$X^n + \omega^2, \quad X^n + \omega,$$

because $\text{ord}_r(4) = n$. Therefore

$$X^{2n} + X^n + 1 \mid F(X). \quad (9)$$

But $\deg F \leq s_I + s_J \leq 2n$. In the only possible equality case $s_I = s_J = n$, the two leading coefficients are both ω^{2n} and cancel in characteristic two. Thus $\deg F < 2n$, and (9) forces $F = 0$.

The constant coefficient first gives $\omega^{s_I} = \omega^{s_J}$. Cancel that scalar and all factors belonging to $I \cap J$. If j_0 is the least remaining part, comparison of the X^{j_0} -coefficients gives $\omega^{j_0} = 1$: on one

side j_0 occurs in an A -factor, with coefficient ω^{j_0} , and on the other in a B -factor, with coefficient 1; larger parts cannot contribute. This is impossible because $3 \nmid j_0$. Hence $I = J$.

The $B(n)$ powers in (7) are therefore pairwise distinct, proving the order bound. Meinardus's partition theorem [17], applied to the Euler product

$$G(x) := \prod_{3 \nmid j} (1 + x^j) = \frac{\prod_{j \geq 1} (1 + x^j)}{\prod_{j \geq 1} (1 + x^{3j})}, \quad \log G(e^{-t}) \sim \frac{\pi^2}{18t},$$

gives (B). Finally, if $\eta_k \in U_k^\ell$, all $B(n)$ displayed powers lie in the subgroup of order N/ℓ . The strict inequality rules this out; the cyclic-group argument of Proposition 6.19 then gives the full ℓ -primary assertion. $\square$

Remark 6.22 (ceiling of the height-budgeted amplification). The preceding construction is scale-optimal within the following precise class of degree arguments. For $J = (\mathbb{Z}/3n\mathbb{Z})^\times / \{\pm 1\}$, assign an atom indexed by $j \in J$ the height

$$h(j) := \min\{|b| : b \equiv \pm j \pmod{3n}\}.$$

Suppose a family of integer exponent vectors is certified pairwise distinct only by bounding the Laurent span after cross-multiplication with the triangle estimate

$$w(e - e') := \sum_{j \in J} h(j) |e_j - e'_j| \leq 2n$$

and then invoking the two degree- n minimal polynomials as above (including the verified leading-term cancellation at the boundary). Then the family has cardinality $\exp(O(\sqrt{n}))$.

Indeed, fix one vector e_0 . Every difference $e - e_0$ lies in the displayed weighted ℓ^1 -ball. Each positive height occurs only $O(1)$ times, so the number of such lattice points is bounded by the coefficients through degree $2n$ of a fixed power of

$$\prod_{m \geq 1} (1 + 2x^m + 2x^{2m} + \cdots) \preccurlyeq \prod_{m \geq 1} (1 - x^m)^{-2}.$$

Colored-partition bounds give $\exp(O(\sqrt{n}))$. Polynomially many Frobenius or cyclic shifts do not change that scale, whereas $|U_k| = \exp((2 \log 2/3 + o(1))n)$. Thus more triangle-budgeted subset encodings cannot prove primitivity. This does *not* rule out systematic factor cancellations or exact reduction modulo Φ_{3n} ; those require new arithmetic input rather than the generic degree budget.

6.5 The exact remaining character test

Proposition 6.23 (no transfer and the exact character test). *The kernel orders are pairwise coprime: for $1 \leq i < k$,*

$$\gcd(\Phi_{3^i}(2), \Phi_{3^k}(2)) = 1.$$

For each prime $\ell \mid |U_k|$, let $\chi_{k,\ell} : U_k \rightarrow \mu_\ell$ be a character of exact order ℓ . Then

$$\mathrm{Hom}\left(\mathbb{F}_{2^{3k-1}}^\times, \mu_\ell\right) = 1; \tag{\dagger_0}$$

no multiplicative character or power-residue datum from a previous level can transfer an ℓ -part to level k . Moreover,

$$C_k \iff C_{k-1} \text{ and } \chi_{k,\ell}(\eta_k) \neq 1 \text{ for every prime } \ell \mid |U_k|. \tag{\dagger}$$

Equivalently, with

$$\rho_{k,\ell}(A) = \chi_{k,\ell}(A^{q_k-1}) \quad (A \in \mathbb{F}_{q_k}^\times),$$

where $\rho_{k,\ell}$ is trivial on $\mathbb{F}_{q_k}^\times$, the missing assertion is $\rho_{k,\ell}(\gamma_k) \neq 1$ for every such ℓ . This is the smallest power-residue form of the cubic obstruction.

Proof. If a prime r divided two displayed cyclotomic values, then $r \neq 3$ and the standard cyclotomic-divisor criterion would give both $\text{ord}_r(2) = 3^i$ and $\text{ord}_r(2) = 3^k$, a contradiction. Thus every $\ell \mid |U_k|$ is coprime to $|\mathbb{F}_{2^{3k-1}}^\times| = \prod_{j < k} \Phi_{3^j}(2)$, proving $(\dagger_0)$. In the cyclic group U_k , the kernel of an exact order- ℓ character is U_k^ℓ . If $|U_k| = \ell^a m$ with $\ell \nmid m$, then an element has the full ℓ^a in its order exactly when it is not in U_k^ℓ . Applying this for every prime divisor and using $(**)$ proves $(\dagger)$. The formula for $\rho_{k,\ell}$ and $\eta_k = \gamma_k^{q_k-1}$ give the final equivalence. $\square$

Proposition 6.24 (short cyclotomic-period form of the cubic obstruction). *Let $k \geq 1$, put $n = 3^k$, $q = 2^{n/3}$, and let ℓ be a prime divisor of $\Phi_{3^k}(2)$. Let $\rho = \rho_{k,\ell}$ be the exact ℓ -character from Proposition 6.23, realized with values in the complex roots of unity. In $\mathbb{F}_{2^{2n}}$ let μ_{3n} denote the roots of unity of order dividing $3n = 3^{k+1}$, and set*

$$\mathcal{S}_{k,\ell} := \sum_{\substack{t \in \mu_{3n} \\ t \neq 1}} \rho(t + t^{-1}). \quad (\text{S})$$

Every argument of ρ in this sum is a nonzero element of $\mathbb{F}_{2^n}$. If $\rho(\gamma_k) = \zeta_\ell^a$, then

$$\mathcal{S}_{k,\ell} = (n-1) + 2 \sum_{i=0}^{n-1} \zeta_\ell^{a2^i}. \quad (\text{P})$$

Consequently

$$\chi_{k,\ell}(\eta_k) = 1 \iff \mathcal{S}_{k,\ell} = 3n - 1. \quad (\text{C})$$

Thus the surviving datum is exactly a short Gauss period over the order- n subgroup generated by 2 in $(\mathbb{Z}/\ell\mathbb{Z})^\times$.

Proof. Since $2^n \equiv -1 \pmod{3n}$, every $t + t^{-1}$ in (S) is fixed by 2^n and lies in $\mathbb{F}_{2^n}$; it is nonzero unless $t = 1$. The $n-1$ nonidentity elements of μ_n give real cyclotomic elements in the lower field $\mathbb{F}_{2^{n/3}}^\times$, on which ρ is trivial.

The remaining $2n$ elements are the primitive $3n$ -th roots. Pairing them by inversion leaves the n Frobenius conjugates $\gamma_k^{2^i}$, each twice. This proves (P). Since $q-1$ is invertible modulo ℓ , $\chi_{k,\ell}(\eta_k) = 1$ exactly when $\rho(\gamma_k) = 1$, which gives the forward implication in (C). Conversely every summand in (S) has complex absolute value one. Equality with the number of summands $3n-1$ forces equality in the triangle inequality and hence forces every summand to be 1, in particular $\rho(\gamma_k) = 1$. $\square$

Remark 6.25 (why a full Singer character sum is too coarse). The Singer difference-set character sum controls a set of size $q+1$ and has the usual square-root magnitude. Equation (P) instead localizes to only $3n-1$ cyclotomic terms, with the unknown part supported on the order- n subgroup $\langle 2 \rangle$ modulo ℓ . Generic Weil or difference-set bounds do not evaluate this localized scalar; an all-level proof must rule out its extremal value in (C).

Proposition 6.26 (cyclotomic-secant saturation). *Retain $n = 3^k$, put $m = 3n$, and let $K = \mathbb{F}_{2^{2n}}$ contain a primitive m -th root ζ . For every prime $\ell \mid \Phi_{3^k}(2)$, the following are equivalent:*

- (i) $\eta_k \in U_k^\ell$;
- (ii) $\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell$;
- (iii) $1 + \zeta \in (K^\times)^\ell$;
- (iv) for every $a \not\equiv b \pmod{m}$, $\zeta^a + \zeta^b \in (K^\times)^\ell$.

Thus, for an exact order- ℓ character χ of $K^\times$, failure is also equivalent to the extremal identity

$$\sum_{\substack{t \in \mu_m \\ t \neq 1}} \chi(1 - t) = m - 1. \quad (\text{CS})$$

It saturates every cyclotomic secant, not only the Frobenius orbit of the distinguished Gaussian period.

Proof. The first equivalence follows from

$$\eta_k = \gamma_k^{2^{n/3}-1},$$

because $\ell \nmid 2^{n/3} - 1$. Next,

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2.$$

The order of ζ is coprime to ℓ , and squaring is invertible on the ℓ -primary quotient. Moreover the inclusion $\mathbb{F}_{2^n}^\times \subset K^\times$ preserves that quotient, since $\ell \nmid 2^n + 1$. This proves the equivalence with (iii).

Write $d = b - a$. The factor ζ^a is automatically an ℓ -th power. If $3 \nmid d$, then (2) generates $(\mathbb{Z}/m\mathbb{Z})^\times$, so $d \equiv 2^j \pmod{m}$ and

$$1 + \zeta^d = (1 + \zeta)^{2^j}.$$

If $3 \mid d$, then $1 + \zeta^d \in \mathbb{F}_{2^{2n/3}}^\times$. This group has order coprime to ℓ , because $\text{ord}_\ell(2) = n$, so all its elements are ℓ -th powers in K . This proves (iii) $\Rightarrow$ (iv); the converse is the case $a = 0, b = 1$.

Finally, all terms in (CS) have complex absolute value one. Part (iv) makes them all one; conversely equality with the positive real number $m - 1$ forces every term to be one by equality in the triangle inequality. $\square$

Proposition 6.27 (cyclic parity form of the cubic obstruction). *Put $n = 3^k$, $m = 3n$, with $k \geq 1$, and let the prime ℓ divide $\Phi_n(2)$. Define*

$$t = \frac{2^n - 1}{\ell}, \quad s = \frac{m + 1}{2}, \quad r \equiv st \pmod{m}, \quad 0 \leq r < m,$$

and let $R_{k,\ell}(X)$ be the degree- $< m$ reduction of $(1 + X)^t$ modulo $X^m - 1$. Then, independently of cubic failure,

$$\frac{X^n - 1}{X - 1} \mid R_{k,\ell}(X) + X^r. \quad (\text{CP}_1)$$

Moreover, for a primitive m -th root ζ and $\gamma_k = \zeta + \zeta^{-1}$,

$$\gamma_k \in (\mathbb{F}_{2^n}^\times)^\ell \iff R_{k,\ell}(X) + X^r = 1 + X + \cdots + X^{m-1}. \quad (\text{CP}_2)$$

Equivalently, write the binary expansion

$$t = \sum_{i \in I} 2^i.$$

Then cubic failure at ℓ is exactly the group-algebra identity

$$\prod_{i \in I} (1 + X^{2^i}) = \sum_{\substack{0 \leq j < m \\ j \neq r}} X^j \quad \text{in } \mathbb{F}_2[X]/(X^m - 1). \quad (\text{CP}_3)$$

Thus, if

$$c_j = \# \left\{ J \subseteq I : \sum_{i \in J} 2^i \equiv j \pmod{m} \right\} \pmod{2},$$

failure says

$$c_r = 0, \quad c_j = 1 \quad (j \neq r). \quad (\text{CP}_4)$$

The remaining cubic theorem is therefore the universal impossibility of this complement-of-one parity pattern for the particular binary set I coming from $(2^n - 1)/\ell$.

Proof. Let $\xi \neq 1$ be an n -th root of unity, of order $e = 3^{a+1}$, and put $d = 3^a < n$. The cyclotomic-divisor criterion gives $\text{ord}_\ell(2) = n$, so $\gcd(\ell, 2^d - 1) = 1$. Since $d \mid n$, it follows that

$$T := \frac{t}{2^d - 1} \in \mathbb{Z}.$$

Now $2^d \equiv -1 \pmod{e}$, and hence

$$(1 + \xi)^{2^d - 1} = \xi^{-1}.$$

Also $t = T(2^d - 1) \equiv -2T \pmod{e}$. Because $2s \equiv 1 \pmod{m}$, this gives $-T \equiv st \equiv r \pmod{e}$, and therefore

$$R_{k,\ell}(\xi) = (1 + \xi)^t = \xi^{-T} = \xi^r.$$

The polynomial $(X^n - 1)/(X - 1)$ is squarefree, since n is odd, and its roots are exactly these nonidentity n -th roots. This proves (CP_1) .

At the primitive m -th root ζ , one has

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2.$$

The element γ_k is an ℓ -th power exactly when $\gamma_k^t = 1$. Since squaring is bijective in characteristic two, this is equivalent to

$$(1 + \zeta)^t = \zeta^{st} = \zeta^r,$$

and hence to

$$\Phi_m(X) \mid R_{k,\ell}(X) + X^r. \quad (\text{CP}_5)$$

The two factors in (CP_1) and (CP_5) are coprime and satisfy

$$\frac{X^n - 1}{X - 1} \Phi_m(X) = \frac{X^m - 1}{X - 1} = 1 + X + \cdots + X^{m-1}.$$

The polynomial $R_{k,\ell} + X^r$ has degree $< m$. Under failure it is therefore a scalar multiple of the displayed degree- $m - 1$ polynomial. Evaluation at $X = 1$, which is preserved by reduction modulo $X^m - 1$, gives

$$(R_{k,\ell} + X^r)(1) = (1 + 1)^t + 1 = 1;$$

the all-ones polynomial also has value $m = 1$ in characteristic two. The scalar is consequently one, proving (CP_2) ; the converse is immediate.

Finally, in the group algebra,

$$(1 + X)^t = \prod_{i \in I} (1 + X^{2^i}).$$

The coefficient of X^j after cyclic reduction is the parity of the subsets with exponent sum congruent to j . Rewriting (CP_2) as $R_{k,\ell} = \sum_{j \neq r} X^j$ proves (CP_3) – (CP_4) . $\square$

Corollary 6.28 (augmentation-ideal normalization of cubic failure). *Retain Proposition 6.27, and put*

$$A_m = \mathbb{F}_2[X]/(X^m - 1), \quad J_m = \sum_{j=0}^{m-1} X^j, \quad e_m = 1 + J_m.$$

The element e_m is the identity of the augmentation ideal $e_m A_m$. For $0 \leq i < n$, define

$$h_i = X^{-s2^i} (1 + X^{2^i}) = X^{s2^i} + X^{-s2^i} \in e_m A_m.$$

If I is the binary support of $t = (2^n - 1)/\ell$, then cubic failure at ℓ is exactly

$$\prod_{i \in I} h_i = e_m. \tag{CA_1}$$

On the other hand, the full-product relation

$$\prod_{i=0}^{n-1} h_i = e_m \tag{CA_2}$$

holds unconditionally. Thus failure makes both the proper digit subset I and its complement multiply to the identity in $e_m A_m$.

Likewise, raising (CP_3) to the ℓ -th power yields only the unconditional identity

$$(1 + X)^{2^n - 1} = J_m + X^{-1} \quad \text{in } A_m, \tag{CA_3}$$

because $r\ell \equiv -1 \pmod{m}$. Hence neither the full product nor the strongest immediate exponent check can exclude failure. The remaining cubic theorem is precisely the absence of the extra proper-subproduct relation (CA_1) for this selected binary support.

Proof. For $y \in A_m$, one has $J_m y = \text{aug}(y) J_m$. Since m is odd, e_m has augmentation zero and acts as the identity on the augmentation ideal.

Write $P_I = \prod_{i \in I} (1 + X^{2^i})$. Under (CP_3) , $P_I = J_m + X^r$, and therefore

$$\prod_{i \in I} h_i = X^{-st} P_I = X^{-r} (J_m + X^r) = e_m.$$

Every step is reversible, proving (CA_1) .

For the full product, let ξ be an m -th root of unity in an algebraic closure. If $\xi \neq 1$, then

$$\prod_{i=0}^{n-1} (1 + \xi^{2^i}) = (1 + \xi)^{2^n - 1} = \frac{1 + \xi^{2^n}}{1 + \xi} = \xi^{-1},$$

because $2^n \equiv -1 \pmod{m}$. Also $s(2^n - 1) \equiv -1 \pmod{m}$, so the monomial factor in the product of the h_i contributes ξ . At $\xi = 1$ the product vanishes. These are exactly the values of e_m ; since m is odd, $X^m - 1$ is squarefree, proving (CA₂). It follows that every h_i is a unit of $e_m A_m$, so (CA₁) also holds for the complementary subset.

Finally $J_m + X^r = X^r e_m$, and e_m is idempotent. Raising $(1 + X)^t = X^r e_m$ to the ℓ -th power gives

$$(1 + X)^{2^n - 1} = X^{r\ell} e_m = X^{-1} e_m = J_m + X^{-1},$$

which proves (CA₃). □

Proposition 6.29 (cubic block compression and forced midpoint). *Retain Proposition 6.27. Put*

$$d = \frac{n}{3}, \quad q = 2^d, \quad N = q^2 + q + 1, \quad u = \frac{N}{\ell},$$

so that $t = (q - 1)u$, and set

$$B(X) = 1 + X + \cdots + X^{n-2} \in A_m, \quad b = X^{-s(q-1)} B.$$

Then

$$(1 + X)^{q-1} = B(X) \quad \text{in } A_m, \tag{CB_1}$$

and cubic failure at ℓ is exactly

$$B^u = X^{st} e_m, \quad \text{equivalently} \quad b^u = e_m. \tag{CB_2}$$

The normalized element satisfies, unconditionally,

$$b(X^{-1}) = b(X), \quad b^N = e_m. \tag{CB_3}$$

At a primitive m -th root ζ ,

$$b(\zeta)^2 = \eta_k. \tag{CB_4}$$

Thus (CB₂) is the original selected current-factor order question in compressed form, rather than a weaker consequence.

Moreover, if C_j is the number of binary submasks of t whose sum is $j \pmod{m}$, then

$$C_j = C_{t-j}. \tag{CB_5}$$

The unique fixed residue of $j \mapsto t - j$ is $r = st$, and C_r is even. Hence complementation forces the exceptional zero required by (CP₄) into exactly the right position; it cannot contradict cubic failure.

Proof. The lifted congruence in Proposition 6.18 gives

$$q \equiv n - 1 \pmod{3n}, \quad \frac{q - (n - 1)}{3n} \equiv 0 \pmod{2}.$$

Since q is a power of two, $(1 + X)^{q-1} = \sum_{a=0}^{q-1} X^a$. Cyclic reduction modulo $X^m - 1$ cancels the even number of complete residue blocks and leaves exactly B , proving (CB₁). Substitution in (CP₂), followed by multiplication by X^{-st} , proves (CB₂).

Because $2s \equiv 1 \pmod{m}$ and $q - 1 \equiv n - 2 \pmod{m}$, reversing B proves $b(X^{-1}) = b(X)$. Also $(q - 1)N = 2^n - 1$. At every nonidentity m -th root ξ ,

$$(1 + \xi)^{2^n - 1} = \xi^{-1}, \quad s(2^n - 1) \equiv -1 \pmod{m},$$

so $b(\xi)^N = 1$; at $\xi = 1$, $B(1) = n - 1 = 0$. Squarefree interpolation proves $b^N = e_m$. Finally $(1 + \zeta)^2 = \zeta\gamma_k$, whence

$$b(\zeta)^2 = \zeta^{-(q-1)}(1 + \zeta)^{2(q-1)} = \gamma_k^{q-1} = \eta_k.$$

The binary integer t is odd. Complementing a submask is therefore fixed-point-free and changes its sum from j to $t - j \pmod{m}$, which proves (CB₅). Since m is odd, the involution on residues has the unique fixed point $r = st$; all submasks above that point occur in complementary pairs. $\square$

Proposition 6.30 (distinguished Wendt-factor obstruction). *Retain Proposition 6.29, and let $P_{\eta,k}$ be the degree- n minimal polynomial of η_k over $\mathbb{F}_2$. Define*

$$G_u(X) = \gcd(X^u + 1, (X + 1)^u + 1) \in \mathbb{F}_2[X].$$

Then cubic failure at ℓ is equivalent to

$$P_{\eta,k} \mid G_u. \tag{CW_1}$$

Under failure, the two distinct factors

$$P_{\eta,k}(X), \quad P_{\eta,k}(X + 1) \tag{CW_2}$$

both divide G_u , and the second is the reciprocal partner of the first. Every irreducible factor of G_u has degree exactly n ; translation pairs those factors without fixed points. Consequently

$$G_u = 1 \quad \text{or} \quad 2n \mid \deg G_u. \tag{CW_3}$$

The usual six-element group generated by $X \mapsto X + 1$ and $X \mapsto X^{-1}$ acts freely on the roots of G_u . Thus the surviving assertion is specifically that the distinguished pair (CW₂), not merely some Wendt pair, never occurs.

Proof. Failure says $\eta_k^u = 1$. Since $\eta_k + 1 = \eta_k^{q+1}$, it also gives $(\eta_k + 1)^u = 1$, and Frobenius gives the same two identities for every conjugate. This proves (CW₁) and (CW₂); the converse uses the first factor.

The relative trace recursion in Proposition 6.18 gives

$$\mathrm{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(\eta_k) = 0.$$

As n is odd, the trace of $\eta_k + 1$ is one, so the two factors in (CW₂) are distinct. Also $\eta_k^{-1} = \eta_k^q + 1$; hence the translate is the minimal polynomial of η_k^{-1} , namely the reciprocal partner.

Let x be a root of G_u . The nontrivial orders of both x and $x + 1$ divide $u \mid \Phi_n(2)$. Every prime $p \mid u$ satisfies $\mathrm{ord}_p(2) = n$: the only prime dividing n is three, and $3 \nmid \Phi_n(2)$. Thus x has degree at least n , while its order divides $2^n - 1$, so its degree is exactly n . Translation changes the trace by one and therefore pairs the degree- n irreducible factors without fixed points, proving (CW₃).

Finally none of a root x , its inverse, or its translate is fixed by an involution: the only possible values would be zero or one. A fixed point of the order-three map satisfies $x^2 + x + 1 = 0$, hence has order three, but $3 \nmid u$. The six-element action is therefore free. $\square$

Proposition 6.31 (Singer factorization of the Wendt gcd). *Retain Proposition 6.30. In $L = \mathbb{F}_{q^3}$, put*

$$U = \ker(\mathrm{N}_{L/\mathbb{F}_q}) = \mu_N, \quad H = U^\ell = \mu_u,$$

and let

$$D = \{x \in U : x^{q+1} + x + 1 = 0\}.$$

Thus D is the planar Singer $(N, q + 1, 1)$ difference set from Proposition 6.18. Then

$$\text{Roots}(G_u) = (D \cap H) \dot{\cup} ((D \cap H) + 1). \quad (\text{SW}_1)$$

If $\mathcal{O}$ runs over the absolute-Frobenius orbits in $D \cap H$, and $P_{\mathcal{O}}$ is the corresponding minimal polynomial, then

$$G_u(X) = \prod_{\mathcal{O} \subset D \cap H} P_{\mathcal{O}}(X)P_{\mathcal{O}}(X + 1), \quad \deg G_u = 2|D \cap H|. \quad (\text{SW}_2)$$

Every orbit $\mathcal{O}$ has size n , and $P_{\mathcal{O}}(X + 1)$ is its reciprocal partner.

More precisely, choose a generator g of U , and put

$$a_j = |D \cap g^j H| \quad (j \in \mathbb{Z}/\ell\mathbb{Z}).$$

The quotient intersection numbers obey the exact moment identities

$$\begin{aligned} \sum_j a_j &= q + 1, & \sum_j a_j^2 &= q + u, \\ \sum_j a_j a_{j+r} &= u \quad (r \neq 0), & a_{2j} &= a_j. \end{aligned} \quad (\text{SW}_3)$$

In particular, if

$$\ell - 1 < \sqrt{q} + \frac{1}{\sqrt{q}}, \quad (\text{SW}_4)$$

then every a_j is positive. In that range $a_0 \geq n$ and $\deg G_u \geq 2n$: Wendt factors of the correct degree and symmetry are forced to exist even though the distinguished factor may still be absent.

Proof. A root x of G_u satisfies $x, x + 1 \in H \subset U$. Put $y = x^{q+1}$. Since

$$x^q = \frac{y}{x}, \quad x^{q^2} = \frac{1}{y},$$

the condition $N(x + 1) = 1$, after clearing xy , factors as

$$(y + x + 1)((x + 1)y + x) = 0. \quad (\text{SW}_5)$$

The first factor says $x \in D$. In the second case $x + 1 \in D$. Conversely, if $x \in D \cap H$, then $x + 1 = x^{q+1} \in H$, so both x and $x + 1$ are roots of G_u . The two sets in (SW_1) are disjoint: if both x and $x + 1$ lay in D , their two Frobenius equations would give $x^2 + x + 1 = 0$, impossible because $3 \nmid N$.

Every nonidentity element of H has absolute degree n , by the cyclotomic-divisor argument in Proposition 6.30. The polynomial G_u is squarefree, and (SW_1) therefore gives (SW_2) . For $x \in D$,

$$(x + 1)^q = x^{-1},$$

so translation of its absolute orbit is the inverse orbit, proving the reciprocal assertion.

The first identity in (SW_3) is $|D| = q + 1$. For the second, $\sum a_j^2$ counts ordered pairs of points of D whose quotient lies in H . The identity quotient contributes $q + 1$, and each of the other $u - 1$ elements occurs once because D has difference parameter one. If $r \neq 0$, the corresponding coset has u nonidentity quotients, proving the third identity. Squaring stabilizes D and sends the j -th coset to the $2j$ -th, which proves the last.

If some a_j vanished, Cauchy's inequality applied to the remaining $\ell - 1$ entries would give

$$(q + 1)^2 \leq (\ell - 1)(q + u). \quad (\text{SW}_6)$$

But a direct simplification using $u = N/\ell$ gives

$$\ell((q + 1)^2 - (\ell - 1)(q + u)) = q \left[\left(\sqrt{q} + \frac{1}{\sqrt{q}} \right)^2 - (\ell - 1)^2 \right],$$

which is positive under (SW_4) , a contradiction. Finally $D \cap H$ is a union of size- n orbits, so positivity of a_0 implies $a_0 \geq n$. $\square$

Proposition 6.32 (the selected S_3 -quotient of the Wendt pair). *For $x \in D$, put*

$$J(x) = \frac{(x^2 + x + 1)^3}{x^2(x + 1)^2}, \quad a = \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x).$$

Then

$$J(x) = a^2 + a + 1. \quad (\text{SJ}_1)$$

Let $d = 3^{k-1}$, let W_{k-1} be the degree- d minimal polynomial of

$$z_{k-1} = \eta_{k-1}^2 + \eta_{k-1} + 1,$$

and retain the selected polynomial $P_{\eta,k}$. Then

$$\begin{aligned} P_{\eta,k}(X)P_{\eta,k}(X + 1) &= [X(X + 1)]^{2d} \\ &\cdot W_{k-1} \left(\frac{(X^2 + X + 1)^3}{X^2(X + 1)^2} \right). \end{aligned} \quad (\text{SJ}_2)$$

Thus quotienting by the free S_3 -action lowers the selected degree- $2n$ pair to one degree- d trace factor. Cubic failure is still the assertion that this particular factor, rather than an unselected Singer intersection with the same orbit structure, occurs in the Wendt quotient.

Proof. The cubic for $x \in D$ is

$$T^3 + aT^2 + (a + 1)T + 1.$$

Substitution of x gives

$$a = \frac{x^3 + x + 1}{x(x + 1)}.$$

Putting this expression over a common denominator proves

$$a^2 + a + 1 = \frac{(x^2 + x + 1)^3}{x^2(x + 1)^2},$$

which is (SJ_1) .

The element η_{k-1} has odd degree d , while it is at most quadratic over z_{k-1} . Hence the two fields are equal and W_{k-1} has degree d . The invariant J is constant on the six-element orbit generated by $X \mapsto X + 1$ and $X \mapsto X^{-1}$. On D , its order-three subgroup is relative Frobenius, while translation gives the reciprocal absolute orbit. Consequently the roots of the two sides of (SJ_2) are the same. Both sides are monic of degree $6d = 2n$, which proves the identity. $\square$

Corollary 6.33 (reciprocal Conway selector and the exact order seam). *For $j \geq 1$, put*

$$z_j = \eta_j^2 + \eta_j + 1, \quad z_0 = 1.$$

Then

$$z_j = \gamma_j^{-2}. \tag{RS_1}$$

If $q = 2^{3^j-1}$, then z_j has the irreducible cubic

$$Z^3 + z_{j-1}Z^2 + z_{j-1} = 0 \tag{RS_2}$$

over $\mathbb{F}_q$. In particular

$$\mathrm{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(z_j) = \mathrm{N}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(z_j) = z_{j-1}, \quad z_{j-1} = \frac{z_j^3}{z_j^2 + 1}. \tag{RS_3}$$

If P_j is the minimal polynomial of γ_j , the minimal polynomial of z_j is its reciprocal

$$W_j(X) = X^{3^j} P_j(X^{-1}). \tag{RS_4}$$

Most importantly,

$$z_j^{q-1} = \eta_j^{-2}, \quad \mathrm{ord}(z_j) = \mathrm{ord}(z_{j-1}) \mathrm{ord}(\eta_j). \tag{RS_5}$$

Thus W_j is primitive exactly when P_j is. For every current prime $\ell \mid q^2 + q + 1$,

$$z_j \in (\mathbb{F}_{q^3}^\times)^\ell \iff \eta_j \in U_j^\ell. \tag{RS_6}$$

The selected S_3 -quotient is therefore a lossless reciprocal formulation of C_j , not an independent induction from C_{j-1} .

Proof. Put $t = \zeta_j^2$ and $M(t) = (\omega + \omega^2 t)/(1 + t) = \eta_j$. Direct calculation in $\mathbb{F}_4(t)$ gives

$$M(t)^2 + M(t) + 1 = \frac{t}{(1+t)^2}, \quad \gamma_j^2 = t + t^{-1} = \frac{(1+t)^2}{t},$$

which proves (RS₁).

Write $b = \eta_{j-1}$, $c = b^2 + b + 1 = z_{j-1}$, and let a_i be the three relative conjugates of η_j . By Proposition 6.18,

$$e_1(a_i) = b, \quad e_2(a_i) = b + 1, \quad e_3(a_i) = 1.$$

For $w_i = a_i^2 + a_i + 1$, these identities give

$$e_1(w_i) = c, \quad e_2(w_i) = 0.$$

Moreover, over $\mathbb{F}_4$,

$$\prod_i w_i = \prod_i (a_i + \omega) \prod_i (a_i + \omega^2) = (b + \omega)(b + \omega^2) = c.$$

This proves (RS₂). Since η_j is at most quadratic over z_j , while its absolute degree 3^j is odd, the two elements generate the same field. Thus the cubic is irreducible, and its coefficients give (RS₃).

Squaring permutes the absolute conjugates of γ_j , while inversion sends them to the conjugates of $z_j = \gamma_j^{-2}$. The constant term of P_j is one, proving (RS₄). The first equality in (RS₅) follows from $\eta_j = \gamma_j^{q-1}$.

Now

$$\mathbb{F}_{q^3}^\times \simeq \mathbb{F}_q^\times \times U_j, \quad \gcd(q-1, q^2+q+1) = 1.$$

Relative norm detects the $\mathbb{F}_q^\times$ -component of z_j without changing its order, because it cubes that component and $3 \nmid q-1$. Raising to $q-1$ detects the U_j -component without changing its order. Equations (RS₃) and (RS₅), together with the oddness of $|U_j|$, therefore give the order product. Finally ℓ -powering is an automorphism on $\mathbb{F}_q^\times$, while both $q-1$ and 2 are invertible modulo ℓ ; this proves (RS₆). $\square$

Proposition 6.34 (the selected Singer trace is the whole remaining phase). *Retain Proposition 6.31. For $a \in \mathbb{F}_q$ put*

$$D_a = \{x \in D : \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x) = a\}.$$

Each D_a is empty or one three-element relative-Frobenius orbit. Let $\chi : U \rightarrow \mu_\ell$ have exact order ℓ , where $\text{ord}_\ell(q) = 3$, and put $H = \ker \chi$. If $D_a \neq \emptyset$, define

$$\Theta_\chi(a) = \sum_{x \in D_a} \chi(x).$$

Then

$$D_a \subset H \iff \Theta_\chi(a) = 3. \tag{ST_1}$$

In particular, current failure is exactly

$$\Theta_\chi(\eta_{j-1}) = 3. \tag{ST_2}$$

For any nontrivial additive character ψ of $\mathbb{F}_q$, Fourier inversion gives

$$\Theta_\chi(a) = \frac{1}{q} \sum_{b \in \mathbb{F}_q} \psi(-ba) \sum_{x \in D} \chi(x) \psi\left(b \text{Tr}_{\mathbb{F}_{q^3}/\mathbb{F}_q}(x)\right). \tag{ST_3}$$

Whenever the moment criterion (SW₄) forces $D \cap H \neq \emptyset$, some trace fibre actually attains the extremal value 3. Hence a uniform character bound cannot exclude (ST₂); one must distinguish the selected Conway fibre.

Proof. The trace- a elements of D are the roots of

$$T^3 + aT^2 + (a+1)T + 1.$$

Relative Frobenius has no fixed point on D : an element of $D \cap \mathbb{F}_q$ would lie in $\mathbb{F}_q^\times \cap U = \{1\}$, while $1 \notin D$. Thus every nonempty fibre is one orbit of length three.

For $x \in D_a$, put $\theta = \chi(x)$. Then

$$\Theta_\chi(a) = \theta + \theta^q + \theta^{q^2}.$$

The fibre lies in H exactly when $\theta = 1$. Conversely, equality to 3 in the displayed sum forces all three unit complex numbers to equal one, by equality in the triangle inequality. This proves (ST₁). The distinguished fibre is the relative orbit of η_j , whose trace is η_{j-1} , proving (ST₂). Character orthogonality on the additive group of $\mathbb{F}_q$ gives (ST₃). Finally, if $x \in D \cap H$, its whole relative orbit lies in H , so its trace fibre has $\Theta_\chi = 3$. $\square$

Remark 6.35 (the natural secant Kummer cover is rational). Over $\mathbb{F}_4$, the Möbius function

$$M(T) = \frac{\omega + \omega^2 T}{1 + T}$$

which expresses the cyclotomic secant has Kummer cover

$$Z^\ell = M(T), \quad T = \frac{Z^\ell + \omega}{Z^\ell + \omega^2}.$$

Its function field is therefore $\mathbb{F}_4(Z)$, of genus zero. A positive-genus Weil estimate on this cover cannot rule out the distinguished torsion fibre; the remaining information is its actual Artin value.

Proposition 6.36 (Capelli form of the surviving lemma). *Put*

$$A(X) = X^3 + X + 1, \quad B(X) = X^2 + X, \quad R(X) = \frac{A(X)}{B(X)}.$$

Let $Q_0(X) = X$ and, for $k \geq 1$, define the monic binary polynomial

$$Q_k(X) = B(X)^{3^{k-1}} Q_{k-1}(R(X)). \quad (\star)$$

Then Q_k is the minimal polynomial of η_k over $\mathbb{F}_2$ and is irreducible of degree 3^k . Its first four values are

$$\begin{aligned} Q_1(X) &= X^3 + X + 1, \\ Q_2(X) &= X^9 + X + 1, \\ Q_3(X) &= X^{27} + X^{25} + X^{24} + X^{19} + X^{18} + X^{16} + X^{10} + X^9 + X^3 + X + 1, \\ Q_4(X) &= X^{81} + X^{64} + X^{16} + X + 1. \end{aligned}$$

For every prime $\ell \mid \Phi_{3^k}(2)$, the following are equivalent:

- (i) $\chi_{k,\ell}(\eta_k) \neq 1$;
- (ii) η_k is not an ℓ -th power in $\mathbb{F}_{2^{3^k}}^\times$;
- (iii) $X^\ell - \eta_k$ is irreducible over $\mathbb{F}_{2^{3^k}}$;
- (iv) $Q_k(X^\ell)$ is irreducible over $\mathbb{F}_2$.

Thus the surviving character lemma is equivalently the assertion that all the explicitly recursive compositions $Q_k(X^\ell)$, for $\ell \mid \Phi_{3^k}(2)$, are irreducible. At the first composite level $k = 4$, the three targets are the monomial compositions of the displayed pentanomial Q_4 by $\ell \in \{2593, 71119, 97685839\}$.

Proof. Proposition 6.18 gives $R(\eta_k) = \eta_{k-1}$ and an irreducible cubic for η_k over $\mathbb{F}_{2^{3^{k-1}}}$. Since $\eta_{k-1} \in \mathbb{F}_2(\eta_k)$, induction gives $[\mathbb{F}_2(\eta_k) : \mathbb{F}_2] = 3^k$. Clearing the denominator in the minimal polynomial of η_{k-1} gives $(\star)$, a monic polynomial of degree 3^k having η_k as a root: its leading term is $A(X)^{3^{k-1}} = X^{3^k} + \dots$. It is therefore the minimal polynomial. Direct substitution in $(\star)$ gives the four displayed polynomials.

Here $\Phi_{3^k}(2) = q_k^2 + q_k + 1 \equiv 1 \pmod{3}$, so a prime $\ell \mid \Phi_{3^k}(2)$ is not 3. The cyclotomic-divisor criterion therefore gives $\text{ord}_\ell(2) = 3^k$, and hence $\mu_\ell \subset \mathbb{F}_{2^{3^k}}$. Since ℓ is prime, Kummer theory says that $X^\ell - \eta_k$ either splits off a linear factor or is irreducible, according as η_k is or is not an ℓ -th power. This proves the equivalence of (ii) and (iii), while Capelli's lemma gives the equivalence of (iii) and (iv). The direct product $\mathbb{F}_{2^{3^k}}^\times \simeq \mathbb{F}_{q_k}^\times \times U_k$ and $\ell \nmid (q_k - 1)$ show that being an ℓ -th power in the full field is equivalent, for $\eta_k \in U_k$, to lying in U_k^ℓ . This is the equivalence with (i). $\square$

Remark 6.37 (Singer membership is not primitivity). The difference-set identification cannot replace the character test. Indeed, exponentiation

$$s_k : U_k \longrightarrow U_k, \quad s_k(y) = y^{q_k+1},$$

is an automorphism of order 6: one has $\gcd(q_k + 1, |U_k|) = 1$ and $(q_k + 1)^3 \equiv -1 \pmod{|U_k|}$, while $(q_k + 1)^2 \equiv q_k \not\equiv 1$. The Singer equation is simply $s_k(y) = y + 1$. Thus its consequence $\text{ord}(y + 1) = \text{ord}(y)$ is automatic order preservation by an automorphism and supplies no order divisor by itself. For example, take $q = 32$ and let y be a root of $X^3 + X^2 + 1$ over $\mathbb{F}_2$. Then y has order 7 and $q + 1 \equiv 5 \pmod{7}$; the defining cubic gives $y^5 + y + 1 = 0$. Hence $y \in \{x : x^{q+1} + x + 1 = 0\}$, but y does not generate the norm-one group of order $q^2 + q + 1 = 7 \cdot 151$. What must be used in the Conway tower is the recursive coefficient η_{k-1} in (*), not Singer membership alone.

Proposition 6.38 (prime-step closure). *If $\Phi_{3^k}(2)$ is prime, then*

$$C_k \iff C_{k-1}.$$

Consequently C_1, C_2, C_3 hold analytically: the three new kernel orders are respectively

$$7, \quad 73, \quad 262657,$$

and all are prime.

Proof. Proposition 6.18 proves that η_k has degree 3 over $\mathbb{F}_{q_k}$, so in particular $\eta_k \neq 1$. A nonidentity element of a group of prime order generates it, and (**) gives the first claim. The primality of 7 and 73 is immediate. For completeness, $262657 - 1 = 2^9 \cdot 3^3 \cdot 19$, and base 5 gives the Pocklington certificate

$$5^{262656} \equiv 1 \pmod{262657}, \quad \gcd\left(5^{262656/r} - 1, 262657\right) = 1 \quad (r = 2, 3, 19),$$

which proves the third primality assertion. □

Proposition 6.39 (cubic-arm equivalence). *For primes r with $f(r) \mid 3^k$ a 3-power:*

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \dots, 3^k\}.$$

The family formula is the 0/1/4 candidate of Section 1 restricted to the 3-power column.

Proposition 6.40 (cyclotomic-unit residue reduction). *Let $h = 3^k$ and let ℓ be any prime divisor of $2^h - 1$. Then*

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2$$

is an ℓ -th power in $\mathbb{F}_{2^h}$ if and only if $1 + \zeta$ is an ℓ -th power in $\mathbb{F}_{2^{2h}}$. The same equivalence holds at every power $\ell^a \mid 2^h - 1$; non- ℓ -power already implies that γ_k has the full ℓ -primary part of $2^h - 1$ in its order.

Proof. Because $\ell \mid 2^h - 1$, one has $\ell \nmid 2^h + 1$, so extension from $\mathbb{F}_{2^h}^\times$ to $\mathbb{F}_{2^{2h}}^\times$ preserves the ℓ -primary quotient. Also $\ell \neq 3$, hence ζ^{-1} is an ℓ -th power in the 3^{k+1} -group generated by ζ ; squaring is an automorphism of every odd ℓ -primary quotient. The displayed factorization proves both assertions. □

General multiplicative-order bounds for Gaussian periods do not imply maximal order [2]. Here the norm recursion $\gamma_k^3 + \gamma_k = \gamma_{k-1}$ propagates the old factors, but a proof must still show that each new prime-power factor of $\Phi_{3^k}(2)$ divides $\text{ord}(\gamma_k)$. Irreducibility of the corresponding Dickson polynomial is likewise weaker than primitivity [15]. Proposition 6.18 supplies the appropriate cubic replacement and places the root on a Singer difference set, but Remark 6.37 shows that Singer membership alone cannot prove generation.

6.6 Open problem C

The universal cubic arm is Conjecture 6.1. The prime-step argument proves C_k analytically for $k \leq 3$, and Proposition 6.21 proves the required primary order above an explicit partition threshold at every level. The remaining task is the selected Kummer-class nonvanishing at the smaller new primary factors. Finite order checks, their factorization dependencies, and the distinction between local certificates and external primality records are isolated in Appendix A.3.

7 The singleton-even Conway–Fermat arm

The candidate contains strong explicit maximal-order assertions even before the non-cyclotomic singleton chains are reached. This prevents a proof from following from field degree or irreducibility alone.

7.1 Maximal projective order

Proposition 7.1 (the singleton-even arm is Conway–Fermat maximality). *For $n \geq 0$, put*

$$h_n = 2^{n+1}, \quad c_n = \kappa_{h_n} = 2^{2^n}, \quad E_n = \mathbb{F}_{2^{h_n}}, \quad F_n = 2^{2^n} + 1.$$

With $E_{-1} = \mathbb{F}_2$, let

$$\delta_n = \text{ord}(c_n E_{n-1}^\times) \quad \text{in} \quad E_n^\times / E_{n-1}^\times.$$

Then $1 < \delta_n \mid F_n$, and the $m = 0$ arm of Conjecture 1.1 for every prime $p \mid F_n$ is equivalent to

$$\delta_n = F_n.$$

Proof. The Conway finite-nimber tower satisfies

$$c_n^2 + c_n = c_{n-1} c_{n-2} \cdots c_0$$

(with the empty product 1). Thus E_n/E_{n-1} is quadratic and $c_n \notin E_{n-1}$, so the coset order is nontrivial. Its ambient quotient has order

$$\frac{2^{2^{n+1}} - 1}{2^{2^n} - 1} = F_n,$$

which proves $1 < \delta_n \mid F_n$; equivalently the relative norm is $c_n^{F_n} = c_{n-1} \cdots c_0$.

If $p \mid F_n$, then $2^{2^n} \equiv -1 \pmod{p}$, hence $f(p) = 2^{n+1} = h_n$. Fermat numbers are pairwise coprime, so the entire p -primary part of $|E_n^\times|$ lies in the quotient by $E_{n-1}^\times$. The full power criterion therefore gives $m_p = 0$ iff $v_p(\delta_n) = v_p(F_n)$. Requiring this for every prime divisor of F_n is exactly $\delta_n = F_n$. $\square$

Corollary 7.2 (exact primitive-product equivalence). *Put $a_n = \prod_{j=0}^n c_j$. For every $n \geq 2$,*

$$\text{ord}(c_n) = \text{ord}(a_n) = \prod_{j=1}^n \delta_j. \tag{PP_1}$$

Consequently

$$c_0 c_n \text{ is primitive in } E_n \iff \delta_j = F_j \quad (1 \leq j \leq n). \tag{PP_2}$$

Thus the all-level Conway–Fermat arm is exactly the all-level primitive-product problem studied by Popovych [21], in both directions, rather than merely an analogous maximal-order question.

Proof. Write $q = 2^{2^n}$. Since $\gcd(q-1, q+1) = 1$, the cyclic group $E_n^\times$ is the direct product of $E_{n-1}^\times$, of order $q-1$, and its subgroup of order $F_n = q+1$. Write accordingly $c_n = g_n u_n$, where $u_n \in E_{n-1}^\times$ and $\text{ord}(g_n) = \delta_n$. The tower norm gives

$$a_{n-1} = c_n^{q+1} = u_n^2,$$

so $\text{ord}(u_n) = \text{ord}(a_{n-1})$. The two factors have coprime orders, and therefore

$$\text{ord}(c_n) = \delta_n \text{ord}(a_{n-1}), \quad a_n = a_{n-1} c_n = g_n u_n^3.$$

At $n = 1$, the quotient has prime order $F_1 = 5$, so $\delta_1 = 5$; the same decomposition gives $u_1^3 = 1$ and therefore $a_1 = g_1$, of order 5. Inductively $\text{ord}(u_n) = \prod_{j=1}^{n-1} \delta_j$, which is prime to three because every $\delta_j \mid F_j$ for $j \geq 1$. Cubing therefore preserves the order of u_n , proving (PP₁).

The element c_0 has order three, coprime to $\text{ord}(c_n)$, while

$$|E_n^\times| = 3 \prod_{j=1}^n F_j.$$

Hence $c_0 c_n$ is primitive exactly when $\prod_{j=1}^n \delta_j = \prod_{j=1}^n F_j$. Since the Fermat numbers are pairwise coprime and $\delta_j \mid F_j$, this is equivalent to all the displayed equalities in (PP₂). $\square$

Proposition 7.3 (Fibonacci-polynomial criterion). *Fix $n \geq 0$, put $m = 2^n$, $q = 2^m$, and write $K = E_{n-1} = \mathbb{F}_q$, $L = E_n = \mathbb{F}_{q^2}$ and $a = a_{n-1} = \prod_{j < n} c_j$. If*

$$S_0(X) = 0, \quad S_1(X) = 1, \quad S_{r+2}(X) = S_{r+1}(X) + X S_r(X),$$

then

$$\delta_n = \min\{r > 0 : S_r(a) = 0\},$$

and hence

$$\delta_n = F_n \iff S_{F_n/\ell}(a) \neq 0 \quad \text{for every prime } \ell \mid F_n. \quad (\text{F}_n)$$

More precisely, for odd $d > 1$ define the fibotomic polynomial

$$\mathfrak{F}_d(X) = \prod_{\substack{\{v, v^{-1}\} \\ \text{ord}(v)=d}} \left(X - \frac{v}{(v+1)^2} \right),$$

where the product takes one representative of each inverse-pair of primitive d -th roots. Then $\mathfrak{F}_d \in \mathbb{F}_2[X]$,

$$S_r(X) = \prod_{\substack{d \mid r \\ d > 1}} \mathfrak{F}_d(X) \quad (r \text{ odd}), \quad (\text{B}_r)$$

and, if A_{n-1} is the minimal polynomial of a_{n-1} over $\mathbb{F}_2$,

$$\delta_n = d \iff A_{n-1} \mid \mathfrak{F}_d.$$

Every irreducible factor of every $\mathfrak{F}_d$ with $1 < d \mid F_n$ has the same degree m . Thus the Conway–Fermat conjecture is exactly the assertion $A_{n-1} \mid \mathfrak{F}_{F_n}$; field degree cannot distinguish that stratum from any proper-divisor stratum.

Proof. Put $c = c_n$ and $u = (c + 1)/c$. Since $c^2 + c = a$ and $c^q = c + 1$, the element u lies in the norm-one group of order $q + 1 = F_n$, and $\delta_n = \text{ord}(u)$. The two roots $c, c + 1$ of $T^2 + T + a$ give the Binet identity

$$S_r(a) = c^r + (c + 1)^r = \frac{1 + u^r}{(1 + u)^r}. \quad (\text{B})$$

This proves the first criterion. In a cyclic group of order F_n an element has full order precisely when its F_n/ℓ -th power is nontrivial for every prime $\ell \mid F_n$, proving $(\mathbf{F}_n)$.

For odd r , the nontrivial r -th roots v occur in inverse pairs, and the common value $v/(v + 1)^2$ of a pair is, by (B), a root of S_r . Frobenius permutes the pairs, so the displayed factors have coefficients in $\mathbb{F}_2$. Both sides of (B_r) are monic of degree $(r - 1)/2$, which proves the factorization and the divisibility criterion.

Finally $2^m \equiv -1 \pmod{d}$ for every nontrivial $d \mid F_n$. Because m is a power of two, $\text{ord}_d(2) = 2m$, and the least $j > 0$ with $2^j \equiv \pm 1 \pmod{d}$ is m . This is exactly the Frobenius orbit length of $v/(v + 1)^2$: indeed

$$\frac{v}{(v + 1)^2} = \frac{w}{(w + 1)^2} \iff (v + w)(vw + 1) = 0,$$

so only the inverse pair $\{v, v^{-1}\}$ has the same image. This proves the equal-degree assertion. $\square$

7.2 The selected factor and the limits of one-step induction

Proposition 7.4 (coarse top-step data cannot force maximality). *For every divisor $d > 1$ of F_n there are elements $a_d \in \mathbb{F}_q$ and $c_d \in \mathbb{F}_{q^2}$ such that*

$$[\mathbb{F}_2(a_d) : \mathbb{F}_2] = m, \quad [\mathbb{F}_2(c_d) : \mathbb{F}_2] = 2m, \quad c_d^q = c_d + 1, \quad c_d^2 + c_d = a_d,$$

$$\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(a_d) = 1, \quad \text{N}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(c_d) = a_d,$$

but the order of $c_d \mathbb{F}_q^\times$ is exactly d . In particular all proper-divisor strata obey the same degree, trace, Artin–Schreier and relative-norm data as the Conway top step.

Proof. Choose v of order d in the norm-one subgroup of $\mathbb{F}_{q^2}^\times$ and put

$$c_d = \frac{1}{v + 1}, \quad a_d = \frac{v}{(v + 1)^2}.$$

Then $v^q = v^{-1}$, so $c_d^q = c_d + 1$; the remaining displayed identities follow directly. The orbit calculation in the proof of Proposition 7.3 gives the two field degrees. Consequently $T^2 + T + a_d$ is irreducible over $\mathbb{F}_q$, which is equivalent to the stated absolute trace, and $(c_d + 1)/c_d = v$ gives quotient order d . $\square$

Proposition 7.5 (the exact recursively selected factor). *Set $a_{-1} = 1$, $a_i = \prod_{j \leq i} c_j$, and let A_i be its minimal polynomial over $\mathbb{F}_2$, with $A_{-1}(Y) = Y + 1$. Then*

$$a_i^2 + a_{i-1}a_i + a_{i-1}^3 = 0$$

and

$$A_i(X) = \text{Res}_Y(A_{i-1}(Y), X^2 + YX + Y^3). \quad (\text{R}_i)$$

Hence the unresolved Conway–Fermat lemma is the precise recursive claim that the factor selected by (R_i) lies in the full-order fibotomic stratum, rather than in any equal-degree $\mathfrak{F}_d$ with $d \mid F_{i+1}$ and $d < F_{i+1}$.

Proof. Multiplying $c_i^2 + c_i = a_{i-1}$ by a_{i-1}^2 gives the first identity. We prove the degree assertion and the resultant formula inductively from $a_{-1} = 1$. The resultant in (R_i) is monic of degree $2 \deg A_{i-1}$ and vanishes at a_i . Since $c_i = a_i/a_{i-1} \notin E_{i-1}$, one has $a_i \notin E_{i-1}$. But $E_i/\mathbb{F}_2$ has degree 2^{i+1} , and every proper subfield of E_i is contained in its unique index-two subfield E_{i-1} . Therefore a_i has full degree $2^{i+1} = 2 \deg A_{i-1}$, so the resultant is its minimal polynomial. $\square$

Proposition 7.6 (one-step maximality is not an inductive invariant). *Let*

$$C(Y) = Y^{16} + Y^{15} + Y^{12} + Y^{11} + Y^{10} + Y^8 + 1$$

and apply the exact Conway resultant operator of Proposition 7.5:

$$\begin{aligned} B(X) &= \text{Res}_Y(C(Y), X^2 + YX + Y^3) \\ &= X^{32} + X^{31} + X^{30} + X^{27} + X^{26} + X^{24} + X^{23} + X^{22} \\ &\quad + X^{19} + X^{15} + X^{14} + X^{11} + X^8 + X^7 + 1. \end{aligned} \tag{NI_1}$$

Then C is an irreducible factor of the full-order fibotomic stratum $\mathfrak{F}_{F_4}$, where $F_4 = 65537$, whereas B is an irreducible factor of the proper-order stratum

$$\mathfrak{F}_{6700417}, \quad 6700417 = F_5/641. \tag{NI_2}$$

Moreover, if y is a root of C , then

$$\text{ord}(y) = 21845 = \frac{2^{16} - 1}{3} = 5 \cdot 17 \cdot 257. \tag{NI_{2a}}$$

Thus the countermodel also satisfies the exact multiplicative-order consequence that all preceding Conway–Fermat maximalities impose on the distinguished degree-sixteen element. Even that strengthened predecessor hypothesis, together with the exact resultant transition, does not force target maximality.

Thus the exact one-step resultant can send a full-order predecessor factor to a proper-order target factor, already at the first composite Fermat level. In particular, predecessor maximality plus the transition formula alone is not an induction theorem. This is not a counterexample to the distinguished Conway path: C is an alternative factor at the A_3 level and need not have the earlier ancestry of the actual A_3 .

Proof. First, exact modular squaring gives

$$Y^{2^{16}} \equiv Y \pmod{C},$$

and

$$Y^{2^{56}} + Y \equiv R := Y^{15} + Y^{14} + Y^{13} + Y^9 + Y^8 + Y^2 + Y \pmod{C}.$$

The Bézout identity $UC + VR = 1$ holds for

$$\begin{aligned} U &= Y^{14} + Y^{13} + Y^{12} + Y^{11} + Y^7 + Y^6 + Y^5 + Y^4 \\ &\quad + Y^3 + Y + 1, \\ V &= Y^{15} + Y^{14} + Y^{12} + Y^{11} + Y^9 + Y^8 + Y^5 + Y^3 + Y + 1. \end{aligned}$$

Rabin's criterion, for which two is the only prime divisor of sixteen, therefore proves that C is irreducible.

The Fibonacci polynomials satisfy, in characteristic two,

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + Y S_r^2,$$

and hence

$$S_{2^{m+1}}(Y) = 1 + \sum_{j=0}^{m-1} Y^{2^j}. \quad (\text{NI}_3)$$

For a root y of C , the coefficient of Y^{15} gives $\text{Tr}_{\mathbb{F}_{2^{16}}/\mathbb{F}_2}(y) = 1$. Equation (NI₃) therefore gives $S_{65537}(y) = 0$. Since 65537 is prime, y belongs to the full-order stratum $\mathfrak{F}_{65537}$.

The predecessor's own multiplicative order is certified independently by the exact reductions

$$\begin{aligned} Y^{21845} &\equiv 1, \\ Y^{4369} &\equiv 7576, \quad Y^{1285} \equiv 1653, \quad Y^{85} \equiv 2464 \pmod{C}, \end{aligned}$$

where hexadecimal bit j is the coefficient of Y^j . The last three residues are not one. Since $21845 = 5 \cdot 17 \cdot 257$, these identities prove (NI_{2a}). Under $\delta_j = F_j$ for $1 \leq j \leq 3$, Corollary 7.2 gives the same order for a_3 .

Now choose z with $z^2 + z = y$ and put $x = yz$. The trace just computed makes $Z^2 + Z + y$ irreducible over $\mathbb{F}_{2^{16}}$. Thus $x \notin \mathbb{F}_{2^{16}}$; because every proper subfield of $\mathbb{F}_{2^{32}}$ lies in $\mathbb{F}_{2^{16}}$, x has degree thirty-two over $\mathbb{F}_2$. Moreover

$$x^2 + yx + y^3 = y^2(z^2 + z + y) = 0. \quad (\text{NI}_4)$$

Modulo the relation in (NI₄), direct reduction gives

$$B(X) = C(Y)(Q_1(Y)X + Q_0(Y)), \quad (\text{NI}_5)$$

where

$$\begin{aligned} Q_1 &= Y^{29} + Y^{28} + Y^{25} + Y^{24} + Y^{21} + Y^{18} + Y^{17} + Y^{16} \\ &\quad + Y^{15} + Y^{14} + Y^{11} + Y^{10} + Y^9 + Y^6, \\ Q_0 &= Y^{32} + Y^{31} + Y^{29} + Y^{28} + Y^{27} + Y^{26} + Y^{25} + Y^{23} \\ &\quad + Y^{20} + Y^{14} + Y^{12} + Y^{11} + Y^{10} + Y^9 + 1. \end{aligned}$$

Thus $B(x) = 0$. Its displayed degree is thirty-two, so it is the minimal polynomial of x , and consequently the stated resultant.

It remains to identify the target stratum. From a pair $(f, g) = (S_r, S_{r+1})$, put

$$A = f^2, \quad D = g^2 + Xf^2.$$

Reading one further binary digit sends the pair to (A, D) for a zero and to (D, g^2) for a one. Repeated reduction modulo B along the binary word

$$6700417 = (11001100011110110000001)_2$$

ends at the exact residue pair

$$(S_{6700417}, S_{6700418}) \equiv (00000000, 1875a343) \pmod{B}, \quad (\text{NI}_6)$$

where a hexadecimal bit in position j is the coefficient of X^j . This is an $O(\log 6700417)$ polynomial-identity certificate, not field sampling. Hence $S_{6700417}(x) = 0$. The integer 6700417 is prime, so x belongs to the exact order-6700417 fibotomic stratum. Finally, if $v = (z+1)/z$, then

$$y = \frac{v}{(v+1)^2}, \quad x = yz = \frac{v}{(v+1)^3},$$

showing directly that this is the same fibotomic/elliptic transition as the Conway resultant. $\square$

7.3 Ancestry refinements and exact no-go results

Proposition 7.7 (reciprocal-trace ancestry and a trace-refined no-go). *Every distinguished Conway factor satisfies*

$$\text{Tr}_{E_i/\mathbb{F}_2}(a_i^{-1}) = 1. \quad (\text{RT}_1)$$

Equivalently, the constant and linear coefficients of every A_i are both one. More generally, this reciprocal-trace bit is preserved by every full-degree transition of Proposition 7.5.

This extra ancestry invariant still does not make one-step maximality inductive. Put

$$C_1(Y) = Y^{16} + Y^{15} + Y^{12} + Y^{11} + Y^6 + Y + 1.$$

Then C_1 is an irreducible full-order factor of $\mathfrak{F}_{65537}$, has constant and linear coefficients one, and its exact Conway resultant is

$$\begin{aligned} B_1(X) = & X^{32} + X^{31} + X^{30} + X^{28} + X^{27} + X^{26} + X^{24} + X^{23} + X^{22} \\ & + X^{20} + X^{19} + X^{18} + X^{16} + X^{15} + X^{14} + X^{13} + X^{12} + X^{11} \\ & + X^9 + X^8 + X^7 + X^5 + X^2 + X + 1. \end{aligned} \quad (\text{RT}_2)$$

The polynomial B_1 is irreducible but belongs to the proper-order stratum

$$\mathfrak{F}_{6700417}, \quad 6700417 = F_5/641. \quad (\text{RT}_3)$$

Thus predecessor maximality, the exact transition, and the first reciprocal coefficient together still do not force target maximality.

Proof. The two E_i/E_{i-1} -conjugates of a_i^{-1} have sum

$$\text{Tr}_{E_i/E_{i-1}}(a_i^{-1}) = \frac{a_{i-1}}{a_{i-1}^3} = a_{i-1}^{-2}.$$

Taking absolute traces and using compatibility with squaring gives

$$\text{Tr}(a_i^{-1}) = \text{Tr}(a_{i-1}^{-1})^2.$$

Starting with $a_{-1} = 1$ proves (RT_1) . The absolute norms are also one, so the constant coefficient is one and the linear coefficient is norm times reciprocal trace.

For the more general preservation statement, let A be a monic irreducible polynomial with constant coefficient one and root t . Its transition satisfies

$$\text{Res}_Y(A(Y), X^2 + YX + Y^3) = \text{Norm}_{\mathbb{F}_2(t)/\mathbb{F}_2}(1 + t^{-2}X + t^{-3}X^2).$$

The coefficient of X is therefore $\text{Tr}(t^{-2}) = \text{Tr}(t^{-1})^2$, the same bit as the linear coefficient of A .

We now certify the countermodel. Modulo C_1 ,

$$Y^{2^{16}} = Y, \quad Y^{2^{56}} + Y = R := Y^{13} + Y^9 + Y^7 + Y^4 + Y,$$

and the exact Bézout identity $UC_1 + VR = 1$ holds for

$$\begin{aligned} U &= Y^{10} + Y^9 + Y^7 + Y^6 + Y^5 + Y^4 + Y^3 + 1, \\ V &= Y^{13} + Y^{11} + Y^{10} + Y^7 + Y^3 + Y^2 + 1. \end{aligned}$$

Rabin's criterion proves C_1 irreducible. If y is a root, its trace is one, and the identity

$$S_{2^m+1}(Y) = 1 + \sum_{j=0}^{m-1} Y^{2^j}$$

gives $S_{65537}(y) = 0$. Since 65537 is prime, this is the full-order stratum.

Direct expansion of the resultant gives (RT_2) . If $z^2 + z = y$ and $x = yz$, then $x^2 + yx + y^3 = 0$. The trace-one condition makes $Z^2 + Z + y$ irreducible over $\mathbb{F}_{2^{16}}$, so x has degree thirty-two. Hence the displayed degree-thirty-two resultant is irreducible.

Finally, repeated use of

$$S_{2r} = S_r^2, \quad S_{2r+1} = S_{r+1}^2 + XS_r^2$$

along $6700417 = (11001100011110110000001)_2$ gives the exact residues

$$(S_{6700417}, S_{6700418}) \equiv (0, 3407\text{b}7\text{e}4) \pmod{B_1},$$

where hexadecimal bit j is the coefficient of X^j . The second residue is nonzero, and 6700417 is prime. The fibotomic criterion proves (RT_3) . $\square$

Proposition 7.8 (nine absolute moments do not force one-step maximality). *The distinguished degree-sixteen factor is*

$$A_3 = X^{16} + X^{15} + X^{11} + X^{10} + X^8 + X^7 + X^6 + X^3 + X^2 + X + 1.$$

Put

$$C = X^{16} + X^{15} + X^{11} + X^8 + X^5 + X^3 + X^2 + X + 1.$$

Then $A_3 + C = X^{10} + X^7 + X^6 + X^5$. If a and y are roots of A_3 and C , respectively, then

$$(\text{Tr}(a^k))_{k=1}^5 = (\text{Tr}(y^k))_{k=1}^5 = (1, 1, 1, 1, 0),$$

and

$$\text{Tr}(a^{-k}) = \text{Tr}(y^{-k}) = 1 \quad (1 \leq k \leq 4).$$

The factor C is irreducible and lies in the full-order stratum $\mathfrak{F}_{65537}$. Nevertheless its exact Conway resultant is

$$\begin{aligned} B &= X^{32} + X^{31} + X^{30} + X^{20} + X^{19} + X^{16} + X^{15} + X^{14} \\ &\quad + X^{11} + X^8 + X^5 + X + 1, \end{aligned}$$

which is irreducible and lies in the proper stratum $\mathfrak{F}_{6700417}$, where $6700417 = F_5/641$. Thus predecessor maximality, the exact transition, the first five forward power traces, and the first four inverse-power traces still do not force target maximality. These are absolute $\mathbb{F}_2$ -moment data only: C does not have the distinguished relative trace-norm ancestry of A_3 .

Proof. For a monic polynomial $P = X^{16} + e_1X^{15} + \cdots + e_{16}$, Newton's identities in characteristic two read

$$p_k + e_1p_{k-1} + \cdots + e_{k-1}p_1 + (k \bmod 2)e_k = 0, \quad p_k = \sum_{P(r)=0} r^k.$$

Although the final term vanishes for even k , equality of $e_1, \dots, e_s$ still inductively forces equality of $p_1, \dots, p_s$. The leading five coefficient tuples of A_3 and C are both $(1, 0, 0, 0, 1)$, giving $(p_1, \dots, p_5) = (1, 1, 1, 1, 0)$. Apply the same identities to $X^{16}P(X^{-1})/P(0)$: the first four coefficient tuples are both $(1, 1, 1, 0)$, giving the asserted inverse traces.

We give exact polynomial certificates for the remaining claims. In the following hexadecimal strings, bit j is the coefficient of degree j . Modulo $C = 1892\mathbf{f}$,

$$Y^{2^{16}} = Y, \quad R_C := Y^{2^8} + Y = 3877,$$

and

$$U_C = \mathbf{cca}, \quad V_C = 73\mathbf{b1}, \quad U_C C + V_C R_C = 1.$$

Rabin's criterion proves that C is irreducible. Its Y^{15} -coefficient gives $\text{Tr}(y) = 1$, and hence

$$S_{65537}(y) = 1 + \sum_{j=0}^{15} y^{2^j} = 0.$$

Since 65537 is prime, C lies in the full-order stratum.

In the quotient by $X^2 + YX + Y^3$, exact reduction gives

$$B(X) = C(Y)(Q_1(Y)X + Q_0(Y)), \quad Q_1 = 31180083, \quad Q_0 = 1\mathbf{a928cc53}.$$

If $z^2 + z = y$ and $x = yz$, then $x^2 + yx + y^3 = 0$. The trace-one condition makes $Z^2 + Z + y$ irreducible over $\mathbb{F}_{2^{16}}$, so x has degree thirty-two. Hence the monic degree-thirty-two polynomial B is the exact resultant and is irreducible. Independently, its Rabin certificate is

$$X^{2^{32}} = X \pmod{B}, \quad R_B := X^{2^{16}} + X = \mathbf{e55f1c7a},$$

$$U_B = 3295\mathbf{d0c3}, \quad V_B = 6748328\mathbf{a}, \quad U_B B + V_B R_B = 1.$$

Finally the exact Fibonacci residues are

$$(S_{6700417}, S_{6700418}) \equiv (0, \mathbf{bda50b33}) \pmod{B}, \quad S_{641} \equiv 7\mathbf{acf6362} \pmod{B}.$$

As 6700417 is prime, the fibotomic criterion places B in the claimed exact proper-order stratum. If C had the distinguished relative trace-norm ancestry, Proposition 7.10 would force it to be a Frobenius conjugate of a_3 , contrary to $C \neq A_3$; this proves the final scope statement. $\square$

Remark 7.9 (higher reciprocal moments). The full inverse-power trace sequence propagates exactly, but quickly becomes deeper ancestry rather than a short invariant. If $t = a_{i-1}$ and

$$D_0 = 0, \quad D_1 = t^{-2}, \quad D_j = t^{-2}D_{j-1} + t^{-3}D_{j-2},$$

then

$$\text{Tr}_{E_i/\mathbb{F}_2}(a_i^{-j}) = \text{Tr}_{E_{i-1}/\mathbb{F}_2}(D_j(a_{i-1})).$$

For any fixed factor, sufficiently many moments recover its minimal polynomial. What is not known is any uniformly bounded initial segment which forces the full fibotomic stratum throughout the tower.

Proposition 7.10 (complete ancestry and exact trace-fibre exhaustion). *For $i \geq 0$ and $-1 \leq j < i$, the distinguished elements satisfy*

$$\mathrm{Tr}_{E_i/E_j}(a_i) = a_j, \quad \mathrm{Norm}_{E_i/E_j}(a_i) = a_j^{3^{i-j}}. \quad (\mathrm{AN}_1)$$

Conversely, fix $i \geq 0$ and $x_i \in E_i$. Put

$$x_j = \mathrm{Tr}_{E_i/E_j}(x_i) \quad (-1 \leq j < i).$$

If $x_{-1} = 1$ and

$$\mathrm{Norm}_{E_j/E_{j-1}}(x_j) = x_{j-1}^3 \quad (0 \leq j \leq i), \quad (\mathrm{AN}_2)$$

then x_i is an $\mathbb{F}_2$ -Frobenius conjugate of a_i . Thus (AN_1) – (AN_2) are a necessary-and-sufficient ancestry invariant for the actual resultant factor.

Now fix $n \geq 1$, put

$$F = E_{n-1} = \mathbb{F}_q, \quad K = E_n = \mathbb{F}_{q^2},$$

and let $\ell \mid q+1$. Set

$$H = (K^\times)^\ell, \quad U = \ker \mathrm{Norm}_{K/F}.$$

For every $t \in F^\times$,

$$\#\{h \in H : \mathrm{Tr}_{K/F}(h) = t\} = \frac{q+1}{\ell} - 1. \quad (\mathrm{AN}_3)$$

If $t = a_{n-1}$, every element counted in (AN_3) has absolute trace one and is a normal, hence full-degree, element of $K/\mathbb{F}_2$. Thus, whenever $q+1$ is composite, full degree, normality, the complete Frobenius-orbit length, and the actual relative-trace value all occur among ℓ -th powers.

The normalized relative norms of the elements in (AN_3) form exactly

$$\mathcal{B}_\ell = \left\{ \frac{u}{(u+1)^2} : u \in U^\ell, u \neq 1 \right\}, \quad |\mathcal{B}_\ell| = \frac{1}{2} \left(\frac{q+1}{\ell} - 1 \right). \quad (\mathrm{AN}_4)$$

More precisely, an element h of relative trace t is an ℓ -th power and has relative norm t^3 if and only if $t \in \mathcal{B}_\ell$. For $t = a_{n-1}$ this is equivalent to

$$S_{(q+1)/\ell}(a_{n-1}) = 0. \quad (\mathrm{AN}_5)$$

Finally $[a_n] = [c_n]$ in $K^\times/K^{\times\ell}$. Consequently the entire earlier ancestry is recoverable exactly, but after the last quadratic step its power-residue content is precisely the original fibotomic membership (AN_5) . Trace, separate norm, normality, and Frobenius-orbit invariants give no additional obstruction.

Proof. The relation

$$a_i^2 + a_{i-1}a_i + a_{i-1}^3 = 0$$

is the minimal polynomial of a_i over E_{i-1} . It gives immediate trace a_{i-1} and norm a_{i-1}^3 ; transitivity gives (AN_1) .

For the converse, transitivity of trace gives $\mathrm{Tr}_{E_j/E_{j-1}}(x_j) = x_{j-1}$. Together with (AN_2) this says that x_j is a root of

$$X^2 + x_{j-1}X + x_{j-1}^3.$$

Starting from $x_{-1} = a_{-1} = 1$, induct on j . If x_{j-1} is a Frobenius conjugate of a_{j-1} , the displayed polynomial is the corresponding Frobenius conjugate of the polynomial for a_j ; its two roots are

the two lifts of that conjugacy to E_j . Hence x_j , and finally x_i , is a Frobenius conjugate of the distinguished element.

Since $\gcd(\ell, q-1) = 1$, one has $F^\times \subset H$. Multiplication by any $t \in F^\times$ bijects the trace-one and trace- t fibres in H . In characteristic two, $\text{Tr}_{K/F}(h) = 0$ is equivalent to $h^q = h$, so the nonzero trace-zero elements are exactly $F^\times$, all of which lie in H . If the common nonzero-fibre size is M , then

$$\frac{q^2 - 1}{\ell} = |H| = (q-1) + (q-1)M,$$

which gives (AN₃).

The distinguished a_{n-1} has absolute trace one by (AN₁). Thus every h in its relative-trace fibre also has absolute trace one. To justify normality, let $N = [K : \mathbb{F}_2]$, a power of two, and let ρ generate its cyclic Galois group. After choosing one normal generator, K is the regular module over

$$\mathbb{F}_2[\langle \rho \rangle] \simeq \mathbb{F}_2[X]/((X+1)^N),$$

a local ring. An element is a normal generator exactly when its group-ring coordinate is a unit, equivalently when multiplication by $(\rho+1)^{N-1}$ does not kill it. Since

$$(\rho+1)^{N-1} = 1 + \rho + \cdots + \rho^{N-1},$$

this last value is its absolute trace. Hence absolute trace one is equivalent to normality.

Write $h = tz$ in a nonzero trace fibre. Then $\text{Tr}_{K/F}(z) = 1$, so $z^q = z + 1$ and

$$\text{Norm}_{K/F}(h) = t^2 z(z+1).$$

Put $u = (z+1)/z \in U$; then $z = 1/(u+1)$ and

$$z(z+1) = \frac{u}{(u+1)^2}. \quad (\text{AN}_6)$$

We claim $z \in H$ if and only if $u \in U^\ell$. The forward implication follows from $u = z^{q-1}$. Conversely, if $u \in U^\ell$, then $u \in H$. The right side of (AN₆) lies in $F^\times \subset H$, so in $K^\times/H$ one has

$$[u] = 2[u+1].$$

Because ℓ is odd, multiplication by two is invertible; hence $[u] = 0$ forces $[u+1] = 0$, and $z = (u+1)^{-1} \in H$. The map $u \mapsto u/(u+1)^2$ identifies precisely the inverse pair $\{u, u^{-1}\}$, proving (AN₄).

The equality $\text{Norm}(h) = t^3$ is, after division by t^2 , exactly $t = z(z+1)$. This proves the first assertion following (AN₄). The set U^ℓ has order $(q+1)/\ell$, and the Fibonacci-polynomial factorization identifies its nontrivial inverse pairs with the roots of $S_{(q+1)/\ell}$, proving (AN₅). Finally $a_n = a_{n-1}c_n$ and $a_{n-1} \in F^\times \subset H$, so $[a_n] = [c_n]$. $\square$

Proposition 7.11 (ℓ -th-root ancestry and the exact Dickson obstruction). *For variables S, P , define the characteristic-two Dickson power sums by*

$$\mathcal{D}_0(S, P) = 0, \quad \mathcal{D}_1(S, P) = S, \quad \mathcal{D}_{r+2}(S, P) = S\mathcal{D}_{r+1}(S, P) + P\mathcal{D}_r(S, P).$$

Thus, whenever $x + y = S$ and $xy = P$,

$$\mathcal{D}_r(S, P) = x^r + y^r.$$

Fix $n \geq 2$ and a prime $\ell \mid F_n$. Put $r_{-1} = 1$, and for $0 \leq j < n$ let $r_j \in E_j^\times$ be the unique element such that

$$r_j^\ell = a_j.$$

Then, for every $0 \leq j < n$,

$$N_{E_j/E_{j-1}}(r_j) = r_{j-1}^3, \quad \mathcal{D}_\ell(\text{Tr}_{E_j/E_{j-1}}(r_j), r_{j-1}^3) = r_{j-1}^\ell. \quad (\text{RA})$$

Set $b = r_{n-1}$. Failure of the ℓ -primary Conway–Fermat condition at the top step has the exact descended form

$$c_n \in (E_n^\times)^\ell \iff \text{there exists } s \in E_{n-1} \text{ such that } \mathcal{D}_\ell(s, b) = 1. \quad (\text{RD})$$

If additionally $\delta_j = F_j$ for every $1 \leq j < n$, then, with $q = |E_{n-1}| = 2^{2^n}$,

$$\text{ord}(b) = \text{ord}(a_{n-1}) = \frac{q-1}{3}.$$

Consequently all exact lower trace, norm, and multiplicative-order data are automatically compatible with a hypothetical top failure; the single selected equation (RD) is the remaining obstruction.

Proof. Because $\ell \mid F_n$, it is coprime to $|E_j^\times|$ for every $j < n$. Hence the ℓ -power map is an automorphism on every lower multiplicative group and the elements r_j are unique. Applying the ℓ -power map to the norm and using Proposition 7.10 gives

$$N(r_j)^\ell = N(a_j) = a_{j-1}^3 = (r_{j-1}^3)^\ell.$$

Injectivity of the ℓ -power map on $E_{j-1}^\times$ proves the first identity in (RA). If r'_j is the relative conjugate of r_j , then

$$\begin{aligned} \mathcal{D}_\ell(r_j + r'_j, r_j r'_j) &= r_j^\ell + (r'_j)^\ell \\ &= \text{Tr}_{E_j/E_{j-1}}(a_j) = a_{j-1} = r_{j-1}^\ell, \end{aligned}$$

which proves the second.

The group $E_{n-1}^\times$ is uniquely ℓ -divisible, while $E_n^\times/E_{n-1}^\times$ is cyclic. Hence the coset of c_n misses the full ℓ -primary part of the quotient if and only if it is an ℓ -th power there, equivalently if and only if $c_n \in (E_n^\times)^\ell$.

Suppose first that $c_n = t^\ell$. Since $\text{Tr}_{E_n/E_{n-1}}(c_n) = 1$ and $N_{E_n/E_{n-1}}(c_n) = a_{n-1}$, injectivity below gives

$$N(t) = b.$$

For $s = \text{Tr}(t)$, the power-sum identity therefore gives

$$\mathcal{D}_\ell(s, b) = \text{Tr}(t^\ell) = 1.$$

Conversely, suppose $\mathcal{D}_\ell(s, b) = 1$, and let t, t' be the roots of

$$X^2 + sX + b.$$

They lie in the unique extension of E_{n-1} of degree at most two, hence in E_n . Their ℓ -th powers satisfy

$$t^\ell + (t')^\ell = 1, \quad t^\ell (t')^\ell = b^\ell = a_{n-1}.$$

Thus $t^\ell, (t')^\ell$ are the two roots of

$$X^2 + X + a_{n-1},$$

namely $c_n, c_n + 1$. In particular c_n is an ℓ -th power, proving (RD).

Finally, prior maximality, Corollary 7.2, and the $n = 1$ base case established in its proof give

$$\text{ord}(a_{n-1}) = \prod_{j=1}^{n-1} F_j = \frac{q-1}{3}.$$

Taking the unique ℓ -th root preserves this order because ℓ is coprime to $q-1$. □

Proposition 7.12 (Dickson all-or-nothing factorization). *Retain Proposition 7.11. Thus $n \geq 2$,*

$$K = E_{n-1} = \mathbb{F}_q, \quad L = E_n = \mathbb{F}_{q^2}, \quad a = a_{n-1}, \quad b = r_{n-1}, \quad b^\ell = a,$$

where the prime ℓ divides $F_n = q+1$. Put $c = c_n$, so that

$$c^2 + c = a, \quad c^q = c + 1,$$

and define

$$f_{\ell,n}(X) = \mathcal{D}_\ell(X, b) + 1 \in K[X].$$

Then $f_{\ell,n}$ is separable of degree ℓ , and exactly one of the following occurs:

- (i) $c \in L^{\times \ell}$, in which case $f_{\ell,n}$ splits into ℓ distinct linear factors over K ;
- (ii) $c \notin L^{\times \ell}$, in which case $f_{\ell,n}$ is irreducible over K .

Consequently the Dickson equation in (RD) never has an intermediate number of roots: it has either ℓ roots in K or none.

More explicitly, choose $t \in \overline{K}$ with $t^\ell = c$, and put

$$\vartheta = \frac{t^{q+1}}{b} \in \mu_\ell.$$

This element is independent of the chosen ℓ -th root, while the standard finite-field Kummer symbol is

$$c^{(q^2-1)/\ell} = \vartheta^{-2}.$$

With the convention $\text{Res}(f, g) = \prod_{f(s)=0} g(s)$, one has

$$\text{Res}_X(f_{\ell,n}(X), X^q - X) = (1 + \vartheta)^\ell. \tag{DF}$$

Thus the resultant vanishes precisely in the Conway–Fermat failure case.

Finally, if A_{n-1} is the minimal polynomial of a_{n-1} over $\mathbb{F}_2$, then

$$C_n(Y) := A_{n-1}(Y^2 + Y)$$

is the degree- 2^{n+1} minimal polynomial of c_n . The absolute composition

$$C_n(X^\ell) = A_{n-1}(X^{2\ell} + X^\ell) \tag{AC}$$

is irreducible of degree $2^{n+1}\ell$ in case (ii), while in case (i) it is the product of exactly ℓ distinct irreducible polynomials, each of degree 2^{n+1} .

Proof. Because $\ell \mid q + 1$, one has $\mu_\ell \subset L$ and $q \equiv -1 \pmod{\ell}$. Fix a primitive ℓ -th root ζ , choose $t^\ell = c$, and put $\tilde{t} = b/t$. Since

$$\tilde{t}^\ell = \frac{b^\ell}{c} = \frac{a}{c} = c + 1,$$

the ℓ elements, indexed by $\mathbb{Z}/\ell\mathbb{Z}$,

$$s_i = t\zeta^i + \tilde{t}\zeta^{-i}$$

satisfy

$$\mathcal{D}_\ell(s_i, b) = (t\zeta^i)^\ell + (\tilde{t}\zeta^{-i})^\ell = c + (c + 1) = 1.$$

They are distinct. Indeed, equality $s_i = s_j$ identifies the unordered root pairs of the corresponding polynomials $X^2 + s_iX + b$. Besides $i = j$, the only possible pairing would identify $t\zeta^i$ with $\tilde{t}\zeta^{-j}$; taking ℓ -th powers would give $c = c + 1$. Hence the s_i are all the roots of the monic degree- ℓ polynomial $f_{\ell,n}$.

There is a unique $k \in \mathbb{Z}/\ell\mathbb{Z}$ such that

$$t^q = \tilde{t}\zeta^k.$$

Since $\zeta^q = \zeta^{-1}$, direct substitution gives

$$s_i^q = s_{i-k}, \tag{DF_1}$$

and applying q -Frobenius once more gives

$$t^{q^2} = t\zeta^{-2k}. \tag{DF_2}$$

As ℓ is odd, $k = 0$ if and only if $t \in L$. Because all ℓ -th roots of c differ by an element of $\mu_\ell \subset L$, this is equivalent to $c \in L^{\times\ell}$. If $k = 0$, every s_i is fixed by q -Frobenius. If $k \neq 0$, translation by $-k$ is one ℓ -cycle because ℓ is prime. Equation (DF₁) therefore proves the split/irreducible dichotomy.

The quotient t^{q+1}/b is ζ^k , and replacing t by another ℓ -th root does not change it because $\ell \mid q + 1$. Equation (DF₂) yields

$$c^{(q^2-1)/\ell} = t^{q^2-1} = \vartheta^{-2}.$$

For the resultant, characteristic two and (DF₁) give

$$\text{Res}(f_{\ell,n}, X^q - X) = \prod_i (s_{i-k} + s_i).$$

Moreover

$$s_{i-k} + s_i = (1 + \vartheta) (\vartheta^{-1}t\zeta^i + \tilde{t}\zeta^{-i}).$$

After pulling out the factors ζ^{-i} , whose product is one, the fact that multiplication by two permutes $\mathbb{Z}/\ell\mathbb{Z}$ gives

$$\prod_i (\vartheta^{-1}t\zeta^i + \tilde{t}\zeta^{-i}) = (\vartheta^{-1}t)^\ell + \tilde{t}^\ell = c + (c + 1) = 1.$$

This proves (DF).

The element a has absolute trace one, so $Y^2 + Y + a$ is irreducible over K . Since $a = c^2 + c$ and $[K : \mathbb{F}_2] = 2^n$, the element c has degree 2^{n+1} over $\mathbb{F}_2$; hence $\mathcal{C}_n$ is its minimal polynomial. Capelli's lemma now says that $\mathcal{C}_n(X^\ell)$ is irreducible over $\mathbb{F}_2$ if and only if $X^\ell - c$ is irreducible over L . Since $\mu_\ell \subset L$ and ℓ is prime, this holds exactly when $c \notin L^{\times\ell}$.

In the other case every root x of $\mathcal{C}_n(X^\ell)$ lies in L , because its ℓ -th power is a Frobenius conjugate of c and all its ℓ -th roots lie in L . That conjugate has degree 2^{n+1} , so

$$2^{n+1} \leq [\mathbb{F}_2(x) : \mathbb{F}_2] \leq [L : \mathbb{F}_2] = 2^{n+1}.$$

Separability then partitions the $2^{n+1}\ell$ roots into exactly ℓ distinct Frobenius orbits of length 2^{n+1} , proving the final assertion. $\square$

Proposition 7.13 (reverse-Dickson trace divisor). *Retain Proposition 7.12, and put*

$$m = 2^n, \quad q = 2^m, \quad K = \mathbb{F}_q.$$

Let $A_{n-1} \in \mathbb{F}_2[T]$ be the degree- m minimal polynomial of $a = a_{n-1}$, and define

$$\begin{aligned} R_\ell(Z) &= Z^\ell \mathcal{D}_\ell(Z^{-1}, 1) = \mathcal{D}_\ell(1, Z)^2, \\ P_{\ell,a}(Z) &= Z^\ell + a R_\ell(Z), \\ H_m(Z) &= 1 + \sum_{j=0}^{m-1} Z^{2^j}, \\ \mathcal{P}_{\ell,n}(Z) &= \prod_{r=0}^{m-1} P_{\ell,a^{2^r}}(Z) = R_\ell(Z)^m A_{n-1}\left(\frac{Z^\ell}{R_\ell(Z)}\right). \end{aligned} \tag{TD}$$

The displayed rational expression is a polynomial in $\mathbb{F}_2[Z]$. Moreover

$$R_\ell \in \mathbb{F}_2[Z^2], \quad R_\ell(0) = 1, \quad [Z^{\ell-1}]R_\ell = 1,$$

and the following are equivalent:

- (i) $c_n \in E_n^{\times \ell}$, so the singleton-even arm fails at ℓ ;
- (ii) $P_{\ell,a}$ splits completely over K ;
- (iii)

$$\mathcal{P}_{\ell,n}(Z) \mid H_m(Z) \quad \text{in } \mathbb{F}_2[Z]. \tag{TD_1}$$

In the failure case every root of $P_{\ell,a}$ has absolute trace one, and

$$\sum_{P_{\ell,a}(z)=0} z = \prod_{P_{\ell,a}(z)=0} z = a. \tag{TD_2}$$

There is also an absolute all-or-nothing alternative: $\mathcal{P}_{\ell,n}$ is irreducible over $\mathbb{F}_2$ of degree $m\ell$ on success, while on failure it is the product of exactly ℓ distinct irreducible polynomials of degree m .

Consequently failure forces

$$m\ell \leq \frac{q}{2}. \tag{TD_3}$$

Equivalently, for the complementary divisor $d = (q+1)/\ell$, one must have $d \geq 2m+1$.

Proof. The Dickson polynomial $\mathcal{D}_\ell(X, 1)$ contains only odd powers because ℓ is odd. Termwise comparison gives the first identity in (TD) and the three assertions about R_ℓ . In particular

$$P'_{\ell,a}(Z) = Z^{\ell-1};$$

since $P_{\ell,a}(0) = a \neq 0$, this polynomial is separable.

For $s \neq 0$, set $z = b/s^2$. Dickson homogeneity gives

$$\mathcal{D}_\ell(1, z) = s^{-\ell} \mathcal{D}_\ell(s, b),$$

and hence

$$P_{\ell,a}(z) = \frac{a}{s^{2\ell}} (1 + \mathcal{D}_\ell(s, b)^2). \quad (\text{TD}_4)$$

Thus $s \mapsto b/s^2$ is a Frobenius-equivariant bijection between the roots of $\mathcal{D}_\ell(X, b) + 1$ and the roots of $P_{\ell,a}$; its inverse sends z to the unique square root of b/z . The split/irreducible dichotomy in Proposition 7.12 therefore proves the equivalence of (i) and (ii).

Suppose failure holds, and let $s \in K$ be one of the Dickson roots. If t, t' are the roots of $X^2 + sX + b$, then

$$\{t^\ell, (t')^\ell\} = \{c_n, c_n + 1\}.$$

Neither right-hand element belongs to K , so the quadratic is irreducible over K . The Artin–Schreier criterion now gives

$$\text{Tr}_{K/\mathbb{F}_2}(b/s^2) = 1.$$

Thus the ℓ roots of $P_{\ell,a}$ are distinct trace-one elements of K .

The Frobenius conjugates of a are distinct. The root sets of two different conjugate polynomials $P_{\ell,a^{2^r}}$ are disjoint: a common root z would force $R_\ell(z) = 0$, then $z^\ell = 0$, contrary to $R_\ell(0) = 1$. Therefore failure supplies $m\ell$ distinct trace-one roots of $\mathcal{P}_{\ell,n}$. The roots of H_m are exactly the trace-one elements of K , proving (iii). Conversely, (iii) puts every root of $P_{\ell,a}$ in K ; equation (TD₄) then puts every Dickson root in K , so Proposition 7.12 gives failure. The coefficient and constant terms of $P_{\ell,a}$ give (TD₂).

On success $P_{\ell,a}$ is irreducible over K . If z is one of its roots, then

$$a = \frac{z^\ell}{R_\ell(z)} \in \mathbb{F}_2(z),$$

so $K = \mathbb{F}_2(a) \subseteq \mathbb{F}_2(z)$, and z has absolute degree $m\ell$. Hence $\mathcal{P}_{\ell,n}$ is its irreducible polynomial. On failure every root $z \in K$ still has absolute degree m , by the same displayed identity, and the disjoint conjugate fibres form exactly ℓ Frobenius orbits. This proves the factorization alternative. Finally $\deg H_m = 2^{m-1} = q/2$, giving (TD₃); the equivalent form follows from $q \geq 2m$. $\square$

Proposition 7.14 (the reverse-Dickson divisor is a transported Kummer coset). *Retain Proposition 7.13, put $N = q + 1$, and write*

$$U = \ker\left(N_{\mathbb{F}_{q^2}/\mathbb{F}_q}\right) = \mu_N.$$

For odd r , define

$$T_r(Z) = \mathcal{D}_r(1, Z), \quad F_r(Z) = \frac{Z^r}{T_r(Z)^2}, \quad \Phi(u) = \frac{u}{(u+1)^2}.$$

Then Φ induces a bijection

$$(U \setminus \{1\})/(u \sim u^{-1}) \xrightarrow{\sim} \{z \in \mathbb{F}_q : \text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(z) = 1\}, \quad (\text{KC}_1)$$

and one has the semiconjugacy

$$T_r(\Phi(u)) = \frac{1+u^r}{(1+u)^r}, \quad F_r(\Phi(u)) = \Phi(u^r). \quad (\text{KC}_2)$$

Consequently

$$H_m(Z) = T_N(Z) = \prod_{\substack{\{u, u^{-1}\} \\ u \in U \setminus \{1\}}} (Z + \Phi(u)). \quad (\text{KC}_3)$$

For the selected Conway point put

$$w = \frac{c_n + 1}{c_n} \in U, \quad a = \Phi(w) = a_{n-1}.$$

Failure at ℓ is equivalent to $w \in U^\ell$. If $u_0^\ell = w$, then

$$P_{\ell,a}(Z) = Z^\ell + aT_\ell(Z)^2 = \prod_{v \in \mu_\ell} (Z + \Phi(u_0v)). \quad (\text{KC}_4)$$

Thus the reverse-Dickson fibre is exactly one multiplicative μ_ℓ -coset transported through Φ . Furthermore

$$T_\ell \mid H_m, \quad \gcd(\mathcal{P}_{\ell,n}, T_\ell) = 1,$$

so failure gives the sharper factor statement

$$\mathcal{P}_{\ell,n}(Z)T_\ell(Z) \mid H_m(Z). \quad (\text{KC}_5)$$

There are exactly

$$\frac{N/\ell - 1}{2}$$

distinct trace-one values $\Phi(w)$ with $w \in U^\ell \setminus \{1\}$, each realizing the same transported coset factorization.

Proof. For $u \in U \setminus \{1\}$, put $c = (u + 1)^{-1}$. Since $u^q = u^{-1}$,

$$c^q = c + 1, \quad c^2 + c = \Phi(u).$$

The Artin–Schreier trace criterion gives $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(\Phi(u)) = 1$. Direct simplification also gives

$$\Phi(u) = \Phi(v) \iff (u + v)(1 + uv) = 0,$$

so the fibres are precisely inverse pairs. Both sides of (KC_1) have $q/2$ elements.

The roots of $X^2 + X + \Phi(u)$ are $(1 + u)^{-1}$ and $u(1 + u)^{-1}$. The Dickson power-sum identity therefore proves the first formula in (KC_2) , and the second follows on squaring the denominator. Taking $r = N$ shows that the monic degree- $q/2$ polynomials T_N and H_m have the same roots, proving (KC_3) .

One has $w = c_n^{q-1}$. In the coprime decomposition $\mathbb{F}_{q^2}^\times \simeq \mathbb{F}_q^\times \times U$, the ℓ -power map is an automorphism on $\mathbb{F}_q^\times$. Hence c_n is an ℓ -th power exactly when $w \in U^\ell$, by Proposition 7.11. For $v \in \mu_\ell$, semiconjugacy gives

$$F_\ell(\Phi(u_0v)) = \Phi((u_0v)^\ell) = \Phi(w) = a.$$

These ℓ values are distinct. Indeed, an inverse collision inside $u_0\mu_\ell$ would imply $w^2 = 1$, hence $w = 1$ because $|U|$ is odd; but $w = 1$ is impossible. The monic degree- ℓ polynomial $P_{\ell,a}$ therefore has precisely the roots in (KC_4) .

The same argument on $\mu_\ell \setminus \{1\}$ gives $T_\ell \mid T_N$. Also

$$P_{\ell,a} \equiv Z^\ell \pmod{T_\ell}, \quad T_\ell(0) = 1,$$

and the same congruence holds for every Frobenius conjugate. This proves coprimality and (KC_5) . Finally the image U^ℓ has order N/ℓ ; its nonidentity elements form $(N/\ell - 1)/2$ inverse pairs, and (KC_1) proves the count. $\square$

Remark 7.15 (exact fibotomic support of a hypothetical failure). Use the fibotomic polynomials $\mathfrak{F}_s$ of Proposition 7.3. Put

$$e = v_\ell(N), \quad \delta = \text{ord}(w) = \ell^f \delta_0, \quad (\ell, \delta_0) = 1.$$

Under failure $0 \leq f \leq e - 1$. If $f = 0$, the coset $u_0\mu_\ell$ contains exactly one element of order δ and $\ell - 1$ elements of order $\ell\delta$. Accordingly $\mathcal{P}_{\ell,n}$ has exactly one degree- m irreducible factor in $\mathfrak{F}_\delta$ and exactly $\ell - 1$ in $\mathfrak{F}_{\ell\delta}$. If $f \geq 1$, every element of the coset has order $\ell\delta$, and all ℓ factors lie in $\mathfrak{F}_{\ell\delta}$. This includes the cases $v_\ell(N) > 1$.

These support constraints do not close the arm. Every trace-one element of $\mathbb{F}_q$ has absolute degree m : if it lay in a proper subfield $\mathbb{F}_{2^d}$, then m/d would be even and its absolute trace would vanish. Hence the only resulting capacity inequality is

$$\varphi(\delta) \geq 2m.$$

It is automatic, because $\delta > 1$ divides $2^m + 1$, while m is a power of two; therefore $\text{ord}_\delta(2) = 2m$. The full divisor, derivative, Hasse-derivative, Newton-sum, coefficient, and fibotomic-support packages are thus realized by transported nonselected cosets. Excluding the selected one still requires the recursive Conway ancestry.

Remark 7.16 (the reverse-Dickson gap is selected but not yet excluded). Put $e = v_2(\ell - 1)$. For $e \geq 2$, substitution $X = W + W^{-1}$ gives

$$\mathcal{D}_\ell(X, 1) + X = (W^{\ell-1} + 1)(W + W^{-\ell}).$$

At $W = 1$ the two factors vanish to orders 2^e and 2, while X has order 2. Therefore

$$\mathcal{D}_\ell(X, 1) \equiv X \pmod{X^{2^{e-1}+1}}. \quad (\text{TG})$$

Every prime divisor of the Fermat number $F_n = q + 1$ has $e \geq n + 2$. Hence

$$\begin{aligned} P_{\ell,a}(Z) &= Z^\ell + aZ^{\ell-1} + \text{terms of degree at most } \ell - (2m + 1), \\ \mathcal{P}_{\ell,n}(Z) &= Z^{m(\ell-1)} A_{n-1}(Z) + \text{terms of degree at most } m\ell - (2m + 1). \end{aligned} \quad (\text{TG}_1)$$

This long leading gap still does not contradict (TD₁). For $n > 2$, every nontrivial complementary divisor $d \mid F_n$ satisfies $d \equiv 1 \pmod{4m}$, hence $d \geq 4m + 1$ and $m\ell < q/4$. A hypothetical quotient $H_m/\mathcal{P}_{\ell,n}$ is therefore long enough to absorb the leading gap. The divisor criterion retains the actual selected polynomial A_{n-1} ; its elementary degree and coefficient consequences alone do not decide it.

7.4 Elliptic and generalized-Jacobian reformulations

Proposition 7.17 (supersingular elliptic form of the Conway transition). *For $i \geq 0$ put*

$$u_i := \frac{c_i + 1}{c_i}.$$

Fix $n \geq 1$, set $Q = 2^{2^{n-1}}$, $v = u_{n-1}$ and $u = u_n$. Then the exact Conway recursion is

$$\frac{u}{(u+1)^2} = \frac{v}{(v+1)^3}, \quad (\text{E})$$

or equivalently

$$u(v+1)^3 = v(u+1)^2, \quad u + u^{-1} = v^2 + v + 1 + v^{-1}.$$

The normalization of this correspondence is the fixed supersingular elliptic curve over $\mathbb{F}_2$

$$\mathcal{E} : \quad y^2 + y = x^3 + x^2. \quad (\text{W})$$

The maps to the v - and u -lines have degrees two and three, respectively; in particular the transition has no rational section recovering u from v .

On the original (u, v) correspondence, the rational map

$$\mathcal{A}(u, v) = \left(\frac{u+v}{uv+1}, v^{-1} \right)$$

extends to an automorphism of order four, with

$$\mathcal{A}^2(u, v) = (u^{-1}, v).$$

For the distinguished Conway point $P_n = (u_n, u_{n-1})$ one has

$$\text{Frob}_Q(P_n) = \mathcal{A}(P_n) \quad \text{or} \quad \mathcal{A}^{-1}(P_n). \quad (\text{Q})$$

Thus, at P_n , relative Frobenius is one of the two quarter-turns $\mathcal{A}^{\pm 1}$ on the elliptic correspondence.

Proof. Write $a_i = \prod_{j \leq i} c_j$. The identities

$$a_{n-1} = \frac{u}{(u+1)^2}, \quad a_{n-1} = a_{n-2}c_{n-1} = \frac{v}{(v+1)^2} \frac{1}{v+1}$$

give (E); inversion gives the second displayed form.

Put $U = u + 1$, $V = v + 1$, and blow up the affine singularity $(U, V) = (0, 0)$ by $X = U/V$, $Y = V$. After division by V^2 the equation becomes

$$XY^2 + (X^2 + 1)Y + X^2 = 0.$$

Its homogenization

$$H = XY^2 + (X^2 + Z^2)Y + X^2Z$$

has partial derivatives $(Y^2, X^2 + Z^2, X^2)$ and is therefore smooth. The point $[1 : 1 : 0]$ is a flex. The linear change

$$(x, y, z) = (Z, X, X + Y + Z)$$

takes $H = 0$ to $y^2z + yz^2 = x^3 + x^2z$, which is (W). This curve has five $\mathbb{F}_2$ -points, hence trace -2 , and is supersingular. The original equation has degrees two in u and three in v , proving the degrees of the two projections. A rational section of the nontrivial degree-two finite cover would induce a retraction of its degree-two function-field extension, impossible for a proper finite field extension.

Direct substitution in (E) gives

$$\frac{\mathcal{A}_u}{(\mathcal{A}_u + 1)^2} = v \frac{u}{(u+1)^2} = \frac{v^{-1}}{(v^{-1} + 1)^3},$$

so $\mathcal{A}$ preserves the correspondence. A second substitution gives $\mathcal{A}^2(u, v) = (u^{-1}, v)$, proving the order assertion. Thus this birational self-map and its birational inverse extend over the smooth projective normalization.

Finally $v^Q = v^{-1}$ and

$$\left(\frac{u}{(u+1)^2}\right)^Q = \left(\frac{v}{(v+1)^3}\right)^Q = v \frac{u}{(u+1)^2}.$$

The fibres of $z \mapsto z/(z+1)^2$ are the inverse-pairs $\{z, z^{-1}\}$. The first coordinate of $\text{Frob}_Q(P_n)$ is therefore either $\mathcal{A}_u(P_n)$ or its inverse, while the second coordinate is v^{-1} , proving (Q). $\square$

Proposition 7.18 (new-prime character neutrality and the exact Kummer target). *Retain Proposition 7.17 and let ℓ be a prime divisor of $Q^2 + 1 = F_n$. If $\chi : E_n^\times = \mathbb{F}_{Q^4}^\times \rightarrow \mu_\ell$ has exact order ℓ , and*

$$\alpha = \chi(u), \quad \beta = \chi(u+1),$$

then

$$\alpha = \beta^2, \quad \beta^{Q^2} = \beta^{-1}. \quad (\text{K})$$

The second identity is tautological because $Q^2 \equiv -1 \pmod{\ell}$; in particular the exact lower transition and the quarter-turn Frobenius symmetry do not by themselves exclude $\beta = 1$.

The following are equivalent:

- (i) the order δ_n contains the full ℓ -primary part of F_n ;
- (ii) $u_n \notin U_n^\ell$, where $U_n = \ker N_{E_n/E_{n-1}}$;
- (iii) $c_n \notin (E_n^\times)^\ell$;
- (iv) $X^\ell - c_n$ is irreducible over E_n .

Consequently the all-level Conway–Fermat theorem is exactly the nontriviality, for every $\ell \mid F_n$, of this new Kummer symbol at the distinguished point of (W). No lower-field multiplicative character can supply its value.

Proof. The lower field has order Q^2 , and $\gcd(\ell, Q^2 - 1) = 1$, so χ is trivial on all of $E_{n-1}^\times$, in particular on every expression in v . Applying χ to (E) gives $\alpha = \beta^2$. Since $u^{Q^2} = u^{-1}$,

$$(u+1)^{Q^2} = u^{-1} + 1 = \frac{u+1}{u},$$

which gives the other identity in (K). It reduces to $\beta^{-1} = \beta^{-1}$ modulo ℓ .

In the cyclic group U_n of order F_n , nonmembership in U_n^ℓ is equivalent to having the full ℓ -primary order. Here

$$U_n^\ell = U_n \cap (E_n^\times)^\ell. \quad (\text{K}_0)$$

Indeed, if $u = x^\ell \in U_n$, then $N_{E_n/E_{n-1}}(x)^\ell = 1$; because $\gcd(\ell, Q^2 - 1) = 1$, this forces $N(x) = 1$, so $x \in U_n$. The reverse inclusion is immediate. Moreover (E) lies in the lower field, hence is an ℓ -th power in E_n . In $E_n^\times / (E_n^\times)^\ell$ it therefore says

$$[u] = 2[u+1].$$

Since ℓ is odd, u is an ℓ -th power exactly when $u+1$ is. But $u+1 = c_n^{-1}$, proving the equivalence of the first three assertions. Finally $\mu_\ell \subset E_n$, so the prime-degree Kummer criterion proves the equivalence with irreducibility of $X^\ell - c_n$. $\square$

Proposition 7.19 (four-point saturation of a Fermat failure). *Fix $n \geq 2$, put*

$$c = c_n, \quad d = c_{n-1}, \quad b = a_{n-2}, \quad Q = |E_{n-2}| = 2^{2^{n-1}}.$$

Then, for one $\epsilon \in \mathbb{F}_2$,

$$c^Q = c + d + \epsilon, \quad c^{Q^2} = c + 1. \quad (\text{FP}_1)$$

Consequently, if $\ell \mid F_n$ and c is an ℓ -th power in E_n , the complete affine plane

$$c + \mathbb{F}_2 + \mathbb{F}_2 d = \{c, c + 1, c + d, c + d + 1\} \quad (\text{FP}_2)$$

consists of ℓ -th powers. Its product is nevertheless the old-field element

$$N_{E_n/E_{n-2}}(c) = b^3, \quad (\text{FP}_3)$$

which is automatically an ℓ -th power. Thus the exact four-point saturation and its norm are compatible with failure; they do not supply the missing nonresidue.

Proof. The tower equations give

$$d^2 + d = b, \quad c^2 + c = bd = d^3 + d^2, \quad d^Q = d + 1.$$

For $s = c^Q + c$, subtraction of the equation for c from its Q -th power gives $s^2 + s = b$. Hence $s \in \{d, d + 1\}$, proving the first identity in (FP_1) ; applying Q -Frobenius once more gives the second. Frobenius preserves the subgroup of ℓ -th powers, so one power value forces all four values in (FP_2) . Finally,

$$c(c + 1) = bd, \quad (c + d)(c + d + 1) = b(d + 1),$$

and their product is $b^2 d(d + 1) = b^3$. Since $\ell \mid Q^2 + 1$ is coprime to $|E_{n-2}^\times| = Q - 1$, every old-field element is an ℓ -th power in E_n . $\square$

Proposition 7.20 (CM form and the Kummer-cover obstruction). *On $\mathcal{E} : y^2 + y = x^3 + x^2$, let O be the origin and $T = (0, 0)$. Then $\mathcal{E}(\mathbb{F}_2) = \langle T \rangle \cong \mathbb{Z}/5\mathbb{Z}$, with*

$$-T = (0, 1), \quad 2T = (1, 1), \quad -2T = (1, 0).$$

On the normalized Conway correspondence, the function $\mathfrak{c} = 1/(u + 1)$ is

$$\mathfrak{c} = \frac{x}{x^2 + x + y}, \quad \text{div}(\mathfrak{c}) = [-T] + 2[O] - 3[-2T]. \quad (\text{EC}_1)$$

The automorphism $\mathcal{A}$ satisfies

$$\mathcal{A}^2(P) = T - P, \quad \mathfrak{c}(\mathcal{A}^2 P) = \mathfrak{c}(P) + 1. \quad (\text{EC}_2)$$

If π is 2-power Frobenius and $i = \pi + 1$, then $i^2 = -1$, $i(T) = 2T$, and, after possibly replacing $\mathcal{A}$ by its inverse,

$$\mathcal{A} = \tau_{2T} \circ i.$$

Both orientations fix $R_0 = -2T$.

For $n \geq 4$, set

$$Q = 2^{2^{n-1}}, \quad R = \sqrt{Q} = 2^{2^{n-2}},$$

and let P_n be the distinguished Conway point. If $X_n = P_n - R_0$, then

$$\text{Frob}_Q(X_n) = \pm i X_n, \quad X_n \in \ker([R] \mp i). \quad (\text{EC}_3)$$

Each kernel is cyclic of order $R^2 + 1 = Q + 1$.

For every prime $\ell \mid Q^2 + 1$, the Kummer cover

$$\mathcal{C}_\ell : \quad z^\ell = \mathfrak{c}$$

is geometrically connected, ramified exactly above $-T, O, -2T$, and has genus

$$g(\mathcal{C}_\ell) = \frac{3\ell - 1}{2}. \quad (\text{EC}_4)$$

The automorphism $\mathcal{A}$ does not lift to $\mathcal{C}_\ell$. Hence the quarter-turn on the base supplies no action on the Kummer fibre above P_n ; the surviving Conway condition is exactly

$$\mathfrak{c}(P_n) = c_n \notin (\mathbb{F}_{Q^4}^\times)^\ell. \quad (\text{EC}_5)$$

It is the Artin-symbol problem for one distinguished fibre of this ramified cover, not a consequence of the CM action on $\mathcal{E}$.

Proof. In the blow-up coordinates of Proposition 7.17, $U = u + 1$, $V = v + 1$, $X = U/V$, $Y = V$, and $(x : y : z) = (Z : X : X + Y + Z)$. On $z = 1$, homogeneous representatives give

$$V = \frac{1 + x + y}{x}, \quad U = \frac{y(1 + x + y)}{x^2}.$$

Since $y(1 + x + y) = xy + x^3 + x^2 = x(x^2 + x + y)$, this proves the formula for $\mathfrak{c} = 1/U$. With $d = x^2 + x + y$, direct intersection gives

$$\text{div}(x) = [T] + [-T] - 2[O], \quad \text{div}(d) = [T] + 3[-2T] - 4[O].$$

Indeed $d = 0$ gives $y = x^2 + x$, whence the curve equation gives $x(x + 1)^3 = 0$, and d has pole order four at O . Subtraction proves (EC_1) .

The function $v = (y + 1)/x$ has pole divisor $[O] + [T]$, so the deck involution of its degree-two map is $J(P) = T - P$. Since $\mathcal{A}^2$ fixes v and is nontrivial, it equals J . Also $\mathcal{A}^2$ sends u to u^{-1} , and therefore

$$\mathfrak{c}(\mathcal{A}^2 P) = \frac{1}{u(P)^{-1} + 1} = \frac{u(P)}{u(P) + 1} = \mathfrak{c}(P) + 1,$$

proving (EC_2) .

The Frobenius polynomial of $\mathcal{E}/\mathbb{F}_2$ is $\pi^2 + 2\pi + 2 = 0$. Thus $i = \pi + 1$ satisfies $i^2 = -1$, and it acts on $\mathcal{E}(\mathbb{F}_2)$ as multiplication by two. Write $\mathcal{A} = \tau_S \circ \alpha$, with $\alpha(O) = O$. As $\mathcal{A}$ is defined over $\mathbb{F}_2$, α commutes with π . The centralizer of π in the supersingular endomorphism algebra is $\mathbb{Q}(i)$, whose integral units are $\{\pm 1, \pm i\}$. The linear part of $\mathcal{A}^2$ is $[-1]$, so $\alpha = \pm i$. Comparing translation parts gives

$$\mathcal{A} = \tau_{2T} \circ i \quad \text{or} \quad \mathcal{A} = \tau_{-T} \circ (-i);$$

these are inverse and both fix $R_0 = -2T$.

For $n \geq 4$, 2^{n-1} is divisible by eight. Since $\pi^8 = [16]$,

$$\pi^{2^{n-1}} = [2^{2^{n-2}}] = [R].$$

Translation of R_0 to the origin makes $\mathcal{A}^{\pm 1}$ the linear maps $\pm i$, proving (EC₃). The separable endomorphisms $[R] \mp i$ have degree $R^2 + 1$. For every odd prime p , the Tate module $T_p(\mathcal{E})$ is free of rank one over $\mathbb{Z}_p[i]$ (the polynomial $X^2 + 1$ is separable away from 2). Hence the p -primary kernel is the additive quotient $\mathbb{Z}_p[i]/(R \mp i)$. The matrix of $R \mp i$ over $\mathbb{Z}_p$ has entries with gcd one and determinant $R^2 + 1$, so its first Smith invariant is one. Combining the odd primary parts shows that the geometric kernel is cyclic of order $R^2 + 1$.

Every prime $\ell \mid Q^2 + 1$ is coprime to 15. Thus the valuations 1, 2, -3 in (EC₁) are nonzero modulo ℓ . Kummer theory makes $z^\ell = \mathfrak{c}$ geometrically connected and totally ramified at exactly those three points. Riemann–Hurwitz gives

$$2g(\mathcal{C}_\ell) - 2 = 3(\ell - 1),$$

which proves (EC₄).

It remains to prove non-lifting. Choose $\mathcal{A} = \tau_{2T} \circ i$, put $B_j = \mathcal{A}^j(O)$, and retain $R_0 = -2T$. Then

$$(B_0, B_1, B_2, B_3) = (O, 2T, T, -T).$$

For $D_j = \text{div}(\mathfrak{c} \circ \mathcal{A}^{-j})$, pullback gives

$$D_j = 2[B_j] + [B_{j+3}] - 3[R_0]. \quad (\text{EC}_6)$$

If $\sum_{j=0}^3 r_j D_j \equiv 0 \pmod{\ell}$, comparison at the four B_j 's gives $2r_j + r_{j+1} = 0$ cyclically. Iteration gives $15r_0 = 0$, and $\ell \nmid 15$ forces every $r_j = 0$. Hence

$$\mathfrak{c}, \quad \mathfrak{c} \circ \mathcal{A}, \quad \mathfrak{c} \circ \mathcal{A}^2, \quad \mathfrak{c} \circ \mathcal{A}^3$$

are linearly independent in

$$\overline{\mathbb{F}}_2(\mathcal{E})^\times / \overline{\mathbb{F}}_2(\mathcal{E})^{\times \ell}.$$

A lift of $\mathcal{A}$ would normalize the cyclic deck group, hence send z to gz^r for $r \in (\mathbb{Z}/\ell\mathbb{Z})^\times$, and would require $\mathfrak{c} \circ \mathcal{A} = g^\ell \mathfrak{c}^r$, contradicting this independence.

Finally, set

$$\chi(t) = t^{(Q^4-1)/\ell}, \quad \tau = \chi(\mathfrak{c}(P_n)).$$

Since $\text{Frob}_Q^2(P_n) = \mathcal{A}^2(P_n)$, (EC₂) gives $\mathfrak{c}(P_n)^{Q^2} = \mathfrak{c}(P_n) + 1$. Because $Q^2 \equiv -1 \pmod{\ell}$, the four CM/Frobenius character values are only

$$\tau, \quad \tau^{\pm Q}, \quad \tau^{-1}, \quad \tau^{\mp Q}.$$

All norm identities among them are tautological. This proves neither $\tau = 1$ nor $\tau \neq 1$: the desired assertion remains (EC₅). $\square$

Proposition 7.21 (Miller-unit and generalized-Jacobian location). *Retain Proposition 7.20, put $S = -2T$, $q = Q^2$, and $K_n = \mathbb{F}_{Q^4} = \mathbb{F}_{q^2}$. Then*

$$\mathfrak{c}^{-1} = f_{3,S}$$

up to a nonzero scalar, where $f_{3,S}$ is the Miller tripling function with divisor

$$\text{div}(f_{3,S}) = 3[S] - [3S] - 2[O]. \quad (\text{GJ}_1)$$

For every prime $\ell \mid q + 1$,

$$\#\mathcal{E}(K_n) = (q-1)^2, \quad \mathcal{E}(K_n)[\ell] = 0, \quad [\ell] : \mathcal{E}(K_n) \xrightarrow{\sim} \mathcal{E}(K_n). \quad (\text{GJ}_2)$$

Let $B = \{-T, O, -2T\}$ and $U = \mathcal{E} \setminus B$. The Kummer class of $\mathfrak{c}$ on U has nonzero tame residue vector

$$(1, 2, -3) \pmod{\ell}. \quad (\text{GJ}_3)$$

It is therefore ramified and does not come from an unramified μ_ℓ -class on $\mathcal{E}$. In the generalized Jacobian

$$1 \longrightarrow \mathbb{G}_m^2 \longrightarrow J_B \longrightarrow \mathcal{E} \longrightarrow 0, \quad (\text{GJ}_4)$$

Hilbert 90 and (GJ_2) give

$$J_B(K_n)/\ell J_B(K_n) \cong (K_n^\times/K_n^{\times\ell})^2. \quad (\text{GJ}_5)$$

Thus the residue vector has a genuinely ramified toric projection; the ordinary K_n -rational elliptic quotient cannot detect that projection. Ordinary K_n -rational Tate–Lichtenbaum and formal-group data cannot by themselves decide (EC_5) . This does not assert that the ramified class evaluates nontrivially at the distinguished point, nor does it exclude a calculation using non-rational $\mathcal{E}[\ell]$ or the full extension structure of J_B . An elliptic completion must evaluate that generalized-Jacobian Miller-unit residue.

Proof. The point T has order five, so $3S = 3(-2T) = -T$. Equation (EC_1) therefore gives

$$\text{div}(\mathfrak{c}^{-1}) = 3[S] - [3S] - 2[O],$$

which proves (GJ_1) .

For $n \geq 4$, the Frobenius relation used in Proposition 7.20 gives

$$\pi^{2^{n+1}} = [q].$$

This is Frobenius on $K_n = \mathbb{F}_{2^{2^{n+1}}}$, so

$$\#\mathcal{E}(K_n) = \deg([q] - [1]) = (q-1)^2.$$

As $\ell \mid q+1$ is odd, it is coprime to $q-1$. This proves (GJ_2) ; multiplication by ℓ is likewise an automorphism of the characteristic-two formal group because its linear coefficient is a unit.

The divisor (EC_1) gives the residue vector (GJ_3) , which is nonzero because $\ell \nmid 6$. The standard generalized-Jacobian sequence for the three-point reduced rational modulus is (GJ_4) . Its torus is split; Hilbert 90 makes the map on K_n -points surjective, and the kernel created on passing modulo ℓ vanishes because $\mathcal{E}(K_n)[\ell] = 0$. This proves (GJ_5) and the stated scope of the toric obstruction. $\square$

Proposition 7.22 (exact extension character and cohomological exhaustion). *Retain Proposition 7.21 and put*

$$D_1 = [-T] - [-2T], \quad D_2 = [O] - [-2T]$$

in $\text{Div}^0(B)$. The character lattice of the split torus in J_B is

$$X^*(\mathbb{G}_m^2) = \text{Div}^0(B) = \mathbb{Z}D_1 \oplus \mathbb{Z}D_2,$$

and the extension is classified, under $\mathcal{E} \cong \text{Pic}^0(\mathcal{E})$, by

$$\lambda(aD_1 + bD_2) = (a + 2b)T. \quad (\text{GJ}_6)$$

Thus its entire extension class is rational five-torsion. Moreover

$$r := \text{div}(\mathbf{c}) = D_1 + 2D_2, \quad \lambda(r) = 5T = O,$$

so r defines a character $\varphi_r : J_B \rightarrow \mathbb{G}_m$. If $j_T : U \rightarrow J_B$ is the generalized Abel map normalized by $j_T(T) = 0$, then

$$\varphi_r(j_T(P)) = \mathbf{c}(P). \quad (\text{GJ}_7)$$

For every prime $\ell \mid q+1$, even the non-rational geometric $\mathcal{E}[\ell]$ supplies no further rational Kummer class:

$$H^1(K_n, \mu_\ell^2) \xrightarrow{\sim} H^1(K_n, J_B[\ell]), \quad J_B(K_n)/\ell J_B(K_n) \cong (K_n^\times/K_n^{\times\ell})^2. \quad (\text{GJ}_8)$$

In torus coordinates dual to (D_1, D_2) , the character is $(x, y) \mapsto xy^2$, hence the nonzero linear functional $(x, y) \mapsto x + 2y$ modulo ℓ . Naturality of Kummer theory gives

$$\varphi_{r,*}(\delta_J(j_T(P_n))) = P_n^*[\mathbf{c}] = [\mathbf{c}(P_n)] \in K_n^\times/K_n^{\times\ell}. \quad (\text{GJ}_9)$$

Consequently (EC₅) is exactly the assertion that the normalized Abel class of P_n avoids the hyperplane $x + 2y = 0$. Torus translation above any fixed elliptic point realizes every value of this functional. Hence the CM point, the full extension class, and non-rational elliptic ℓ -torsion do not force (EC₅); the remaining datum is the explicit toric coordinate of this one distinguished normalized Abel lift.

Proof. For a reduced three-point rational modulus, the generalized-Jacobian torus is $(\mathbb{G}_m^3)/\mathbb{G}_m$, with character lattice $\text{Div}^0(B)$. The standard extension-class map sends a boundary divisor to its class in $\text{Pic}^0(\mathcal{E})$. Here

$$(-T) - (-2T) = T, \quad O - (-2T) = 2T,$$

which proves (GJ₆). Equation (EC₁) gives $r = D_1 + 2D_2$, and its image is $T + 4T = O$. Characters of J_B are precisely the torus characters whose pushout extension class vanishes, so r yields φ_r . The generalized-Albanese universal property gives (GJ₇) after normalization at T . Indeed, $T \notin B$ and $\mathbf{c}(T) \in \mathbb{F}_2^\times = \{1\}$.

The K_n -Frobenius on $\mathcal{E}$ is $[q]$, and $q \equiv -1 \pmod{\ell}$. It therefore acts as $[-1]$ on $\mathcal{E}[\ell]$. Since ℓ is odd,

$$\mathcal{E}[\ell](K_n) = 0, \quad H^1(K_n, \mathcal{E}[\ell]) = \mathcal{E}[\ell]/(\text{Frob} - 1)\mathcal{E}[\ell] = 0.$$

The cohomology sequence of

$$0 \longrightarrow \mu_\ell^2 \longrightarrow J_B[\ell] \longrightarrow \mathcal{E}[\ell] \longrightarrow 0$$

then proves the first isomorphism in (GJ₈). Lang's theorem and Kummer theory prove the second. The natural Kummer square for φ_r , together with (GJ₇), gives (GJ₉). Finally $\mathbb{G}_m^2(K_n)/\ell \rightarrow J_B(K_n)/\ell$ is an isomorphism: surjectivity uses bijectivity of $[\ell]$ on $\mathcal{E}(K_n)$, and injectivity uses $\mathcal{E}(K_n)[\ell] = 0$. The character $(x, y) \mapsto xy^2$ is surjective on the ℓ -quotient, proving the fixed-fibre statement and the asserted exact boundary. $\square$

Proposition 7.23 (explicit Rosenlicht coordinate and its reciprocity boundary). *Retain Proposition 7.22, write $S = -2T$, $P = P_n$, and put $X = P - S$. For $\ell \mid q+1$, let $Z \in \mathcal{E}(K_n)$ be the unique point with*

$$[\ell]Z = P - T, \quad W = T + Z. \quad (\text{RC}_1)$$

If $e\ell \equiv 1 \pmod{5(Q+1)}$, then

$$Z = [e]X + [2e]T, \quad W = [e]X + [2e+1]T, \quad (\text{RC}_2)$$

so the division point remains an explicit translate of the same cyclic CM kernel as X . Choose a Miller function $F = F_{\ell,Z}$ with

$$\text{div}(F) = \ell[Z] - [[\ell]Z] - (\ell-1)[O].$$

In the Rosenlicht coordinates dual to $D_1 = [-T] - [S]$, $D_2 = [O] - [S]$, the normalized Abel Kummer class of P is

$$\rho_1 = \frac{F(2T)}{F(S)}, \quad \rho_2 = \frac{F(2T)}{F(-T)}, \quad (\text{RC}_3)$$

and its Miller-unit character is exactly

$$[\mathfrak{c}(P)] = [\rho_1 \rho_2^2] \quad \text{in } K_n^\times / K_n^{\times \ell}. \quad (\text{RC}_4)$$

There is also a formula avoiding the ℓ -division point. On $\mathcal{E} : y^2 + y = x^3 + x^2$, set

$$A_5 = 1 + x + x^2 + y + xy, \quad B_5 = 1 + x + xy, \quad g_1 = \frac{A_5}{B_5}, \quad g_2 = \frac{1}{B_5}.$$

Then

$$\text{div}(g_1) = 5D_1, \quad \text{div}(g_2) = 5D_2, \quad \mathfrak{c}^5 = g_1 g_2^2. \quad (\text{RC}_5)$$

If $5\bar{5} \equiv 1 \pmod{\ell}$, the full normalized Abel class in $J_B(K_n)/\ell J_B(K_n)$ has coordinates

$$([g_1(P)]^{\bar{5}}, [g_2(P)]^{\bar{5}}). \quad (\text{RC}_6)$$

More explicitly, for the consecutive Conway variables $u = u_n$, $v = u_{n-1}$, put

$$U = u + 1, \quad V = v + 1, \quad D = V^2 + V + U.$$

Then the normalized point has $x = V/D$, $y = U/D$, and

$$g_1(P) = \frac{D^2 + D(U+V) + V^2 + UV}{D^2 + VD + UV}, \quad g_2(P) = \frac{D^2}{D^2 + VD + UV}, \quad (\text{RC}_7)$$

with $g_1(P)g_2(P)^2 = U^{-5}$.

These formulas compute the missing toric coordinate exactly, but do not force it to be nonzero modulo ℓ . Indeed, Weil reciprocity gives

$$\mathfrak{c}(P) = \mathfrak{c}(W)^\ell \frac{F(2T)^3}{F(S)F(-T)^2}. \quad (\text{RC}_8)$$

Applying reciprocity instead to $G(Y) = \mathfrak{c}(Y+T)$, whose divisor is $[S] + 2[-T] - 3[2T]$, returns precisely the inverse identity

$$\frac{F(S)F(-T)^2}{F(2T)^3} = \frac{\mathfrak{c}(W)^\ell}{\mathfrak{c}(P)}. \quad (\text{RC}_9)$$

Thus the boundary Miller values and the rational five-torsion splitting exhaust the Rosenlicht coordinate, but collapsing them by the evident reciprocity law is exactly equivalent to the original Kummer value $[\mathfrak{c}(P)]$; it supplies no independent nonresidue constraint.

Proof. Since $P = X + 3T$, one has $P - T = X + 2T$. Multiplication by ℓ is invertible both on the cyclic kernel containing X and on $\langle T \rangle$, which proves (RC₂).

Put $h(Y) = F(Y - T)$. Translation of the displayed divisor of F gives

$$\operatorname{div}(h) = \ell[W] - [P] - (\ell - 1)[T].$$

The Rosenlicht evaluation of this principal relation against D_1, D_2 is

$$\frac{h(S)}{h(-T)} = \frac{F(2T)}{F(S)}, \quad \frac{h(S)}{h(O)} = \frac{F(2T)}{F(-T)},$$

which proves (RC₃). Weil reciprocity between h and $\mathfrak{c}$, using

$$\operatorname{div}(\mathfrak{c}) = [-T] + 2[O] - 3[S], \quad \mathfrak{c}(T) = 1,$$

gives (RC₈), and hence (RC₄).

Direct intersection with the Weierstrass equation, including the local orders at the indicated five-torsion points, gives

$$\operatorname{div}(A_5) = 5[-T] - 5[O], \quad \operatorname{div}(B_5) = 5[S] - 5[O].$$

Thus g_1, g_2 have the divisors asserted in (RC₅). Both sides of $\mathfrak{c}^5 = g_1 g_2^2$ have the same divisor and value one at T , proving the exact identity. Since five is invertible modulo ℓ , this proves (RC₆). Substitution of $x = V/D$, $y = U/D$ gives (RC₇); here $\mathfrak{c}(P) = 1/U$, so its last identity is also immediate from (RC₅).

Finally, the divisor of G follows by translating that of $\mathfrak{c}$. Weil reciprocity between F and G is (RC₉), proving the exact neutrality statement. $\square$

Proposition 7.24 (five-point equivariant generalized-Jacobian exhaustion). *Retain the Conway point $P = P_n$, the quarter-turn $\mathcal{A}$, and $S = -2T$. Put*

$$B_0 = O, \quad B_1 = 2T, \quad B_2 = T, \quad B_3 = -T, \quad D = \mathcal{E}(\mathbb{F}_2) = \{S, B_0, B_1, B_2, B_3\}.$$

For $j \in \mathbb{Z}/4\mathbb{Z}$, set

$$e_j = [B_j] - [S], \quad f_j = \mathfrak{c} \circ \mathcal{A}^{-j}, \quad r_j = \operatorname{div}(f_j) = 2e_j + e_{j+3}. \quad (\text{FE}_1)$$

Then D is $\mathcal{A}$ -invariant, and the four classes r_j form a basis of $\operatorname{Div}^0(D) \otimes \mathbb{F}_\ell$ for every prime $\ell \mid F_n$, $n \geq 2$.

Let J_D be the generalized Jacobian of the five-point reduced modulus, let $K_n = \mathbb{F}_{Q^4}$, where $Q = 2^{2^{n-1}}$, and choose any $R \in \mathcal{E}(\mathbb{F}_{16}) \setminus D$. If j_R denotes the generalized Abel map normalized at R , then

$$\mathfrak{c}(P) \in K_n^{\times \ell} \iff [j_R(P)] = 0 \text{ in } J_D(K_n)/\ell J_D(K_n). \quad (\text{FE}_2)$$

Thus even the simultaneous Rosenlicht data of the complete quarter-turn orbit contain no obstruction beyond the original Kummer value.

For $n \geq 3$, put $z = [j_R(P)]$ and let $\sigma = \operatorname{Frob}_Q$. If $\sigma(P) = \mathcal{A}^\epsilon(P)$, with $\epsilon \in \{1, -1\}$, then

$$Qz = \mathcal{A}_*^\epsilon z. \quad (\text{FE}_3)$$

Inside the split rank-four torus quotient this eigenspace is one-dimensional over $\mathbb{F}_\ell$, and evaluation by φ_{r_0} is an isomorphism from it to $K_n^\times/K_n^{\times\ell}$. Under this isomorphism z maps to $[\mathfrak{c}(P)]$. Consequently the full $\mathcal{A}$ -invariant generalized Jacobian, its rational five-torsion extension class, and the exact Frobenius quarter-turn relation determine precisely one scalar—the scalar in (EC₅)—but do not determine whether that scalar vanishes.

Proof. The automorphism $\mathcal{A}$ fixes S and cycles the other four $\mathbb{F}_2$ -points as

$$O \mapsto 2T \mapsto T \mapsto -T \mapsto O.$$

Equation (EC₆) gives (FE₁). In the basis (e_0, e_1, e_2, e_3) , the four rows r_j form the circulant matrix $2I + C$, where C is a four-cycle. Hence

$$\det(2I + C) = 2^4 - 1 = 15. \quad (\text{FE}_4)$$

No prime divisor of F_n , $n \geq 2$, divides 15, proving the basis assertion. Notice also that the union of the supports of the four f_j is exactly D .

The Frobenius polynomial of $\mathcal{E}/\mathbb{F}_2$ gives $\#\mathcal{E}(\mathbb{F}_{16}) = 25$, so a point $R \in \mathcal{E}(\mathbb{F}_{16}) \setminus D$ exists. Every $f_j(R)$ is a nonzero element of $\mathbb{F}_{16}$; since $\gcd(\ell, 15) = 1$, it is an ℓ -th power already in $\mathbb{F}_{16}$. The five-point generalized Jacobian fits into

$$1 \longrightarrow \mathbb{G}_m^4 \longrightarrow J_D \longrightarrow \mathcal{E} \longrightarrow 0.$$

As in (GJ₂), $\#\mathcal{E}(K_n) = (Q^2 - 1)^2$, so multiplication by ℓ is bijective on $\mathcal{E}(K_n)$. Hilbert 90 and the absence of rational ℓ -torsion therefore give

$$J_D(K_n)/\ell J_D(K_n) \simeq (K_n^\times/K_n^{\times\ell})^4. \quad (\text{FE}_5)$$

For $r_j = \text{div}(f_j)$, let $\varphi_{r_j} : J_D \rightarrow \mathbb{G}_m$ be the corresponding principal boundary character. Generalized Abel evaluation is

$$\varphi_{r_j}(j_R(P)) = \frac{f_j(P)}{f_j(R)} \quad \text{in } K_n^\times/K_n^{\times\ell}. \quad (\text{FE}_6)$$

Writing $\sigma = \text{Frob}_Q$, if $\mathfrak{c}(P)$ is an ℓ -th power, then $\sigma(P) = \mathcal{A}^{\pm 1}(P)$ and Frobenius stability of $K_n^{\times\ell}$ make every $f_j(P)$ an ℓ -th power. Equations (FE₄)–(FE₆) then give $j_R(P) = 0$ modulo ℓ . The converse follows by evaluating φ_{r_0} , since $f_0 = \mathfrak{c}$. This proves (FE₂).

Suppose now that $n \geq 3$. Then σ fixes $\mathbb{F}_{16}$, and hence R . Moreover

$$\#J_D(\mathbb{F}_{16}) = 15^4 \cdot 25$$

is prime to ℓ . Therefore the affine base-point correction $[j_R(\mathcal{A}^\epsilon R)]$ vanishes modulo ℓ , and

$$\sigma j_R(P) = j_R(\sigma P) = \mathcal{A}_*^\epsilon j_R(P) + j_R(\mathcal{A}^\epsilon R)$$

reduces to (FE₃). On the right side of (FE₅), σ acts on the scalar Kummer quotient as multiplication by Q , while $\mathcal{A}_*^\epsilon$ acts on the rank-four cocharacter lattice as a four-cycle. Since $Q^2 \equiv -1 \pmod{\ell}$, Q has exact order four modulo ℓ . The Q -eigenspace of that four-cycle is consequently one-dimensional.

Finally, if φ_{r_0} vanished on a vector in this eigenspace, its four $\mathcal{A}^*$ -translates r_j would all vanish there. The basis assertion would force the vector to be zero. Thus φ_{r_0} is nonzero, and hence an isomorphism, on the one-dimensional eigenspace. Its value is $[f_0(P)/f_0(R)] = [\mathfrak{c}(P)]$, completing the proof. $\square$

When F_n is prime, irreducibility and $\delta_n > 1$ force $\delta_n = F_n$, recovering Lenstra’s Fermat-prime rule. When F_n is composite, irreducibility forces only one nontrivial divisor: it does not force every prime-power factor. The condition $\delta_n = F_n$ is exactly the maximal-coset-order hypothesis studied for Conway’s quadratic tower by Popovych [21]. It is strictly weaker than primitivity of c_n , which already fails at $n = 2$ since $\text{ord}(c_2) = 5 \cdot 17 = 85 < 255$; Popovych uses the all-level hypothesis instead to prove that products such as $c_0 c_n$ are primitive. Wiedemann’s related iterated quadratic tower poses the analogous full-Fermat-order problem [23]. Thus the apparently easy “singleton even implies zero” clause already contains a longstanding explicit maximal-coset-order problem. Popovych verified the Conway condition only through $n = 11$; the recent analysis of Cagliero–Herman–Szechtman still retains the quotient order as an undetermined divisor [5]. No all-level proof or counterexample is presently known.

7.5 Comparison with the cubic arm

These two embedded arms are instances of the same projective statement. The quadratic arm asks that the class of c_n generate $E_n^\times / E_{n-1}^\times$. In the cubic arm the natural map

$$U_k \longrightarrow \mathbb{F}_{q_k^3}^\times / \mathbb{F}_{q_k}^\times$$

is an isomorphism, and the increment from C_{k-1} to C_k asks that the class of γ_k generate this quotient. Equivalently, the projective class of the multiplication-by- γ_k matrix generates a Singer cycle in $\text{PGL}_3(q_k)$. The 0/1/4 rule therefore contains a quadratic and a cubic maximal-projective-order conjecture among its four exact arms, rather than two unrelated order coincidences.

7.6 Open problem Z

The singleton-even part of the zero arm is exactly the assertion $\delta_n = F_n$ for all n . The results above identify the recursively selected fibotomic factor and exhaust the trace, norm, elliptic, and generalized-Jacobian data currently available without evaluating its Kummer class. Thus the remaining theorem is not another coarse invariant: it is the nonvanishing of the explicit distinguished Miller-unit value in Proposition 7.24.

8 The exceptional $2 \cdot 3^k$ arm

8.1 Why component data do not decide the excess

$\mathcal{Q}(f(p))$ alone does not determine m_p :

$\mathcal{Q}$	$m = 4$	$m = 1$
$\{9\}$	$p = 19, f = 18$	$p = 73, f = 9$
$\{81\}$	$p = 163, f = 162$	$p = 2593, f = 81$
$\{243\}$	$p = 1459, f = 486$	$p = 487, f = 243$

The split for $\mathcal{Q} = \{9\}$ is explained by the order criterion: $\text{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$ and $511 = 7 \cdot 73$, so $73 \mid \text{ord}(\kappa_9 + 1)$ but $19 \nmid \text{ord}(\kappa_9 + 1)$; adding $m = 4$ changes the order and introduces the factor 19.

The lower bound of the exception column is unconditional: $m_p \geq 4$ for all p with $f(p) = 2 \cdot 3^k$ (Proposition 6.5).

[conjectured] The matching upper bound $m_p \leq 4$ on that column; its exact current-level formulation is D'_k in Conjecture 8.4, not the Gaussian-period conjecture C_k .

8.2 The corrected norm and current-factor target

The $m = 4$ translate $\kappa_{3^k} + 4$ involves the number $4 \in \mathbb{F}_{16}$, which does not lie in $\mathbb{F}_2(\zeta)$ (as $4 \nmid 2 \cdot 3^k$); the translate therefore lies in the degree-4 $\cdot 3^k$ compositum $\mathbb{F}_{2^{4h}}$, where $h = 3^k$.

Proposition 8.1 (the corrected norm). *Let $\sigma = \text{Frob}^{2^h}$ generate $\text{Gal}(\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}})$. Then $\sigma(4) = 5$, and with $\omega := \zeta^h = \kappa_2$,*

$$N := N_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega.$$

Proof. $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$ because $\gcd(4, 2 \cdot 3^k) = 2$, so $\sigma|_{\mathbb{F}_{16}}$ is the nontrivial element of $\text{Gal}(\mathbb{F}_{16}/\mathbb{F}_4)$, which swaps the two roots 4, 5 of the Artin–Schreier minimal polynomial $y^2 + y + \omega$ of 4 over $\mathbb{F}_4$. (Equivalently: $2h \equiv 2 \pmod{4}$ and the Frobenius orbit of 4 is $4 \rightarrow 6 \rightarrow 5 \rightarrow 7$, so $\sigma(4) = 4^{2^2} = 5$.) Then $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + (4 \otimes 5)$ and $4 \otimes 5 = 4^2 + 4 = \omega$ by the same Artin–Schreier relation. $\square$

The corrected norm collapses the $m = 4$ test into the *same* sparse trinomial model $\mathbb{F}_2[x]/(x^{2^h} + x^h + 1)$ as the rest of the 3-power analysis: no compositum arithmetic is needed.

Corollary 8.2 (in-field criterion). *For a prime p with $f(p) = 2 \cdot 3^k$ and $v_p(2^{2^h} - 1) = 1$, write $\bar{N} = N^{2^h}$ and $M_k := \bar{N}/N = N^{2^h-1} \in U$. Then*

$$m_p = 4 \iff p \mid \text{ord}(M_k) \iff M_k^{(2^h+1)/p} \neq 1.$$

Proof. $m_p \geq 4$ is Proposition 6.5, so $m_p = 4$ iff $\kappa + 4$ has no p -th root. By Propositions 3.1 and 3.4 with $E = 4h$, $f = 2h$ (and $v_p(2^{4h} - 1) = v_p(2^{2h} - 1) = 1$ by lifting the exponent, since $p \nmid 2$), that holds iff $N^{(2^h-1)/p} \neq 1$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders, $p \mid 2^h + 1$, and $(2^{2h} - 1)/p = (2^h - 1) \cdot ((2^h + 1)/p)$, the test equals $M_k^{(2^h+1)/p} \neq 1$. $\square$

Proposition 8.3 (exact current-level form of the exception conjecture). *Let $k \geq 1$. Retain the notation above, put $q = 2^h$, $n = 3^{k+1} = 3h$, and write*

$$\zeta := \kappa_{3^k} \in K := \mathbb{F}_{2^{2h}}, \quad \tilde{K} := \mathbb{F}_{2^{4h}}, \quad \beta := \zeta + 4.$$

Thus ζ is a primitive n -th root of unity. Define

$$N := N_{\tilde{K}/K}(\beta) = \zeta^2 + \zeta + \zeta^h, \quad M_k := N^{q-1}, \quad \Psi_k := \frac{\Phi_{2 \cdot 3^k}(2)}{3},$$

and

$$R_k(Y) := \text{Res}_X \left(X^{2h} + X^h + 1, Y + X^h + X^2 + X \right) \in \mathbb{F}_2[Y].$$

Then R_k is the irreducible minimal polynomial of N over $\mathbb{F}_2$ and has degree $2h$. The prime divisors of Ψ_k are exactly the primes $\ell \neq 3$ with $\text{ord}_\ell(2) = 2 \cdot 3^k$, with full valuations

$$v_\ell(\Psi_k) = v_\ell(2^{2h} - 1) = v_\ell(2^h + 1).$$

Moreover the following are equivalent:

- (a) $m_\ell = 4$ for every prime ℓ with $f(\ell) = 2 \cdot 3^k$;
- (b) $\Psi_k \mid \text{ord}(M_k)$;
- (c) $N \notin (K^\times)^\ell$ for every prime $\ell \mid \Psi_k$;

(d) $R_k(X^\ell)$ is irreducible over $\mathbb{F}_2$ for every prime $\ell \mid \Psi_k$.

Proof. The order identity $\text{ord}_{3^{k+1}}(2) = 2h$ makes $X^{2h} + X^h + 1$ the irreducible minimal polynomial of ζ . We first prove that $N = \zeta^h + \zeta^2 + \zeta$ also has degree $2h$. Suppose $N^{2^j} = N$, and put $a \equiv 2^j \pmod{3h}$. The resulting vanishing polynomial, of degree less than $3h$, has support equal to the symmetric difference of

$$\{ah, a, 2a\} \quad \text{and} \quad \{h, 1, 2\}. \quad (1)$$

Every multiple of $X^{2h} + X^h + 1$ of degree less than $3h$ has support a disjoint union of triples $\{r, r+h, r+2h\}$. If $a \equiv 2 \pmod{3}$, the support in (1) contains h and $2h$ but not 0, a contradiction. If $a \equiv 1 \pmod{3}$, the h -terms cancel and each residue class modulo h occurs at most twice; a nonempty union of such triples is again impossible. Hence the support is empty, so $\{a, 2a\} = \{1, 2\}$ modulo $3h$. Since $h \geq 3$, this forces $a = 1$. Thus the Frobenius orbit of N has length $2h$, and the displayed resultant is its irreducible minimal polynomial.

Writing $A = 2^{3^{k-1}}$ gives $\Phi_{2 \cdot 3^k}(2) = A^2 - A + 1 \equiv 3 \pmod{9}$, so division by 3 removes exactly the prime dividing the cyclotomic index. The standard cyclotomic-divisor criterion now says that the prime divisors of Ψ_k are precisely the ℓ with $\text{ord}_\ell(2) = 2h$. Their entire primary contribution to $2^{2h} - 1$ lies in this cyclotomic factor and, since $\ell \nmid 2^h - 1$, in $2^h + 1$. This proves the valuation statement.

Fix $\ell \mid \Psi_k$. Propositions 6.4 and 6.5 make the translates $\zeta, \zeta + 1, \zeta + 2, \zeta + 3$ into ℓ -th powers, so $m_\ell \geq 4$. The corrected norm and the exact power criterion give

$$\beta^{(2^{4h}-1)/\ell} = N^{(2^{2h}-1)/\ell},$$

and therefore

$$m_\ell = 4 \iff N \notin (K^\times)^\ell. \quad (2)$$

This uses the full ℓ -primary valuation and has no squarefreeness or non-Wieferich hypothesis. Since $q - 1 \equiv -2 \pmod{\ell}$, raising to $q - 1$ is invertible on $K^\times / (K^\times)^\ell$, so

$$N \in (K^\times)^\ell \iff M_k \in (K^\times)^\ell. \quad (3)$$

The entire ℓ -primary part of $K^\times$ lies in its norm-one factor of order $q + 1$. In a cyclic group, nonmembership in the subgroup of ℓ -th powers is equivalent to the order containing the full ℓ -primary part. Equations (2)–(3) prove the equivalence of the first three assertions.

Finally $\mu_\ell \subset K$. Kummer theory says that $X^\ell - N$ is irreducible over K exactly when $N \notin (K^\times)^\ell$. Since $K = \mathbb{F}_2(N)$, Capelli's lemma says in turn that this is equivalent to irreducibility of $R_k(X^\ell)$ over $\mathbb{F}_2$, completing the proof. $\square$

Conjecture 8.4 (D'_k : the exact current exception factor). *[conjectured] One has*

$$D'_k : \quad \Psi_k \mid \text{ord}(M_k).$$

By Proposition 8.3, this is exactly the $m = 4$ upper bound for the primes at level $f = 2 \cdot 3^k$.

8.3 Projection, fibotomic, and translated-binomial forms

Proposition 8.5 (exact current-layer projection and dominant-factor sieve). *Let $k \geq 2$, put*

$$s = 3^{k-1}, \quad h = 3s, \quad a = 2^s, \quad q = a^3 = 2^h, \quad \Lambda_k = a^2 - a + 1 = 3\Psi_k,$$

and retain $N, M_k, \zeta, \omega = \zeta^h$ from Proposition 8.3. Define

$$P_k := M_k^{a+1}.$$

Then $P_k^{\Lambda_k} = 1$, and for every prime $\ell \mid \Psi_k$,

$$v_\ell(\text{ord } P_k) = v_\ell(\text{ord } M_k). \quad (\text{D}_1)$$

Moreover

$$P_k = \frac{\zeta^4 + \zeta + \omega}{\omega^2 \zeta^4 + \zeta^3 + 1}. \quad (\text{D}_2)$$

$$[\mathbb{F}_2(P_k) : \mathbb{F}_2] = 2h. \quad (\text{D}_3)$$

and

$$v_3(\text{ord } P_k) = \begin{cases} 1, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases} \quad (\text{D}_4)$$

Set $c_k = 3$ for odd k and $c_k = 1$ for even k . Then

$$\text{ord}(P_k) \geq c_k(2h + 1). \quad (\text{D}_5)$$

Consequently, if $\ell \mid \Psi_k$ and

$$\ell > \frac{\Lambda_k}{c_k(2h + 1)}, \quad (\text{D}_6)$$

then P_k , and hence M_k , contains the full ℓ -primary part of Λ_k in its order. In particular D'_k follows whenever Ψ_k is prime; this proves D'_2 and D'_3 analytically, since $\Psi_2 = 19$ and $\Psi_3 = 87211$.

Proof. The factorization

$$q + 1 = (a + 1)(a^2 - a + 1) = (a + 1)\Lambda_k$$

and $M_k^{q+1} = 1$ give $P_k^{\Lambda_k} = 1$. Also $\gcd(\Lambda_k, a + 1) = 3$. Since $k \geq 2$, one has $a \equiv -1 \pmod{9}$, so $\Lambda_k \equiv 3 \pmod{9}$ and $\gcd(\Psi_k, a + 1) = 1$. Exponentiation by $a + 1$ therefore preserves all the primary valuations belonging to Ψ_k , proving (D₁).

The lifted congruence

$$2^{3^{k-1}} \equiv 3^k - 1 \pmod{3^{k+1}}$$

gives, modulo $3h$,

$$a \equiv h - 1, \quad a^2 \equiv h + 1, \quad a^3 \equiv -1, \quad a^4 \equiv 2h + 1. \quad (\text{D}_7)$$

As $M_k = N^{a^3-1}$,

$$P_k = \frac{N^{a^4} N^{a^3}}{N^a N}.$$

Using $N = \zeta^2 + \zeta + \omega$ and (D₇), direct Frobenius expansion gives

$$N^a = \omega^2 \zeta^{-2} + \omega \zeta^{-1} + \omega^2,$$

$$N^{a^3} = \zeta^{-2} + \zeta^{-1} + \omega^2,$$

$$N^{a^4} = \omega \zeta^2 + \omega^2 \zeta + \omega.$$

Multiplying N^{a^3} and N^a by the same cancelling factor ζ^2 , and using $1 + \omega + \omega^2 = 0$, yields

$$N^{a^4}(\zeta^2 N^{a^3}) = \zeta^4 + \zeta + \omega,$$

$$(\zeta^2 N^a)N = \omega^2 \zeta^4 + \zeta^3 + 1,$$

which proves (D₂).

The minimal polynomial of ζ over $\mathbb{F}_2$ is $X^{2h} + X^h + 1$. Write the numerator and denominator in (D₂) as C, D . Since $P_k^q = P_k^{-1}$, membership in $\mathbb{F}_q$ would force $P_k = 1$. But $C = D$, after reducing $\zeta^{2h} = \zeta^h + 1$, would give

$$\zeta^{h+4} + \zeta^h + \zeta^3 + \zeta + 1 = 0, \quad (\text{D}_8)$$

a nonzero polynomial of degree $h + 4 < 2h$.

If $P_k \in \mathbb{F}_{a^2}$, its order divides $\gcd(\Lambda_k, a^2 - 1) = 3$, so $P_k \in \{1, \omega, \omega^2\}$. The first case was excluded; the other two respectively give

$$\zeta^{h+3} + \zeta = 0$$

and

$$\zeta^{h+4} + \zeta^{h+3} + \zeta^4 + \zeta^3 + \zeta + 1 = 0.$$

Both are again nonzero polynomials of degree below $2h$. Every proper divisor of $2h = 2 \cdot 3^k$ divides either h or $2h/3$; hence every proper subfield of $\mathbb{F}_{2^{2h}}$ lies in either $\mathbb{F}_q$ or $\mathbb{F}_{a^2}$. This proves (D₃).

For the 3-part, observe that

$$P_k^{\Lambda_k/3} = M_k^{(q+1)/3} = N^{(q^2-1)/3} = N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_4}(N). \quad (\text{D}_9)$$

Choose $u \in \mathbb{F}_{16}$ with $u^2 + u = \omega$. Then $u^4 = u + 1$, $u^5 = \omega$, and u has order 15. The extensions $\mathbb{F}_{2^{2h}}/\mathbb{F}_4$ and $\mathbb{F}_{16}/\mathbb{F}_4$ are linearly disjoint, and over $\mathbb{F}_{16}$ the minimal polynomial of ζ is $X^h + \omega$, irreducible because $\text{ord}_{3h}(16) = h$. Since

$$N = (\zeta + u)(\zeta + u + 1),$$

base change in (D₉) gives

$$N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_4}(N) = (u^h + \omega)((u + 1)^h + \omega).$$

The residues of $h = 3^k$ modulo 15 are 3, 9, 12, 6 as $k \equiv 1, 2, 3, 0 \pmod{4}$; reduction with $u^4 = u + 1$ gives

$$P_k^{\Lambda_k/3} = \begin{cases} \omega^2, & k \text{ odd}, \\ 1, & k \text{ even}. \end{cases}$$

Because $v_3(\Lambda_k) = 1$, this proves (D₄).

Let $e = \text{ord}(P_k)$. From (D₃), $\text{ord}_e(2) = 2h$. Since $e \mid \Lambda_k$ and $e \notin \{1, 3\}$, some prime $r \mid \Psi_k$ divides e ; every such r has $\text{ord}_r(2) = 2h$, so $r \equiv 1 \pmod{2h}$ and $r \geq 2h + 1$. When k is odd, (D₄) also gives $3r \mid e$, which proves (D₅).

If the full ℓ -primary part of Λ_k were missing from e , then $e \mid \Lambda_k/\ell$. This contradicts (D₅) under (D₆). Equation (D₁) transfers the conclusion back to M_k , and the last assertions follow. $\square$

Proposition 8.6 (fibotomic form of the current power-residue test). *Retain Proposition 8.5, put*

$$U = \{x \in \mathbb{F}_{q^2}^\times : x^{q+1} = 1\}, \quad \pi(x) = \frac{x}{(x+1)^2} = \frac{1}{x+x^{-1}},$$

and set $Z = \pi(M_k)$. The map π is two-to-one from $U \setminus \{1\}$ onto

$$\mathcal{T}_q = \{z \in \mathbb{F}_q : \text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(z) = 1\},$$

with fibres $\{x, x^{-1}\}$.

For an odd prime ℓ , define

$$H_\ell(W) = \sum_{j=0}^{(\ell-1)/2} \binom{\ell-j-1}{j} W^j \in \mathbb{F}_2[W].$$

If $\ell \mid \Psi_k$, then

$$M_k \in U^\ell \iff \text{there is } W \in \mathcal{T}_q \text{ with } W^\ell + ZH_\ell(W)^2 = 0. \quad (\text{F})$$

The polynomial

$$\mathcal{F}_{\ell,Z}(W) = W^\ell + ZH_\ell(W)^2$$

either has no root in $\mathbb{F}_q$, or splits there into ℓ distinct linear factors. Finally H_ℓ is squarefree and is a product of exactly $(\ell-1)/(2h)$ irreducible polynomials over $\mathbb{F}_2$, each of degree h . Thus D'_k is equivalently the assertion that $\mathcal{F}_{\ell,Z}$ is rootless over $\mathbb{F}_q$ for every prime $\ell \mid \Psi_k$.

Proof. For $x \in U$, one has $x^q = x^{-1}$, so $\pi(x) \in \mathbb{F}_q$. The equation $\pi(x) = z$ is

$$X^2 + z^{-1}X + 1 = 0.$$

The characteristic-two quadratic criterion says that this polynomial is irreducible over $\mathbb{F}_q$ exactly when $\text{Tr}(z) = 1$. Its roots are inverse and have norm one. Counting, or the same quadratic argument in reverse, proves the stated two-to-one description.

The defining polynomial identity is

$$\frac{X^\ell + 1}{(X+1)^\ell} = H_\ell\left(\frac{X}{(X+1)^2}\right). \quad (\text{H})$$

If $M_k = T^\ell$ with $T \in U$, put $W = \pi(T)$. Identity (H) gives

$$Z = \pi(T^\ell) = \frac{W^\ell}{H_\ell(W)^2},$$

so W satisfies (F). Conversely, lift a trace-one solution W to $T \in U$. Then (H) gives $\pi(T^\ell) = \pi(M_k)$, hence $T^\ell = M_k$ or M_k^{-1} . Replacing T by T^{-1} in the second case proves $M_k \in U^\ell$.

In fact every $\mathbb{F}_q$ -root of $\mathcal{F}_{\ell,Z}$ has trace one. Indeed, $H_\ell(W) \neq 0$ at a root, since otherwise the equation would force $W = 0$, whereas $H_\ell(0) = 1$. Lift a nonzero root through $\pi(T) = W$. If $\text{Tr}(W) = 0$, the quadratic criterion puts T in $\mathbb{F}_q$. Identity (H) then gives $\pi(T^\ell) = Z$, but for $y \in \mathbb{F}_q \setminus \{1\}$,

$$\pi(y) = \left(\frac{y}{y+1}\right)^2 + \frac{y}{y+1}$$

has absolute trace zero, whereas $Z = \pi(M_k)$ has trace one. This is a contradiction.

If one root exists, therefore lift it to $T \in U$ and orient T so that $T^\ell = M_k$. Then the ℓ elements

$$\pi(T\xi), \quad \xi \in \mu_\ell,$$

are roots in $\mathbb{F}_q$. They are distinct: an inverse collision would give $T^2 \in \mu_\ell$, hence $M_k^2 = 1$, whereas $M_k \neq 1$ because $N \notin \mathbb{F}_q$. Since $\mathcal{F}_{\ell,Z}$ has degree ℓ , it splits completely. Its derivative is $W^{\ell-1}$, and trace-one roots are nonzero, so they are simple.

Finally, the roots of H_ℓ are

$$\pi(\xi), \quad \xi \in \mu_\ell \setminus \{1\},$$

with ξ and ξ^{-1} identified. Their Frobenius-orbit length is the least $d > 0$ with $2^d \equiv \pm 1 \pmod{\ell}$. Since $\text{ord}_\ell(2) = 2h$, this least d is h . The number and degrees of the factors follow, and the roots are distinct, proving squarefreeness. $\square$

Proposition 8.7 (translated-binomial form of D'_k). *Retain $k \geq 2$, $h = 3^k$, and choose $u \in \mathbb{F}_{16}$ with $u^2 + u = \omega$. Put*

$$L = \mathbb{F}_{16^h} = \mathbb{F}_{2^{4h}}, \quad \beta = \zeta + u.$$

Then the minimal polynomial of β over $\mathbb{F}_{16}$ is the irreducible degree- h polynomial

$$B_k(Y) = (Y + u)^h + \omega.$$

For every prime $\ell \mid \Psi_k$, the following are equivalent:

- (i) *the full ℓ -primary part occurs in $\text{ord}(M_k)$;*
- (ii) *$\beta \notin (L^\times)^\ell$;*
- (iii) *$B_k(X^\ell) = (X^\ell + u)^h + \omega$ is irreducible over $\mathbb{F}_{16}$.*

More precisely, $B_k(X^\ell)$ is either irreducible of degree $h\ell$, or is a product of exactly ℓ irreducibles of degree h .

Proof. Since

$$\text{ord}_{3h}(16) = \frac{\text{ord}_{3h}(2)}{\gcd(\text{ord}_{3h}(2), 4)} = h,$$

the element ζ has degree h over $\mathbb{F}_{16}$. As $\zeta^h = \omega$, its minimal polynomial there is $X^h + \omega$; translation by u gives B_k .

Let $K = \mathbb{F}_{2^{2h}}$. The corrected norm is

$$N = N_{L/K}(\beta) = \beta^{2^{2h}+1}.$$

Because $2^{2h} \equiv 1 \pmod{\ell}$, this norm induces multiplication by two on $L^\times / (L^\times)^\ell$. Hence it first gives

$$\beta \in (L^\times)^\ell \iff N \in (L^\times)^\ell.$$

The natural map

$$K^\times / (K^\times)^\ell \longrightarrow L^\times / (L^\times)^\ell$$

is injective: restriction followed by the degree-two norm is multiplication by two, which is invertible modulo the odd prime ℓ . Since $N \in K$, this sharpens the equivalence to

$$\beta \in (L^\times)^\ell \iff N \in (K^\times)^\ell,$$

and Proposition 8.3 identifies the negation with the full ℓ -primary assertion. Kummer theory and Capelli's lemma give the irreducibility equivalence.

If $\beta = \alpha^\ell$ with $\alpha \in L$, every root of $B_k(X^\ell)$ lies in L , because $\mu_\ell \subset L$. Its ℓ -th power is a conjugate of β , of degree h over $\mathbb{F}_{16}$; the root therefore has degree both at least and at most h . All irreducible factors consequently have degree h , and there are exactly ℓ of them. $\square$

Corollary 8.8 (short translated cyclotomic period). *Let $\chi : L^\times \rightarrow \mu_\ell$ have exact order ℓ , realize its values as complex roots of unity, and write $\chi(\beta) = \xi_\ell^r$. Then*

$$\mathcal{J}_{k,\ell} := \sum_{\theta^h = \omega} \chi(\theta + u) = \sum_{j=0}^{h-1} \xi_\ell^{r16^j},$$

and

$$\beta \in (L^\times)^\ell \iff \mathcal{J}_{k,\ell} = h.$$

Proof. The roots of $X^h + \omega$ are ζ^{16^j} , $0 \leq j < h$, and $u \in \mathbb{F}_{16}$. Thus

$$\chi(\zeta^{16^j} + u) = \chi((\zeta + u)^{16^j}) = \xi_\ell^{r16^j}.$$

If $r = 0$, every summand is one. Conversely, equality with the positive real number h forces equality in the complex triangle inequality and hence forces every summand, in particular the first, to be one. $\square$

Proposition 8.9 (full $\mathbb{F}_{16}$ -line saturation). *Retain Proposition 8.7, put $s = 3^{k-1}$, and choose u so that it has order fifteen and $\omega = u^5$. For a prime $\ell \mid \Psi_k$, write*

$$\mathcal{C} = L^\times / (L^\times)^\ell, \quad v(c) = [\zeta + c] \quad (c \in \mathbb{F}_{16}), \quad A = 2^s \pmod{\ell},$$

using additive notation for $\mathcal{C}$. Then A has order six, $A^2 - A + 1 = 0$, and $A^3 = -1$. The four base-field classes vanish,

$$v(c) = 0 \quad (c \in \mathbb{F}_4), \tag{L}_{16,1}$$

while for $c \in \mathbb{F}_{16}^\times$,

$$v(\omega c^{-2^s}) = Av(c), \quad v(c^{-2^h}) = -v(c). \tag{L}_{16,2}$$

For the twelve exponents not divisible by five one has

j	$\{m : v(u^m) = A^j v(u)\}$	
0	$\{1, 4\}$	
1	$\{3, 12\}$	
2	$\{11, 14\}$	
3	$\{7, 13\}$	
4	$\{6, 9\}$	
5	$\{2, 8\}$.	(L) _{16,3}

Consequently

$$\zeta + u \in (L^\times)^\ell \iff \zeta + c \in (L^\times)^\ell \quad \text{for every } c \in \mathbb{F}_{16}. \tag{L}_{16,4}$$

If D'_k holds for ℓ , none of the twelve translates with $c \notin \mathbb{F}_4$ is an ℓ -th power. Equivalently, for an exact-order- ℓ complex character χ of $L^\times$ and $t = \chi(\zeta + u)$,

$$\sum_{c \in \mathbb{F}_{16}} \chi(\zeta + c) = 4 + 2 \sum_{j=0}^5 t^{A^j}, \tag{L}_{16}$$

and failure of D'_k is exactly the extremal value sixteen. Thus Frobenius, conjugation, and all $\mathbb{F}_4$ -translates generate only scalar multiples of the one unresolved Kummer class $v(u)$.

Proof. One has $\text{ord}_\ell(2) = 2h = 6s$, proving the assertions about A . For $c = 0$, the element ζ has ℓ -coprime order $3h$. For $c \in \{1, \omega, \omega^2\}$, Propositions 6.4 and 6.5 give order dividing $3h(2^h - 1)$, again coprime to ℓ . This proves $(L_{16,1})$; constants in $\mathbb{F}_{16}^\times$ and ζ also have zero class.

The congruence $2^s \equiv h - 1 \pmod{3h}$ gives $\zeta^{2^s} = \omega\zeta^{-1}$. Discarding scalar factors with zero class,

$$Av(c) = [(\zeta + c)^{2^s}] = [\omega\zeta^{-1} + c^{2^s}] = [\zeta + \omega c^{-2^s}] = v(\omega c^{-2^s}).$$

Likewise $2^h \equiv -1 \pmod{3h}$ and modulo ℓ , so

$$-v(c) = [(\zeta + c)^{2^h}] = [\zeta^{-1} + c^{2^h}] = [\zeta + c^{-2^h}].$$

On exponents modulo fifteen the two transformations are

$$R(m) = 5 - (2^s \bmod 15)m, \quad S(m) = -(2^h \bmod 15)m.$$

For odd k they are $(5 - 2m, 7m)$, and for even k they are $(5 - 8m, 13m)$. Iterating either pair on the twelve residues outside $5\mathbb{Z}/15\mathbb{Z}$ gives the table $(L_{16,3})$. Since every A^j is nonzero modulo ℓ , all twelve classes vanish exactly when $v(u)$ vanishes, proving $(L_{16,4})$ and its nonvanishing counterpart.

Applying χ to the table gives each t^{A^j} twice, while the four $\mathbb{F}_4$ -terms are one. If $t = 1$, the sum is sixteen; conversely equality with the positive real number sixteen forces every unit-modulus summand to be one by equality in the triangle inequality. $\square$

Proposition 8.10 (exceptional gap and Conway-block product blindness). *Retain Proposition 8.9, so $k \geq 2$, and put $K = \mathbb{F}_{2^{2h}}$. For every prime $\ell \mid \Psi_k$,*

$$m_\ell = 4 \quad \text{or} \quad m_\ell \geq 16. \quad (\mathbf{G}_{16})$$

Thus a failure of D'_k cannot produce an excess in the interval $5, \dots, 15$.

There is also no contradiction obtained by multiplying a complete finite-Conway-subfield block. More precisely, for $a \geq 1$, put

$$C_a = \mathbb{F}_{2^{2^a}}, \quad Q_a = |C_a| = 2^{2^a}, \quad L_a = \mathbb{F}_{2^{2^a h}} = KC_a.$$

Then, in L_a ,

$$\prod_{c \in C_a} (\zeta + c) = \zeta^{Q_a} + \zeta = \zeta(1 + \zeta^{Q_a-1}) \in (L_a^\times)^\ell. \quad (\mathbf{G}_{C_a})$$

Consequently even an extension of the affine-line saturation from $\mathbb{F}_{16}$ to any complete finite Conway subfield would have a product identity which is automatically invisible in the unresolved ℓ -Kummer quotient.

Proof. Choose the element u of Proposition 8.9 to be the finite number 4. If $m_\ell \neq 4$, then the lower bound $m_\ell \geq 4$ and the exact power criterion say that $\zeta + 4$ is an ℓ -th power in $L = \mathbb{F}_{2^{4h}}$. Equation $(L_{16,4})$ then makes every $\zeta + c$, $c \in \mathbb{F}_{16}$, an ℓ -th power. The finite numbers $0, 1, \dots, 15$ are exactly the elements of $\mathbb{F}_{16}$, so none of these sixteen translates can be the excess. This proves $(\mathbf{G}_{16})$.

For the second assertion, the roots of $X^{Q_a} + X$ are precisely the elements of C_a , which gives the displayed product formula. Since 2^a is even, lifting the exponent gives

$$v_3(Q_a - 1) = v_3(2^{2^a} - 1) = 1.$$

The element ζ has order $3h = 3^{k+1}$, so ζ^{Q_a-1} has exact order h and lies in

$$\mathbb{F}_{2^{\text{ord}_h(2)}} = \mathbb{F}_{2 \cdot 3^{k-1}}.$$

This lower field embeds in K , hence in every L_a . On the other hand $\text{ord}_\ell(2) = 2h$. Hence ℓ divides neither $3h$ nor the multiplicative order of that lower field. It follows that both ζ and $1 + \zeta^{Q_a-1}$ are ℓ -th powers in L_a , proving (G_{C_a}) . $\square$

Corollary 8.11 (composite-conductor cyclotomic-unit form of D'_k). *Retain Proposition 8.10, put $N_0 = 15h$, and choose $\theta \in L^\times$ so that*

$$\theta^5 = \zeta, \quad \theta^h = u.$$

Then θ has exact order N_0 , and

$$\rho := \frac{\zeta}{u} = \theta^{5-h}$$

also has exact order N_0 . If $\tilde{\rho}$ is a primitive complex N_0 -th root of unity,

$$\varepsilon = 1 - \tilde{\rho} \in \mathbb{Q}(\mu_{N_0})^\times$$

is a global cyclotomic unit. There is a prime $\mathfrak{p} \mid 2$ whose residue field is $L = \mathbb{F}_{2^{4h}}$ and for which

$$\bar{\varepsilon} = 1 + \rho, \quad \zeta + u = u(1 + \rho).$$

For every prime $\ell \mid \Psi_k$,

$$D'_k \text{ holds at } \ell \iff \left(\frac{\varepsilon}{\mathfrak{p}} \right)_\ell \neq 1. \quad (\text{CU}_{15h})$$

Equivalently, the chosen prime over two has nontrivial Frobenius in the Kummer extension obtained by adjoining an ℓ -th root of ε after adjoining μ_ℓ .

Proof. The elements ζ and u have orders $3h$ and 15 , and their h -th and fifth powers agree: $\zeta^h = u^5 = \omega$. The compatible pair therefore lifts in the cyclic group of order $\text{lcm}(3h, 15) = 15h$ to the stated θ . Its two displayed images have their full orders, so θ has order $15h$. Moreover

$$\gcd(5 - h, 15h) = 1$$

because h is a power of three and is not divisible by five; hence ρ is primitive of the same order.

The conductor $15h$ has two distinct prime divisors, so $\Phi_{15h}(1) = 1$, which makes $1 - \tilde{\rho}$ a unit. Also

$$\text{ord}_{15h}(2) = \text{lcm}(\text{ord}_{3h}(2), \text{ord}_5(2)) = \text{lcm}(2h, 4) = 4h.$$

Thus a prime over two has residue field L , and it may be chosen to send $\tilde{\rho}$ to ρ . Reduction gives the two displayed identities. Finally $\ell \nmid 15$, so u is an ℓ -th power. Proposition 8.7 identifies D'_k at ℓ with $\zeta + u$ being a non- ℓ -th-power; the standard power-residue/Frobenius criterion now gives (CU_{15h}) . $\square$

Proposition 8.12 (exceptional failure as simultaneous Jacobi flatness). *Let $k \geq 2$, put*

$$h = 3^k, \quad M = 15h, \quad Q = 2^{4h},$$

and let the prime ℓ divide Ψ_k . Choose the reduction $z \in \mathbb{F}_Q^\times$ of the selected primitive M -th root in Corollary 8.11, choose a generator g with $g^{(Q-1)/M} = z$, and form the mixed Jacobi sums

$$J_j = \sum_{x \in \mathbb{F}_Q} \chi_M(x)^j \chi_\ell(1-x) \quad (1 \leq j < M)$$

of Proposition 5.13. At a chosen ℓ -adic place put $\pi = \zeta_\ell - 1$. Then failure of D'_k at ℓ is equivalent to the simultaneous first-order flatness

$$J_j \equiv -1 \pmod{\pi^2} \quad (1 \leq j < M). \quad (\text{DJ})$$

Thus mixed Jacobi sums see the complete exceptional Kummer vector, but the missing theorem is exactly that this vector is not identically zero; Stickelberger ideal factorization alone does not evaluate its first local phase.

Proof. Write the unique coprime-order decomposition

$$z = ab, \quad \text{ord}(a) = 3h, \quad \text{ord}(b) = 5,$$

and put $q = 2^h$, $L = \mathbb{F}_{q^4} = \mathbb{F}_Q$, and $K_0 = \mathbb{F}_{q^2}$. By Corollary 8.11, failure at ℓ says that $1 + ab$ is an ℓ -th power in L . The norm $L^\times \rightarrow K_0^\times$ acts on the one-dimensional ℓ -Kummer quotient by

$$q^2 + 1 \equiv 2 \pmod{\ell},$$

because $\text{ord}_\ell(2) = 2h$. It is therefore an isomorphism on that quotient, and

$$U = (1 + ab)(1 + ab^{-1})$$

is an ℓ -th power in K_0 .

Since $a^q = a^{-1}$,

$$(1 + a)^{q-1} = a^{-1}, \quad (1 + a^5)^{q-1} = a^{-5}.$$

Here $q - 1 \equiv -2 \pmod{\ell}$, while a has order prime to ℓ . Hence both $1 + a$ and $1 + a^5$ are ℓ -th powers. If

$$U' = (1 + ab^2)(1 + ab^{-2}),$$

then

$$UU' = \Phi_5(a) = \frac{1 + a^5}{1 + a}$$

is an ℓ -th power. Thus U' is an ℓ -th power, and the same norm isomorphism implies that $1 + ab^2$ is an ℓ -th power in L .

Now

$$\text{ord}_M(2) = 4h, \quad \varphi(M) = 8h.$$

The primitive M -th roots consequently form two Frobenius orbits, represented by ab and ab^2 . These representatives are not in the same orbit: a power 2^j fixing the $3h$ -component has $j \equiv 0 \pmod{2h}$, whereas no such exponent sends b to b^2 . Hence $1 + v$ is an ℓ -th power for every primitive M -th root v .

It remains to treat proper-order powers of z . If $d \mid M$ is proper, the divisibility $2h \mid \text{ord}_d(2)$ can occur only for $d = 3h$. For an element v of that order, $v^q = v^{-1}$, and

$$(1 + v)^{q-1} = v^{-1}.$$

As above, $q - 1 \equiv -2 \pmod{\ell}$ and $\ell \nmid \text{ord}(v)$, so $1 + v$ is an ℓ -th power. For every other proper d , the smallest finite field containing v has unit group of order prime to ℓ ; its ℓ -power map is an automorphism. We have therefore proved

$$1 - z^a \in (\mathbb{F}_Q^\times)^\ell \quad (1 \leq a < M).$$

Proposition 5.13 now gives (DJ). Conversely, (DJ) makes every Fourier coefficient C_j vanish, so (MJ₂) at $a = 1$ says that $1 - z$ is an ℓ -th power. This is precisely failure of D'_k at ℓ . $\square$

Proposition 8.13 (binary cyclic-parity form of exceptional failure). *Retain Proposition 8.12, set*

$$E = \frac{Q-1}{\ell}, \quad S_M(X) = 1 + X + \cdots + X^{M-1},$$

and let $R_{k,\ell}(X)$ be the degree- $< M$ reduction of $(1+X)^E$ modulo $X^M - 1$. Then failure of D'_k at ℓ is equivalent to

$$R_{k,\ell}(X) = 1 + S_M(X) \quad \text{in } \mathbb{F}_2[X]/(X^M - 1). \quad (\text{DP}_1)$$

Equivalently, among the binary submasks u of E , the number with $u \equiv 0 \pmod{M}$ is even, while the number in every nonzero residue class modulo M is odd.

Moreover, raising (DP₁) to the ℓ -th power gives only the unconditional identity

$$(1+X)^{Q-1} = 1 + S_M(X) \quad \text{in } \mathbb{F}_2[X]/(X^M - 1). \quad (\text{DP}_2)$$

Thus ordinary exponent, determinant, or product consequences of (DP₁) cannot rule out exceptional failure; the remaining assertion is exactly that this selected binary-residue parity pattern never occurs.

Proof. The proof of Proposition 8.12 established that failure is equivalent to

$$1 - z^a \in (\mathbb{F}_Q^\times)^\ell \quad (1 \leq a < M). \quad (\text{DP}_3)$$

Here $\ell \mid Q - 1$, and in the cyclic group $\mathbb{F}_Q^\times$ an element v is an ℓ -th power exactly when $v^E = 1$. Consequently (DP₃) says that $R_{k,\ell}(\xi) = 1$ at every nonidentity M -th root ξ . At $\xi = 1$, it has value 0. Since M is odd, $X^M - 1$ is squarefree, and $1 + S_M$ has exactly these values. This proves (DP₁), including its converse by evaluation at z .

Lucas's theorem gives

$$(1+X)^E = \sum_{u \subseteq_2 E} X^u,$$

so cyclic reduction proves the submask-parity formulation. Finally $S_M^2 = MS_M = S_M$, hence $1 + S_M$ is idempotent. The right side of (DP₁) is therefore unchanged on taking its ℓ -th power. Identity (DP₂) is unconditional: at $X = 1$ both sides vanish, while at every other M -th root $\xi \in \mathbb{F}_Q$, Fermat gives $(1 + \xi)^{Q-1} = 1$. Squarefree interpolation completes the proof. $\square$

Proposition 8.14 (exceptional half-block norm and its automatic symmetries). *Retain Proposition 8.13, and put*

$$A = 2^{2h}, \quad u = \frac{A-1}{\ell}, \quad R = \mathbb{F}_2[X]/(X^M - 1), \quad e = 1 + S_M.$$

Let $I = eR$, with identity e , and set

$$U = 1 + X \in I^\times, \quad \tau(X) = X^A, \quad F = U^u.$$

Then

$$F^\ell = \frac{\tau(U)}{U}, \quad F\tau(F) = U^{u(A+1)}, \quad (\text{DH}_1)$$

and exceptional failure is exactly

$$F\tau(F) = e. \quad (\text{DH}_2)$$

On the selected primitive M -th-root component of Corollary 8.11, put

$$w = (1 + \rho)(1 + \rho^A) \in \mathbb{F}_A^\times, \quad v = w^u \in \mu_\ell.$$

Then failure is precisely $v = 1$.

There is no further Hilbert–90 or quadratic-norm obstruction to (DH_2) . Indeed $\tau(U)/U$ belongs to the norm-one group of $\mathbb{F}_{A^2}/\mathbb{F}_A$, whose order is $A + 1$; since $\gcd(\ell, A + 1) = 1$, it has a unique ℓ -th root in that group. Failure says only that the distinguished root $F = U^u$ is this norm-one root.

The coefficient symmetries are automatic as well. The τ -fixed residues modulo M are exactly the multiples of five. For such a residue j ,

$$[X^j]F\tau(F) = \sum_{\substack{a \pmod{M} \\ (1+A)a=j}} [X^a]F. \quad (\text{DH}_3)$$

After identifying the fixed subgroup with C_{3h} , the right side is, up to an invertible multiplier of its indices, the cyclic reduction of $(1 + Y)^u$; unconditionally,

$$(1 + Y)^u = e_{3h} \quad \text{in } \mathbb{F}_2[C_{3h}]. \quad (\text{DH}_4)$$

Thus the fixed-residue part of the exceptional parity pattern is forced before the selected Kummer question is asked.

Proof. Since $U^A = \tau(U)$, the first identity in (DH_1) is

$$F^\ell = U^{A-1} = \frac{\tau(U)}{U}.$$

The second follows by multiplying U^u and $\tau(U)^u = U^{Au}$. Since

$$u(A + 1) = \frac{A^2 - 1}{\ell} = \frac{Q - 1}{\ell},$$

Proposition 8.13 proves (DH_2) . On a primitive component, τ is the quadratic Frobenius of $\mathbb{F}_{A^2}/\mathbb{F}_A$, and evaluation gives $F\tau(F) = w^u = v$. The norm-one assertion and uniqueness then follow from $\ell \mid A - 1$ and $\gcd(A - 1, A + 1) = 1$.

One has

$$\gcd(A - 1, M) = 3h, \quad \gcd(A + 1, M) = 5.$$

Hence the fixed residues of multiplication by A are precisely the multiples of five. In the convolution coefficient of such a residue, swapping the two selected submasks preserves the residue. Every off-diagonal pair cancels in characteristic two, leaving only the diagonal pairs; this is (DH_3) . Multiplication by $(A + 1)/5$ is invertible modulo $3h$, so the displayed pushforward is the reduction of $(1 + Y)^u$, up to a permutation of coefficients.

Put $q = 2^h$. If $\xi \neq 1$ is a $3h$ -th root of unity, then $\xi^q = \xi^{-1}$, and therefore

$$(1 + \xi)^{q-1} = \xi^{-1}.$$

Also $u = (q - 1)(q + 1)/\ell$, while $3h \mid (q + 1)/\ell$. It follows that $(1 + \xi)^u = 1$. At $\xi = 1$ it vanishes, so squarefree interpolation proves (DH_4) . $\square$

Proposition 8.15 (one-coordinate trace form of the exceptional phase). *Retain Proposition 8.14, put $s = h/3$, and let*

$$B \equiv 2^s \pmod{\ell}.$$

Then B has order six modulo ℓ and $B^2 - B + 1 = 0$. For $v \in \mu_\ell \subset \mathbb{F}_{2^{6s}}$, define

$$\mathcal{T}(v) = \text{Tr}_{\mathbb{F}_{2^{6s}}/\mathbb{F}_{2^s}}(v) = \sum_{j=0}^5 v^{B^j}.$$

One has the exact equivalence

$$\mathcal{T}(v) = 0 \iff v = 1. \quad (\text{DT}_1)$$

Consequently the exceptional assertion at ℓ is exactly

$$\mathcal{T}\left(\left((1+\rho)(1+\rho^A)\right)^{(A-1)/\ell}\right) \neq 0. \quad (\text{DT}_2)$$

This is a single relative-trace coordinate of the selected residue symbol. It is equivalent to, rather than a generic norm consequence of, the missing nontriviality.

Proof. The order identity $\text{ord}_\ell(2) = 2h = 6s$ proves the assertions about B . Put $x = v$ and $y = v^B$. The six powers of B , together with $B^2 = B - 1$, give the elementary factorization

$$xy\mathcal{T}(v) = (x+y)(x+1)(y+1). \quad (\text{DT}_3)$$

If $v \neq 1$, then $x \neq 1$, $y \neq 1$, and $x \neq y$, because ℓ is prime and $B \not\equiv 1 \pmod{\ell}$. Thus the right side is nonzero. Conversely $\mathcal{T}(1) = 6 = 0$ in characteristic two, proving (DT_1) . Proposition 8.14 identifies failure with $v = 1$, so its negation is (DT_2) . $\square$

8.4 Cyclotomic antiunit and regulator bounds

Proposition 8.16 (quadratic-twist antiunit and the two exceptional zeros). *Retain Corollary 8.11, put $m = 3h$, and write the unique coprime-order decomposition*

$$\tilde{\rho} = \alpha\beta, \quad \text{ord}(\alpha) = m, \quad \text{ord}(\beta) = 5.$$

Set

$$s = \beta + \beta^{-1}, \quad s' = \beta^2 + \beta^{-2},$$

and define the cyclotomic antiunit

$$d_k := \frac{(1 - \alpha\beta)(1 - \alpha\beta^{-1})}{(1 - \alpha\beta^2)(1 - \alpha\beta^{-2})} = \frac{\alpha^2 - s\alpha + 1}{\alpha^2 - s'\alpha + 1} = \frac{\alpha + \alpha^{-1} - s}{\alpha + \alpha^{-1} - s'}. \quad (\text{QT}_1)$$

Then d_k belongs to

$$F_k^{\text{tw}} = \mathbb{Q}(\alpha + \alpha^{-1}, \sqrt{5}),$$

is fixed by complex conjugation, and is inverted by the nontrivial automorphism of $F_k^{\text{tw}}/\mathbb{Q}(\alpha + \alpha^{-1})$. In particular its relative norm is one.

The invariant conductor-five norm loses exactly the information carried by d_k :

$$N_{\mathbb{Q}(\mu_{5m})/\mathbb{Q}(\mu_m)}(1 - \tilde{\rho}) = \Phi_5(\alpha) = \frac{1 - \alpha^5}{1 - \alpha}. \quad (\text{QT}_2)$$

More precisely, reduce at the prime selected in Corollary 8.11, write

$$a = \bar{\alpha}, \quad b = \bar{\beta}, \quad c = b + b^{-1} \in \mathbb{F}_4 \setminus \mathbb{F}_2,$$

and put, in $K = \mathbb{F}_{2^{2h}}$,

$$U_k = a^2 + ca + 1, \quad U'_k = a^2 + c^2a + 1, \quad R_k = \frac{U_k}{U'_k} = \bar{d}_k. \quad (\text{QT}_3)$$

For every prime $\ell \mid \Psi_k$, the element $\Phi_5(a)$ is an ℓ -th power in K , and

$$D'_k \text{ holds at } \ell \iff U_k \notin (K^\times)^\ell \iff R_k \notin (K^\times)^\ell. \quad (\text{QT}_4)$$

Thus the two conductor-five places over the cubic field carry inverse Kummer classes; their invariant product is identically blind.

This antiunit is an exact quadratic lift of a conjugate of the cubic Gaussian period. If

$$q = 2^h, \quad \gamma = a + a^{-1}, \quad A = \gamma^2 + \gamma + 1,$$

then

$$R_k^q = R_k^{-1}, \quad R_k + R_k^{-1} = A^{-1}, \quad \frac{R_k}{(R_k + 1)^2} = A. \quad (\text{QT}_5)$$

After choosing compatible roots $a_k^3 = a_{k-1}$ and keeping c fixed, the degree-three norm swaps the two factors:

$$N(U_k) = U'_{k-1}, \quad N(U'_k) = U_{k-1}, \quad N(R_k) = R_{k-1}^{-1}. \quad (\text{QT}_6)$$

Consequently

$$\mathcal{E}_k := R_k^{(-1)^k} \text{ satisfies } N(\mathcal{E}_k) = \mathcal{E}_{k-1}. \quad (\text{QT}_7)$$

Finally the cubic and exceptional arms are the untwisted and quadratic-modulo-five twists of the same simple exceptional zero. On the current Kummer line the 3-power cyclotomic generator acts as $+2$ for $\ell \mid \Phi_h(2)$, and as -2 for $\ell \mid \Phi_h(-2) = \Phi_{2h}(2)$. With $r = h/3$, the relative norm polynomial

$$g_r(X) = 1 + X^r + X^{2r}$$

has a simple zero at $X = 2$ in the cubic case and at $X = -2$ in the exceptional case. Hence the norm coherence in (QT_7) transports old factors but annihilates the current one. It does not prove the still-open nonvanishing in (QT_4) ; both arms require the distinguished first-order unit regulator in their respective ± 2 -eigenspaces.

Proof. The coprime orders m and 5 give the unique decomposition $\tilde{\rho} = \alpha\beta$. The four automorphisms of $\mathbb{Q}(\mu_{5m})/\mathbb{Q}(\mu_m)$ fix α and send β through the four primitive fifth roots. Therefore

$$\prod_{j=1}^4 (1 - \alpha\beta^j) = \Phi_5(\alpha) = \frac{1 - \alpha^5}{1 - \alpha},$$

which proves (QT_2) . The two values s, s' are the roots of $X^2 + X - 1$, so their interchange is the nontrivial automorphism of $\mathbb{Q}(\sqrt{5})$. It interchanges the numerator and denominator of d_k . Complex conjugation replaces α by α^{-1} and multiplies both quadratic factors by α^{-2} , so their quotient is fixed. This proves all the characteristic-zero assertions preceding (QT_2) .

The residue of α has order m and lies in K , while the residue of β lies in $\mathbb{F}_{16}$. Since

$$2^{2h} \equiv 1 \pmod{m}, \quad 2^{2h} \equiv -1 \pmod{5},$$

the local norm from $L = \mathbb{F}_{2^{4h}}$ to K gives

$$N_{L/K}(1 + ab) = (1 + ab)(1 + ab^{-1}) = U_k.$$

Also $2^{2h} + 1 \equiv 5 \pmod{15}$, so $N_{L/K}(u) = u^5 = \omega$. Hence the corrected norm N of Proposition 8.7 is ωU_k . Because $\ell \neq 3$, it has the same ℓ -Kummer class as U_k .

It remains to show that the invariant product is blind. If $x = 1 + a$, then $a^q = a^{-1}$ and

$$x^{q-1} = a^{-1}.$$

Here $\ell \nmid q - 1$, whereas a has order m , prime to ℓ . Projection to the ℓ -Sylow subgroup of $K^\times$ therefore shows that $1 + a$ is an ℓ -th power. The same argument applies to $1 + a^5$. Thus

$$U_k U'_k = \Phi_5(a) = \frac{1 + a^5}{1 + a} \in (K^\times)^\ell.$$

In additive notation on the one-dimensional Kummer quotient this says $[U'_k] = -[U_k]$, and hence $[R_k] = 2[U_k]$. Since ℓ is odd, Proposition 8.7 now proves (QT_4) .

As h is odd, q -Frobenius sends c to c^2 . Directly,

$$U_k^q = a^{-2} U'_k, \quad (U'_k)^q = a^{-2} U_k,$$

which gives $R_k^q = R_k^{-1}$. Moreover

$$U_k + U'_k = a, \quad U_k U'_k = \Phi_5(a).$$

It follows that

$$R_k + R_k^{-1} = \frac{a^2}{\Phi_5(a)} = \frac{1}{(a + a^{-1})^2 + (a + a^{-1}) + 1}.$$

The last identity in (QT_5) is the characteristic-two identity $R/(R + 1)^2 = 1/(R + R^{-1})$.

The nonvanishing in (QT_4) is unchanged when a is replaced by any primitive m -th root or c is interchanged with c^2 : the former roots form one Frobenius orbit, and odd Frobenius powers merely perform the latter interchange. We may therefore choose $a_k^3 = a_{k-1}$. Factoring

$$U_k = (1 + a_k b)(1 + a_k b^{-1})$$

and, after the harmless quadratic scalar extension adjoining b , taking the product over the three conjugates obtained by multiplying a_k by a cube root of unity gives

$$N(U_k) = (1 + a_{k-1} b^3)(1 + a_{k-1} b^{-3}) = U'_{k-1}.$$

The same calculation with b^2 gives $N(U'_k) = U_{k-1}$, proving (QT_6) – (QT_7) .

For the last assertion, let σ be the 3-power cyclotomic generator. In the untwisted real field its reduction is Frobenius, so it acts as 2 on the current ℓ -Kummer line. In the quadratic twist, rational Frobenius is the product of σ and the $\sqrt{5}$ -involution. The latter reduces to q -Frobenius and acts as $q = -1$ because $\ell \mid q + 1$; hence σ acts as -2 . If $\ell \mid \Phi_h(2)$, then 2^r is a primitive cube

root modulo ℓ . If $\ell \mid \Phi_h(-2)$ and $\ell \neq 3$, then $(-2)^r$ is a primitive cube root instead. Thus g_r vanishes at the asserted scalar. Its derivative there is

$$g'_r(X) = rX^{r-1}(1 + 2X^r) \neq 0 \pmod{\ell} :$$

one has $\ell \nmid r$, and a primitive cube root equal to $-1/2$ would force $\ell \mid 9$. The zero is therefore simple. The final open/non-open distinction follows: the norm operator vanishes on the current line, while its first derivative only identifies the one-dimensional regulator coordinate and does not evaluate d_k in that coordinate. $\square$

Proposition 8.17 (selected exceptional sextic and the Dickson-factor boundary). *Retain Propositions 8.14, 8.15, and 8.16. Put*

$$s = 3^{k-1}, \quad A_0 = 2^s, \quad q = A_0^3 = 2^h, \quad \Lambda = A_0^2 - A_0 + 1 = 3\Psi_k,$$

and write

$$\gamma = \gamma_k, \quad g = \gamma_{k-1}, \quad C = g^2 + g + 1.$$

Choose the orientation $c \in \mathbb{F}_4 \setminus \mathbb{F}_2$ from $(\mathbf{QT}_3)$, and define

$$R = \frac{\gamma + c}{\gamma + c^2}, \quad r = R^{A_0+1}, \quad y = r + r^{-1}.$$

Then $r^\Lambda = 1$. If $\ell \mid \Psi_k$ is prime and $e = \Lambda/\ell$, the half-block phase is exactly

$$v = (r^e)^{-1}. \tag{DS_1}$$

Put $z = \gamma^{A_0^2}$. The selected trace coordinate collapses to

$$y = \frac{(g+1)z + g}{C}, \tag{DS_2}$$

and has the irreducible cubic over $\mathbb{F}_{A_0}$

$$G(Y) = C^2Y^3 + gCY^2 + Y + g^2. \tag{DS_3}$$

The element r has degree six over $\mathbb{F}_{A_0}$, with scaled irreducible reciprocal polynomial

$$\begin{aligned} S(T) = T^3G(T + T^{-1}) &= C^2T^6 + gCT^5 + (C^2 + 1)T^4 + g^2T^3 \\ &\quad + (C^2 + 1)T^2 + gCT + C^2. \end{aligned} \tag{DS_4}$$

Writing $D_n(Y) = \mathcal{D}_n(Y, 1)$, exceptional failure at ℓ is equivalent to each of

$$D_e(y) = 0, \quad G(Y) \mid D_e(Y), \quad S(T) \mid T^e - 1. \tag{DS_5}$$

Moreover the trace in Proposition 8.15 satisfies

$$\mathcal{T}(v) = \mathrm{Tr}_{\mathbb{F}_{A_0^3}/\mathbb{F}_{A_0}}(D_e(y)). \tag{DS_6}$$

There is an equivalent reverse-Dickson form. Put

$$\mathcal{A} = \gamma^2 + \gamma + 1.$$

Then $\mathcal{A}$ and $\mathcal{A}^{-1}$ have the irreducible cubics

$$\begin{aligned}\mathcal{A}^3 + \mathcal{A}^2 + (g+1)\mathcal{A} + C &= 0, \\ CZ^3 + (g+1)Z^2 + Z + 1 &= 0.\end{aligned}\tag{DS7}$$

and

$$y = D_{A_0+1}(\mathcal{A}^{-1}).$$

Thus, with $n = (q+1)/\ell = (A_0+1)e$, failure is also equivalent to

$$CZ^3 + (g+1)Z^2 + Z + 1 \mid D_n(Z).\tag{DS8}$$

Finally the ambient Dickson polynomial has the exact factorization

$$D_e(Y) = Y(Y+1)^2 \prod_{i=1}^{(e-3)/6} Q_i(Y)^2,\tag{DS9}$$

where every Q_i is an irreducible cubic over $\mathbb{F}_{A_0}$. When $e = 3$, this independently recovers the prime- Ψ_k case already proved in Proposition 8.5. For composite Ψ_k , the recursion selects the explicit cubic G among $(e-3)/6$ structurally admissible cubics; excluding that selected factor remains exactly the exceptional theorem.

Proof. Use the notation in the proof of Proposition 8.16. Thus, for the order- $3h$ element α ,

$$U = \alpha^2 + c\alpha + 1 = \alpha(\gamma + c), \quad U' = \alpha^2 + c^2\alpha + 1 = \alpha(\gamma + c^2),$$

and $R = U/U'$. The half-block element w is U . Put

$$u = \frac{A_0^6 - 1}{\ell}, \quad \xi = r^e = R^{(q+1)/\ell}.$$

Since UU' is an ℓ -th power, $(UU')^u = 1$. If $v' = (U')^u$, then $vv' = 1$, and hence

$$R^u = \frac{v}{v'} = v^2.$$

On the other hand $R^u = \xi^{q-1} = \xi^{-2}$, because $\xi \in \mu_\ell$ and $q \equiv -1 \pmod{\ell}$. The group μ_ℓ has odd order, so $(v\xi)^2 = 1$ proves (DS₁). Also $R^{q+1} = 1$, which gives $r^\Lambda = 1$.

Let $x = \gamma$, $x_1 = \gamma^{A_0}$, and $z = \gamma^{A_0^2}$. The Gaussian-period recursion gives

$$x + x_1 + z = 0, \quad z^3 + z + g = 0.$$

Since $c^{A_0} = c^2$,

$$r = RR^{A_0} = \frac{(x+c)(x_1+c^2)}{(x+c^2)(x_1+c)} = \frac{N}{D}.$$

Here $N + D = z$. Also

$$ND = (x^2 + x + 1)(x_1^2 + x_1 + 1).$$

For $f(T) = T^3 + T + g$,

$$\prod_{t \in \{x, x_1, z\}} (t^2 + t + 1) = f(c)f(c^2) = (g+c)(g+c^2) = C.$$

Therefore

$$y = \frac{(N+D)^2}{ND} = \frac{z^2(z^2+z+1)}{C} = \frac{(g+1)z+g}{C},$$

proving (DS₂). Both C and $g+1$ are nonzero, so y generates the same cubic extension of $\mathbb{F}_{A_0}$ as z . Substitution in $z^3+z+g=0$ gives (DS₃).

The quadratic T^2+yT+1 of r over $\mathbb{F}_{A_0}(y)$ is irreducible. Otherwise $r \in \mathbb{F}_{A_0^3}$; but $r^q = r^{-1}$, so $r = 1$, which would force $y = 0$, contrary to (DS₃). This proves the degree-six assertion and (DS₄).

Dickson's identity gives

$$D_e(y) = r^e + r^{-e}.$$

As $\text{ord}(r)$ is odd, this vanishes exactly when $r^e = 1$, which by (DS₁) is failure. Minimal-polynomial divisibility proves (DS₅). Pairing the six A_0 -Frobenius conjugates of $v = \xi^{-1}$ under the inversion A_0^3 -Frobenius gives (DS₆).

The three conjugates x_i of γ satisfy

$$\sum_i (x_i^2 + x_i + 1) = 1, \quad \sum_{i < j} (x_i^2 + x_i + 1)(x_j^2 + x_j + 1) = g + 1,$$

and the product of the three transformed conjugates $x_i^2 + x_i + 1$ is C . This proves the first cubic in (DS₇). Since γ has degree three over $\mathbb{F}_{A_0}$, while it is at most quadratic over $\mathbb{F}_{A_0}(\mathcal{A})$, the odd tower law forces $\mathbb{F}_{A_0}(\mathcal{A}) = \mathbb{F}_{A_0}(\gamma)$; hence the cubic is irreducible. Inversion gives the second cubic. Directly

$$\frac{R}{(R+1)^2} = (\gamma + c)(\gamma + c^2) = \mathcal{A},$$

so the semiconjugacy in Proposition 7.14 gives

$$\Phi(R) = \mathcal{A}, \quad \Phi(r) = F_{A_0+1}(\mathcal{A}) = \frac{1}{D_{A_0+1}(\mathcal{A}^{-1})},$$

and $\Phi(r) = 1/y$. Therefore

$$y = D_{A_0+1}(\mathcal{A}^{-1}).$$

The Dickson composition law, with $n = (A_0 + 1)e$, now proves (DS₈).

It remains to factor D_e . One has $v_3(\Lambda) = 1$, while $\ell \neq 3$; hence $e \equiv 3 \pmod{6}$. Over an algebraic closure,

$$D_e(Y) = Y \prod_{\substack{\{t, t^{-1}\} \\ t \in \mu_e \setminus \{1\}}} (Y - (t + t^{-1}))^2. \quad (\text{DS}_{10})$$

The pair of nontrivial cube roots gives $(Y+1)^2$. For every remaining t , its order has a prime divisor other than three. Since $A_0^3 \equiv -1 \pmod{\Lambda}$, while $\gcd(A_0 - 1, \Lambda) = 1$ and $\gcd(A_0 + 1, \Lambda) = 3$, its inverse-pair Frobenius orbit has length three. Grouping the remaining roots in (DS₁₀) proves (DS₉). $\square$

Proposition 8.18 (the trace-constant shape of every Dickson cubic). *Let Q be a power of two with $Q \equiv 2 \pmod{3}$, and put $\Lambda = Q^2 - Q + 1$. If $t \in \mu_\Lambda \setminus \mu_3$ and $y = t + t^{-1}$, then $[\mathbb{F}_Q(y) : \mathbb{F}_Q] = 3$, and the monic minimal polynomial of y has the form*

$$Y^3 + aY^2 + bY + a^2, \quad (\text{CS}_1)$$

where $a = \text{Tr}_{\mathbb{F}_{Q^3}/\mathbb{F}_Q}(y)$.

Consequently every irreducible cubic Q_i in (DS_9) has this trace-constant shape. The selected cubic G/C^2 has

$$a = \frac{g}{C}, \quad b = \frac{1}{C^2}, \quad \frac{g^2}{C^2} = a^2. \quad (\text{CS}_2)$$

Thus its trace and constant term satisfy the universal Dickson-cubic identity; coefficient shape cannot exclude the selected factor.

Proof. The congruence $Q^3 \equiv -1 \pmod{\Lambda}$ shows that $y \in \mathbb{F}_{Q^3}$. If $y \in \mathbb{F}_Q$, equality of the inverse pairs $\{t^Q, t^{-Q}\} = \{t, t^{-1}\}$ would give $t^{Q-1} = 1$ or $t^{Q+1} = 1$. But

$$\gcd(\Lambda, Q-1) = 1, \quad \gcd(\Lambda, Q+1) = 3,$$

contrary to $t \notin \mu_3$. Hence the degree is three.

Put $y_i = t^{Q^i} + t^{-Q^i}$ for $0 \leq i < 3$, and $a = y_0 + y_1 + y_2$. Since $Q^2 \equiv Q-1 \pmod{\Lambda}$, expanding the eight terms of the product and pairing inverse exponents gives

$$\begin{aligned} y_0 y_1 y_2 &= (t^2 + t^{-2}) + (t^{2Q} + t^{-2Q}) + (t^{2Q^2} + t^{-2Q^2}) + 1 + 1 \\ &= a^2. \end{aligned} \quad (\text{CS}_3)$$

The trace and product are respectively the quadratic and constant coefficients of the monic cubic in characteristic two, proving (CS_1) . Dividing (DS_3) by C^2 gives (CS_2) . $\square$

Proposition 8.19 (the selected cubic's absolute-trace fingerprint). *Put $K_j = \mathbb{F}_{2^{3j}}$, set $\gamma_0 = 1$, and choose the compatible Gaussian periods so that $\gamma_j^3 + \gamma_j = \gamma_{j-1}$. Define*

$$\phi(X) = \frac{X}{X^2 + X + 1}, \quad \psi(X) = \frac{1}{X^2 + X + 1}.$$

Then, for every $j \geq 1$,

$$\begin{aligned} \text{Tr}_{K_j/K_{j-1}}(\phi(\gamma_j)) &= \phi(\gamma_{j-1}), \\ \text{Tr}_{K_j/K_{j-1}}(\psi(\gamma_j)) &= \phi(\gamma_{j-1}) + \psi(\gamma_{j-1}). \end{aligned} \quad (\text{CF}_1)$$

Consequently

$$\text{Tr}_{K_j/\mathbb{F}_2}(\phi(\gamma_j)) = 1, \quad \text{Tr}_{K_j/\mathbb{F}_2}(\psi(\gamma_j)) = j+1 \pmod{2}. \quad (\text{CF}_2)$$

In the notation of Proposition 8.17, this gives the selected coefficient fingerprint

$$\text{Tr}_{\mathbb{F}_{A_0}/\mathbb{F}_2}\left(\frac{g}{C}\right) = 1, \quad \text{Tr}_{\mathbb{F}_{A_0}/\mathbb{F}_2}\left(\frac{1}{C^2}\right) = k \pmod{2}. \quad (\text{CF}_3)$$

These identities distinguish two absolute coordinates of G , but the factorization (DS_9) supplies no incompatible coordinate identity; they do not by themselves exclude $G \mid D_e$.

Proof. Let x_0, x_1, x_2 be the roots of $f(T) = T^3 + T + b$, where $b = \gamma_{j-1}$, and work after the harmless scalar extension adjoining $c \in \mathbb{F}_4 \setminus \mathbb{F}_2$. The denominators below are nonzero: γ_j has odd degree 3^j , so it cannot be a nontrivial element of $\mathbb{F}_4$. The partial fractions

$$\phi(x) = \frac{c}{x+c} + \frac{c^2}{x+c^2}, \quad \psi(x) = \frac{1}{x+c} + \frac{1}{x+c^2}$$

and the logarithmic-derivative identity

$$\sum_{i=0}^2 \frac{1}{x_i + a} = \frac{f'(a)}{f(a)}$$

apply. Since

$$f(c) = b + c^2, \quad f(c^2) = b + c, \quad \text{quad} f'(c) = c, \quad f'(c^2) = c^2,$$

direct simplification gives (CF₁).

The first identity iterates to $\phi(\gamma_0) = \phi(1) = 1$. If $U_j = \text{Tr}_{K_j/\mathbb{F}_2}(\psi(\gamma_j))$, the second gives $U_j = 1 + U_{j-1}$, with $U_0 = \psi(1) = 1$; this proves (CF₂). Finally take $j = k - 1$, use $C = g^2 + g + 1$, and use invariance of the absolute trace under squaring to obtain (CF₃). $\square$

Remark 8.20 (the plus and minus current tori have disjoint orders). At the same cubic step $Q = 2^{3^{k-1}}$, the current rational-point groups in arms (C) and (D) have respective orders

$$Q^2 + Q + 1, \quad \text{quad} Q^2 - Q + 1.$$

Their gcd divides both $2Q$ and two odd integers coprime to Q , so it is one. Hence every homomorphism between these two finite rational-point groups is trivial. The common antiunit formalism does not transfer a current primary order assertion from one arm to the other; the two selected Kummer classes require separate evaluations.

Proposition 8.21 (partition regulator sieve for the exceptional arm). *Retain Proposition 8.5. Let*

$$H_k = \left\lfloor \frac{h-1}{8} \right\rfloor$$

and let $B_-(h)$ be the number of partitions into distinct positive parts congruent to 1 (mod 3) whose total is at most H_k . Then

$$\text{ord}(P_k) \geq B_-(h) = \exp\left(\left(\frac{\pi}{6\sqrt{2}} + o(1)\right) \sqrt{h}\right). \quad (\text{PR}_1)$$

Consequently, if $\ell \mid \Psi_k$ and

$$\ell > \frac{\Lambda_k}{B_-(h)}, \quad (\text{PR}_2)$$

then P_k , and hence M_k , contains the full ℓ -primary part of Λ_k in its order. Thus D'_k holds at every current prime satisfying (PR₂). This replaces the linear denominator in (D₆) by an unconditional quantity exponential in $\sqrt{h}$; current factors at or below the new threshold remain open.

Proof. The element ζ has degree h over $\mathbb{F}_4$, because $\text{ord}_{3h}(4) = h$. The subgroup generated by 4 in $(\mathbb{Z}/3h\mathbb{Z})^\times$ consists exactly of the classes congruent to 1 (mod 3). Hence, for every positive $j \equiv 1$ (mod 3) with $j < 3h$, there is z_j such that $4^{z_j} \equiv j$ (mod $3h$).

Put

$$A_j(X) = X^{4j} + X^j + \omega, \quad B_j(X) = \omega^2 X^{4j} + X^{3j} + 1.$$

Equation (D₂), followed by 4^{z_j} -Frobenius, gives

$$P_k^{4^{z_j}} = \frac{A_j(\zeta)}{B_j(\zeta)}. \quad (\text{PR}_3)$$

For a partition I counted by $B_-(h)$, set $E_I = \sum_{j \in I} 4^{z_j}$. We claim that the powers $P_k^{E_I}$ are pairwise distinct.

Suppose two of them, indexed by I and J , were equal. Cancel the common parts and cross-multiply in (PR₃). The resulting polynomial in $\mathbb{F}_4[X]$ has degree at most

$$4 \left(\sum_{j \in I} j + \sum_{j \in J} j \right) \leq 8H_k < h. \quad (\text{PR}_4)$$

It is nonzero. Indeed $A_j(0)/B_j(0) = \omega$, so different cardinalities modulo three are already separated by the constant term. After dividing each factor by ω , its expansion at zero is

$$\omega^{-1} \frac{A_j(X)}{B_j(X)} = 1 + \omega^2 X^j + O(X^{2j}). \quad (\text{PR}_5)$$

If the constant terms agree, the coefficient belonging to the least part in $I \triangle J$ therefore separates the two products. A nonzero polynomial of degree below h cannot vanish at ζ , proving the claim. The $B_-(h)$ distinct powers give the first inequality in (PR₁).

For the asymptotic, the generating function for distinct parts congruent to one modulo three is

$$\prod_{r \geq 0} (1 + x^{3r+1}).$$

Its logarithm at $x = e^{-t}$ is $\pi^2/(36t) + O(\log(1/t))$. The usual Meinardus/saddle-point estimate [17] therefore gives $\log B_-(h) \sim (\pi/3)\sqrt{H_k}$, which is the second assertion of (PR₁).

Finally P_k lies in the cyclic subgroup of order Λ_k . If its order missed the full ℓ -primary part, then P_k would lie in the subgroup of ℓ -th powers, of order Λ_k/ℓ . All the powers just constructed would lie there, so $B_-(h) \leq \Lambda_k/\ell$, contrary to (PR₂). Equation (D₁) transfers the conclusion to M_k . $\square$

8.5 Exact bad-locus exhaustions

Proposition 8.22 (semiprimitive exhaustion and the saturated planes). *Retain Proposition 8.9, put*

$$h = 3^k, \quad q = 2^h, \quad K = \mathbb{F}_{q^2}, \quad L = \mathbb{F}_{q^4},$$

and, for a prime $\ell \mid \Psi_k$, set

$$H_K = (K^\times)^\ell, \quad H_L = (L^\times)^\ell.$$

Every nontrivial order- ℓ character χ on $K^\times$ has, for the canonical additive character,

$$G_K(\chi^j) = q \quad (1 \leq j < \ell). \quad (\text{SP}_1)$$

Every order- ℓ character on $L^\times$ is the norm lift of one on $K^\times$, and quadratic Hasse–Davenport gives

$$G_L(\tilde{\chi}^j) = -q^2 \quad (1 \leq j < \ell). \quad (\text{SP}_2)$$

Consequently, whenever $a, b, a + b \not\equiv 0 \pmod{\ell}$,

$$J_K(\chi^a, \chi^b) = q, \quad J_L(\tilde{\chi}^a, \tilde{\chi}^b) = -q^2, \quad (\text{SP}_3)$$

while $J(\chi^a, \chi^{-a}) = -1$ in either field.

Failure of D'_k at ℓ has the following two equivalent additive forms:

$$W := \mathbb{F}_{16} + \mathbb{F}_{16}\zeta, \quad W \setminus \{0\} \subset H_L, \quad (\text{SP}_4)$$

and

$$V := \mathbb{F}_4 + \mathbb{F}_4\zeta + \mathbb{F}_4\zeta^2, \quad V \setminus \{0\} \subset H_K. \quad (\text{SP}_5)$$

Thus a bad translate saturates an eight-dimensional $\mathbb{F}_2$ -space in L , and its norm-down form saturates a six-dimensional $\mathbb{F}_2$ -space in K , not merely the original affine line.

The exact semiprimitive Fourier equations nevertheless permit both planes. If $W^\perp$ is the additive annihilator of W , put

$$A_L = |(W^\perp \setminus \{0\}) \cap H_L|, \quad R = q^2, \quad w = 256.$$

Then orthogonality and (SP_2) force

$$A_L = \frac{(R-1)(R+w) - \ell R(w-1)}{\ell w}, \quad |W^\perp \setminus (H_L \cup \{0\})| = \frac{(R-1)(R+w)(\ell-1)}{\ell w}. \quad (\text{SP}_6)$$

Likewise, for $w_0 = 64$,

$$A_K := |(V^\perp \setminus \{0\}) \cap H_K| = \frac{(q+1)(q-w_0) + \ell q(w_0-1)}{\ell w_0}, \quad (\text{SP}_7)$$

and

$$|V^\perp \setminus (H_K \cup \{0\})| = \frac{(\ell-1)(q+1)(q-w_0)}{\ell w_0}. \quad (\text{SP}_8)$$

All four quantities are nonnegative integers with the required total sizes. Hence the complete semiprimitive Gauss and Jacobi table, even together with the dimensions of the saturated spaces, gives no contradiction. A proof of D'_k must use the distinguished cyclotomic placement of W or V , not only the semiprimitive character table or additive-subspace incidence.

Proof. Since $\text{ord}_\ell(2) = 2h$, one has $q \equiv -1 \pmod{\ell}$. Thus every order- ℓ character of $K^\times$ is trivial on $\mathbb{F}_q^\times$. Decompose $K^\times$ into its $q+1$ projective $\mathbb{F}_q$ -lines. On the trace-zero line $\mathbb{F}_q$, the additive sum is $q-1$; on every other line it is -1 . The induced nontrivial projective character sums to zero, giving $G_K(\chi) = q$. The same argument applies to every nontrivial power. The norm induces an isomorphism on the order- ℓ character groups, so Hasse–Davenport proves (SP_2) . The standard quotient formula for Jacobi sums proves (SP_3) , with the inverse case evaluated separately.

If D'_k fails, Proposition 8.9 puts every $\zeta + c$, $c \in \mathbb{F}_{16}$, in H_L . Since $\mathbb{F}_{16}^\times \subset H_L$, every nonzero $a\zeta + b$ is an ℓ -th power, proving (SP_4) ; the converse follows by taking $a = 1, b = u$.

For (SP_5) , every polynomial of degree at most two over $\mathbb{F}_4$ is a product of linear factors or is an irreducible quadratic. In the first case its value at ζ is an ℓ -th power by (SP_4) . In the second, choose a root $c \in \mathbb{F}_{16} \setminus \mathbb{F}_4$; its value is

$$(\zeta + c)(\zeta + c^4) = N_{L/K}(\zeta + c),$$

again an ℓ -th power. Conversely the corrected norm $N = \zeta^2 + \zeta + \omega$ belongs to V ; membership in H_K is exactly failure of D'_k .

Finally the two additive Fourier values of H_L are

$$\frac{-1 - R(\ell - 1)}{\ell} \quad \text{on } H_L, \quad \frac{R - 1}{\ell} \quad \text{off } H_L,$$

by (SP₂). Fourier inversion applied to $|W \cap H_L| = w - 1$ gives (SP₆). The corresponding values for H_K are

$$\frac{-1 + q(\ell - 1)}{\ell} \quad \text{on } H_K, \quad \frac{-1 - q}{\ell} \quad \text{off } H_K,$$

and $|V \cap H_K| = w_0 - 1$ gives (SP₇)–(SP₈). Integrality follows from $\ell \mid q + 1$, $w \mid q^2$, and $w_0 \mid q$; nonnegativity is immediate from $\ell \leq q + 1$ and $q \geq 2^9$. $\square$

Proposition 8.23 (cyclotomic Dirichlet coefficient and subspace-polynomial exhaustion). *Retain $k \geq 2$, put $h = 3^k$, and write*

$$K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \quad \zeta^h = \omega.$$

Let $\ell \mid \Psi_k$, let $\chi : K^\times \rightarrow \mu_\ell$ have exact order ℓ , realized complexly, and let $\tilde{\chi} = \chi \circ \text{Norm}_{\mathbb{F}_{16^h}/K}$. Put

$$s = h/3, \quad A = 2^s \pmod{\ell},$$

choose $u \in \mathbb{F}_{16}$ of order fifteen with $u^5 = \omega$, and set $t = \tilde{\chi}(\zeta + u)$. For $0 \leq r < h$, define

$$\mathcal{A}_r(\chi) = \sum_{\substack{f \in \mathbb{F}_4[X] \text{ monic} \\ \deg f = r}} \chi(f(\zeta)).$$

Then

$$\mathcal{A}_0 = 1, \quad \mathcal{A}_1 = 4, \quad \mathcal{A}_2 = 10 + \sum_{j=0}^5 t^{A_j}. \quad (\text{DP}_1)$$

Failure of D'_k at ℓ is equivalent to the single extremal coefficient

$$\mathcal{A}_2 = 16. \quad (\text{DP}_2)$$

If $\mathcal{A}'_1$ is the degree-one coefficient after the quadratic constant-field extension to $\mathbb{F}_{16}$, then

$$\mathcal{A}'_1 = 2\mathcal{A}_2 - \mathcal{A}_1^2 = 2\mathcal{A}_2 - 16. \quad (\text{DP}_3)$$

Thus the sixteen-term $\mathbb{F}_{16}$ -line extremum and the six-term irreducible-quadratic extremum are the same constraint under constant-field base change, not two independent conditions.

More precisely, let

$$L_\chi(T) = \sum_{r=0}^{h-1} \mathcal{A}_r T^r = (1 - T)L_\chi^*(T), \quad L_\chi^*(T) = \sum_{r=0}^d b_r T^r, \quad d = h - 2,$$

so $b_r = \sum_{j=0}^r \mathcal{A}_j$. The functional equation has root number $+1$ and gives

$$b_{d-r} = 2^{d-2r} \overline{b_r}. \quad (\text{DP}_4)$$

If D'_k fails, then

$$(b_0, b_1, b_2) = (1, 5, 21), \quad b_{d-2} = 21 \cdot 2^{h-6}. \quad (\text{DP}_5)$$

For

$$V = \langle 1, \zeta, \zeta^2 \rangle_{\mathbb{F}_4},$$

the distinguished trace annihilator is

$$V^\perp = \zeta \{f(\zeta) : f \in \mathbb{F}_4[X], \deg f \leq h-4\}. \quad (\text{DP}_6)$$

Consequently, in the bad case,

$$\sum_{x \in V^\perp} \chi(x) = 3b_{d-2} = 63 \cdot 2^{h-6}, \quad (\text{DP}_7)$$

exactly the value forced independently by additive Poisson summation and $G_K(\chi) = 2^h$. Thus the first two Dirichlet coefficients, the exact cyclotomic trace-dual placement, the semiprimitive Gauss sign, the functional equation, and the Riemann hypothesis are mutually compatible with failure.

Finally put

$$D = \zeta^4 + \zeta, \quad E = D^5(D^3 + 1).$$

The $\mathbb{F}_4$ -subspace polynomial of V is

$$\begin{aligned} P_V(Z) &= Z^{64} + ((1 + D^3)^4 + E^3)Z^{16} \\ &\quad + (D^{12} + E^3(1 + D^3))Z^4 + E^3D^3Z. \end{aligned} \quad (\text{DP}_8)$$

Both D and E are ℓ -th powers unconditionally. Hence the derivative coefficient E^3D^3 supplies no missing nonresidue; a subspace-polynomial proof would have to use an additive relation among the Z^{16} - and Z^4 -coefficients rather than merely products of the saturated elements.

Proof. The four monic linears $X + c$, $c \in \mathbb{F}_4$, evaluate to ℓ -th powers, so $\mathcal{A}_1 = 4$. Of the sixteen monic quadratics over $\mathbb{F}_4$, ten are reducible and therefore also have character value one. The six irreducibles are

$$f_c(X) = (X + c)(X + c^4),$$

indexed by the six Frobenius pairs in $\mathbb{F}_{16} \setminus \mathbb{F}_4$. Since

$$\tilde{\chi}(\zeta + c) = \chi(f_c(\zeta)),$$

the orbit table of Proposition 8.9 gives their six values exactly t^{A^j} , $0 \leq j < 6$. This proves (DP₁). The six summands have modulus one, so $\mathcal{A}_2 = 16$ if and only if all are one; the $j = 0$ term then gives $t = 1$, which is exactly failure of D'_k .

The reciprocal roots of L_χ are squared under quadratic constant-field extension. Newton's identity therefore gives

$$\mathcal{A}'_1 = 2\mathcal{A}_2 - \mathcal{A}_1^2,$$

proving (DP₃).

The character is primitive modulo $X^h + \omega$ and even because it is trivial on $\mathbb{F}_4^\times$. Standard function-field Dirichlet theory gives $L_\chi = (1 - T)L_\chi^*$, with $\deg L_\chi^* = d = h - 2$, and the coefficient relation in (DP₄) up to its root number. We determine that sign directly. Let U be the $\mathbb{F}_4$ -space of polynomials in ζ of degree at most $h - 2$. Newton sums for $X^h + \omega$ give

$$\text{Tr}(\zeta^r) = 0 \quad (1 \leq r < h),$$

so $U^\perp = \mathbb{F}_4\zeta$. Additive Poisson summation and $\chi(\mathbb{F}_4^\times\zeta) = 1$ give

$$3b_d = \sum_{x \in U} \chi(x) = \frac{3}{4}G_K(\chi).$$

Proposition 8.22 gives $G_K(\chi) = 2^h$, so

$$b_d = 2^{h-2} = 2^d.$$

The root number is therefore $+1$. Under failure, $b_2 = 1 + 4 + 16 = 21$, and (DP_4) at $r = 2$ gives (DP_5) .

For $x = \sum_{j=0}^{h-1} x_j \zeta^j$, the same Newton-sum calculation gives

$$\text{Tr}(x) = x_0, \quad \text{Tr}(x\zeta) = \omega x_{h-1}, \quad \text{Tr}(x\zeta^2) = \omega x_{h-2}.$$

This proves (DP_6) . Every nonzero polynomial of degree at most $h-4$ is a unique $\mathbb{F}_4^\times$ -multiple of a monic one, while $\chi(\zeta) = 1$. Hence

$$\sum_{x \in V^\perp} \chi(x) = 3b_{h-4} = 3b_{d-2}.$$

Direct Poisson summation under $V \setminus \{0\} \subset \ker \chi$ gives $(63/64)G_K(\chi)$, the same value $63 \cdot 2^{h-6}$. This proves (DP_7) .

It remains to derive the subspace polynomial. First put $U_1 = \langle 1, \zeta \rangle_{\mathbb{F}_4}$. Since $\prod_{c \in \mathbb{F}_4} (Z + c) = Z^4 + Z$,

$$\begin{aligned} P_{U_1}(Z) &= (Z^4 + Z)^4 + D^3(Z^4 + Z) \\ &= Z^{16} + (1 + D^3)Z^4 + D^3Z. \end{aligned}$$

Evaluating at ζ^2 gives

$$E = P_{U_1}(\zeta^2) = D^8 + D^5 = D^5(D^3 + 1).$$

Because P_{U_1} is $\mathbb{F}_4$ -linearized,

$$P_V(Z) = P_{U_1}(Z)^4 + E^3 P_{U_1}(Z),$$

which expands to (DP_8) .

Finally,

$$D = \prod_{c \in \mathbb{F}_4} (\zeta + c),$$

so $\chi(D) = 1$. Also E is the product of all sixteen monic-quadratic values. The ten reducible values have character one, and the other six multiply to

$$t^{\sum_{j=0}^5 A^j} = 1$$

because A has order six modulo ℓ . Thus $\chi(E) = 1$ without assuming failure, which finishes the proof. □

Proposition 8.24 (relative Dickson form and its degree ceiling). *Retain*

$$s = 3^{k-1}, \quad a = 2^s, \quad E = \mathbb{F}_a, \quad K = \mathbb{F}_{a^6}, \quad \Lambda_k = a^2 - a + 1,$$

and set $T_k = \{x \in K^\times : x^{\Lambda_k} = 1\}$. The element $P_k \in T_k$ has degree six over E . Put

$$Y_k = P_k + P_k^{-1} \in \mathbb{F}_{a^3}, \quad G_k(T) = \prod_{j=0}^2 (T + Y_k^{a^j}) \in E[T].$$

Then G_k is irreducible of degree three and $X^3 G_k(X + X^{-1})$ is the irreducible reciprocal minimal polynomial of P_k over E . If $\mathcal{D}_\ell$ is characterized by

$$\mathcal{D}_\ell(v + v^{-1}) = v^\ell + v^{-\ell},$$

then, for every prime $\ell \mid \Psi_k$,

$$P_k \notin T_k^\ell \iff G_k(\mathcal{D}_\ell(X)) \text{ is irreducible over } E. \quad (\text{RD})$$

In the bad case the composition splits over $\mathbb{F}_{a^3}$, and all its irreducible factors over E have degree dividing three. Moreover

$$T_k \cap \mathbb{F}_{a^2}^\times = \mu_3, \quad T_k \cap \mathbb{F}_{a^3}^\times = \{1\}.$$

Thus the elements of T_k of relative degree below six are exactly μ_3 , while the bad subgroup T_k^ℓ contains exactly

$$\frac{\Lambda_k}{\ell} - 3$$

elements of exact degree six. Except in the already settled case $\Lambda_k/\ell = 3$, full relative degree and reciprocal-torus structure cannot prove (D'_k) .

Proof. Proposition 8.5 gives $[\mathbb{F}_2(P_k) : \mathbb{F}_2] = 6s$, so $E(P_k) = K$. Since $a^3 + 1 = (a + 1)\Lambda_k$, one has $P_k^{a^3} = P_k^{-1}$, whence $Y_k \in \mathbb{F}_{a^3}$. The element P_k is quadratic over $E(Y_k)$; its relative degree six forces $[E(Y_k) : E] = 3$, proving the minimal-polynomial assertions.

Suppose $P_k \notin T_k^\ell$. The whole current ℓ -primary part of $K^\times$ lies in T_k , so this is equivalent to $P_k \notin (K^\times)^\ell$. Kummer theory and Capelli make

$$X^{3\ell} G_k(X^\ell + X^{-\ell})$$

irreducible of degree 6ℓ . If $\alpha^\ell = P_k$ and $t = \alpha + \alpha^{-1}$, then $\mathcal{D}_\ell(t) = Y_k$. Since α is at most quadratic over $E(t)$, the degree of t is at least 3ℓ ; it is a root of the degree- 3ℓ polynomial $G_k(\mathcal{D}_\ell(X))$, proving the forward implication in (RD).

Conversely, if $P_k = \alpha^\ell$ with $\alpha \in T_k$, then $\alpha^{a^3} = \alpha^{-1}$, so $t \in \mathbb{F}_{a^3}$. The same holds after multiplying α by any ℓ -th root of unity and after applying the three E -conjugates. Hence every root of the Dickson composition lies in $\mathbb{F}_{a^3}$, proving the bad-case factorization statement and the reverse implication.

Finally the two displayed intersections have orders

$$\gcd(\Lambda_k, a^2 - 1) = 3, \quad \gcd(\Lambda_k, a^3 - 1) = 1.$$

Every proper intermediate field of K/E lies in $\mathbb{F}_{a^2}$ or $\mathbb{F}_{a^3}$, so only μ_3 has relative degree below six. The cyclic group T_k^ℓ has order Λ_k/ℓ and contains μ_3 , because $\ell \neq 3$. Removing those three elements gives the asserted exact count. $\square$

Proposition 8.25 (trace-degree ceiling for the fibotomic criterion). *Retain Proposition 8.6, and let*

$$\mathcal{I}_\ell := \pi(U^\ell \setminus \{1\}) \subseteq \mathbb{F}_q \quad (\ell \text{ a prime divisor of } \Psi_k).$$

Every element of $\mathcal{I}_\ell$ has absolute trace one, and

$$|\mathcal{I}_\ell| = \frac{(q+1)/\ell - 1}{2} \geq \frac{3a+2}{2} > a. \quad (\text{T})$$

At least $a/2 + 1$ elements of $\mathcal{I}_\ell$ have exact degree h over $\mathbb{F}_2$. Consequently, absolute trace one together with full degree h cannot prove that the distinguished Z lies outside $\pi(U^\ell)$.

Proof. The two-to-one trace-one parametrization in Proposition 8.6 restricts to U^ℓ . This cyclic subgroup has order $(q+1)/\ell$, which is odd, so its only self-inverse element is 1. This proves the equality in (T). Since $\ell \leq \Psi_k = \Lambda_k/3$ and $q+1 = (a+1)\Lambda_k$, it also gives

$$|\mathcal{I}_\ell| \geq \frac{3(a+1) - 1}{2} = \frac{3a+2}{2}.$$

Every proper subfield of $\mathbb{F}_q$ lies in $\mathbb{F}_{2^{h/3}} = \mathbb{F}_a$, because $h = 3^k$. Hence at most a elements of $\mathcal{I}_\ell$ have degree smaller than h , leaving at least $(3a+2)/2 - a = a/2 + 1$ of exact degree h .

For completeness, the distinguished Z itself has exact degree h . Indeed, (D_3) and $P_k \in \mathbb{F}_2(M_k)$ force $[\mathbb{F}_2(M_k) : \mathbb{F}_2] = 2h$. The element M_k is quadratic over $\mathbb{F}_2(Z)$, satisfying

$$X^2 + Z^{-1}X + 1 = 0,$$

while $Z \in \mathbb{F}_q$. Therefore $[\mathbb{F}_2(Z) : \mathbb{F}_2] = h$. □

8.6 The limits of norm induction

Proposition 8.26 (all proper-subfield norms are blind). *Retain $k \geq 2$ and the notation of Proposition 8.3. Let ℓ be a prime divisor of Ψ_k , and let d be any proper divisor of $2h$. Then*

$$N_{\mathbb{F}_{2^{2h}}/\mathbb{F}_{2^d}}$$

annihilates the entire ℓ -Sylow subgroup of $\mathbb{F}_{2^{2h}}^\times$. Equivalently, multiplying an element by any current ℓ -primary component leaves every multiplicative norm to a proper subfield unchanged.

Proof. The norm exponent is

$$E_d = \frac{2^{2h} - 1}{2^d - 1}.$$

Since $\text{ord}_\ell(2) = 2h$ and $d < 2h$, one has $\ell \nmid 2^d - 1$. Therefore

$$v_\ell(E_d) = v_\ell(2^{2h} - 1),$$

so exponentiation by E_d kills the full ℓ -Sylow subgroup. □

For $k = 1$, $\Psi_1 = 1$ and D'_1 is vacuous. Exact finite evidence for the subsequent levels, together with its factorization and primality dependencies, is recorded in Appendix A.4.

Proposition 8.27 (twisted norm tower). *With $\eta = \zeta^3 = \kappa_{3^{k-1}}$,*

$$N_{\mathbb{F}_{2^{2 \cdot 3^k}}/\mathbb{F}_{2^{2 \cdot 3^{k-1}}}}(N_k) = \eta^2 + \omega^2 \eta + 1,$$

which is neither $N_{k-1} = \eta^2 + \eta + \omega$, nor one of its Frobenius conjugates (their exponent pattern is $(2^{j+1}, 2^j)$), nor a scaling $N_{k-1}(\lambda\eta)$ with $\lambda \in \mu_3$.

Proof. The generator τ of the degree-3 step sends $\zeta \mapsto \zeta^c$ with $c = 2^{2 \cdot 3^{k-1}}$ a primitive cube root of unity modulo 3^{k+1} ; the cube roots are $1 + a3^k$, so $\tau(\zeta) = \omega^a \zeta$ ($a \neq 0$) while $\tau(\omega) = \omega$. Hence the norm is $\prod_{r^3=\eta} (r^2 + r + \omega)$ over the three cube roots r of η . Writing α, β for the two roots of $T^2 + T + \omega$ (so $\alpha + \beta = 1$, $\alpha\beta = \omega$), each factor is $(r + \alpha)(r + \beta)$, and the product collapses by the resolvent

$$\prod_{r^3=\eta} (r^2 + r + \omega) = (\alpha^3 + \eta)(\beta^3 + \eta) = \eta^2 + (\alpha^3 + \beta^3)\eta + (\alpha\beta)^3 = \eta^2 + \omega^2\eta + 1,$$

since $\alpha^3 + \beta^3 = (\alpha + \beta)^3 + \alpha\beta(\alpha + \beta) = 1 + \omega = \omega^2$ and $(\alpha\beta)^3 = \omega^3 = 1$. Machine-checked at $k = 2, 3, 4$. $\square$

Proposition 8.28 (trace form and norm blindness). *Retain the notation of Proposition 8.3, and put*

$$\gamma_k := \zeta + \zeta^{-1}, \quad A_k := \gamma_k^2 + \gamma_k + 1, \quad Q := 2^{2 \cdot 3^{k-1}} = 2^{2h/3}.$$

Set $\gamma_0 = A_0 = 1$. Then, with $\overline{N} = N^{2^h}$,

$$N + \overline{N} = A_k, \quad N\overline{N} = A_k^Q, \quad M_k + M_k^{-1} = A_k^{2-Q}, \quad Z = A_k^{Q-2}. \quad (4)$$

Moreover

$$N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3^{k-1}}}}(A_k) = A_{k-1}, \quad N_{\mathbb{F}_{2^{3k}}/\mathbb{F}_{2^{3^{k-1}}}}(Z) = A_{k-1}^{-1}. \quad (5)$$

For every prime $\ell \mid \Psi_k$, however, this relative norm is blind to the required ℓ -power residue class.

Proof. Since $2^h \equiv -1 \pmod{3h}$ and $\zeta^h + \zeta^{2h} = 1$, direct expansion gives

$$N + \overline{N} = \zeta^2 + \zeta^{-2} + \zeta + \zeta^{-1} + 1 = A_k.$$

The congruence $Q \equiv h + 1 \pmod{3h}$ follows inductively by cubing $1 + 3^r$ modulo 3^{r+2} . Hence $\gamma_k^Q = \zeta^{h+1} + \zeta^{-(h+1)}$, and another direct expansion yields $N\overline{N} = A_k^Q$. Therefore

$$M_k + M_k^{-1} = \frac{(N + \overline{N})^2}{N\overline{N}} = A_k^{2-Q},$$

and $Z = \pi(M_k) = 1/(M_k + M_k^{-1})$, which proves (4).

Over $B = \mathbb{F}_{2^{h/3}}$, the element γ_k has minimal polynomial $X^3 + X + \gamma_{k-1}$. Base-change this cubic extension by $B_2 = B\mathbb{F}_4 = \mathbb{F}_{2^{2h/3}}$; its compositum is $\mathbb{F}_{2^{2h}}$, and the degree remains three. With $\omega = \zeta^h \in \mathbb{F}_4$ one has $A_k = (\gamma_k + \omega)(\gamma_k + \omega^2)$. Evaluating that minimal polynomial at ω and ω^2 gives

$$N_{\mathbb{F}_{2^{2h}}/B_2}(\gamma_k + \omega) = \gamma_{k-1} + \omega^2, \quad N_{\mathbb{F}_{2^{2h}}/B_2}(\gamma_k + \omega^2) = \gamma_{k-1} + \omega.$$

Their product is A_{k-1} . Since norm commutes with this scalar extension, this is precisely $N_{\mathbb{F}_{2^h}/B}(A_k)$, proving the first identity in (5). Furthermore $Z = A_k^{Q-2}$, so

$$N_{\mathbb{F}_{2^h}/B}(Z) = A_{k-1}^{Q-2} = A_{k-1}^{-1},$$

because $Q - 2 \equiv -1 \pmod{2^{h/3} - 1}$. This proves the second identity.

If $\ell \mid \Psi_k$, then $\ell \nmid 2^h - 1$ and $\ell \nmid 2^{h/3} - 1$. Thus both A_k and A_{k-1} lie in multiplicative groups on which raising to the ℓ -th power is an automorphism, so (5) is tautological on ℓ -power classes. Equivalently, $\text{ord}_\ell(Q) = 3$ and $1 + Q + Q^2 \equiv 0 \pmod{\ell}$: the degree-three norm annihilates the relevant character. The surviving current-level condition is the Capelli test $R_k(X^\ell)$ irreducible from Proposition 8.3. $\square$

So, unlike the γ tower of Proposition 6.7, old-prime full-order parts do *not propagate by that argument*, and D'_k is genuinely per-level for the present method: the certification at each k stands on its own. Structurally, at the degree-3 step with $Q = 2^{2 \cdot 3^{k-1}}$, the relative norm kernel has order

$$Q^2 + Q + 1 = \Phi_{3^k}(2)\Phi_{2 \cdot 3^k}(2).$$

Its gcd with $Q - 1$ is 3, so there is no direct-product splitting at the 3-part as in Proposition 6.8; nevertheless, for every $\ell \neq 3$ the new ℓ -Sylow subgroups of both columns lie entirely in the norm kernel. This is the prime-to-3 analogue of Proposition 6.8 and explains why a lower-level norm identity cannot determine D'_k .

Remark 8.29 (the 3-part). Proposition 8.5 and $v_3(a + 1) = k$ imply

$$v_3(\text{ord } M_k) = k + 1 \quad (k \text{ odd}), \quad v_3(\text{ord } M_k) \leq k \quad (k \text{ even}).$$

Thus the odd-level full 3-part is now explained analytically. **[consistent]** At even levels $2 \leq k \leq 6$, the sharper observed value is $v_3(\text{ord } M_k) = k - 1$; the remaining one-step deficiency is not determined by the projection theorem.

8.7 Open problem D

The exceptional arm is precisely Conjecture 8.4: for each k , the distinguished antiunit M_k must contain the full current-level factor Ψ_k in its order. The projection and partition-regulator theorems settle explicit large primary factors, while the translated-binomial, semiprimitive, Dirichlet, Dickson, and norm calculations prove that the remaining smaller factors survive all present generic criteria. What remains is an arithmetic evaluation of this distinguished cyclotomic Kummer class, not a further degree or norm identity.

9 Consequences and exact frontier

9.1 Boundedness

Lenstra conjectured that m_p is globally bounded [13]. His unconditional rules give positive excess on singleton odd component sets and the lower bound $m_p \geq 4$ when $f(p) = 2 \cdot 3^k$, but do not give an absolute upper bound. The four-arm equivalence of Theorem 1.2 shows exactly what would settle the question: Conjecture 1.1 implies

$$m_p \in \{0, 1, 4\} \quad \text{for every odd prime } p.$$

The theorem also explains why a generic finite-field argument is insufficient. Proposition 5.17 constructs arbitrarily large exact-degree elements whose translates in any fixed finite window are all p -th powers. Any proof must therefore use the distinguished Conway-tower identities.

The available uniform results are substantial but incomplete. The zero arm is proved for primitive support and completely at the first two power-of-two two-spine levels $h = 12, 24$; the ordinary odd-spine arm has projective reductions and a signed conjugate-ball sieve; the cubic and exceptional arms have explicit partition-regulator bounds. Each leaves smaller current primary factors untreated. Consequently neither global boundedness nor the nonexistence of an excess $m_p \geq 5$ is presently proved.

9.2 The four remaining theorem targets

The proof frontier is most cleanly stated in the same four coordinates as Theorem 1.2.

A. Zero arm. There are two exact maximal-order problems. At a singleton-even level, prove

$$\delta_n = F_n$$

in Proposition 7.1. The fibotomic recursion, elliptic correspondence, and equivariant five-point generalized Jacobian reduce this to the nonvanishing of one explicit distinguished Kummer value; they also prove that the available trace, norm, and boundary-character data are circular. Proposition 7.11 pushes the minimal-counterexample hypothesis through unique ℓ -th roots: every lower trace, norm, and maximal-order identity survives automatically, while top failure is exactly one Dickson power-sum equation. Proposition 7.12 proves that its Dickson polynomial has no intermediate factorization: failure is complete splitting and success is irreducibility, with the same Kummer scalar appearing in the exact resultant. Proposition 7.13 then transfers failure to the selected divisibility $\mathcal{P}_{\ell,n} \mid H_m$: the absolute norm of a reverse-Dickson polynomial, containing the actual minimal polynomial of a_{n-1} , must divide the trace-one polynomial. Its factorization is one irreducible of degree $m\ell$ on success and ℓ irreducibles of degree m on failure. Remark 7.16 proves that the resulting degree and coefficient-gap constraints remain compatible with every composite Fermat case. Proposition 7.14 identifies H_m with the inversion quotient of the norm-one torus and $P_{\ell,a}$ with one transported μ_ℓ -coset. Remark 7.15 gives the exact fibotomic strata of that coset and proves their capacity constraints automatic. Thus the selected divisor is sharper but still unevaluated: additive and support data are realized by many nonselected cosets. For multicomponent zero sets, Proposition 4.6 proves that the Conway class is nontrivial in the primitive-support quotient under its stated two-spine hypotheses, but generation of a composite quotient remains open. Remark 4.7 records the exact Hilbert–90 representative and why it yields no order bound by itself.

B. Ordinary odd-spine arm. Evaluate the selected power-residue class $\Theta_{q,s}$ of Proposition 5.1 and prove its full projective order. This includes the 11-chain and the 23, 29, 47, 179, $\dots$ components. The relative-unit formulation and Proposition 5.16 prove every primary factor above an explicit signed-ball threshold, so the unresolved content is confined to the smaller factors. The source-pinned $m_{359} = 1$ row is a concrete instance whose independent trace evaluation remains open. Remark 5.11 shows that the unweighted orbit product in global reciprocity still cancels over the $p - 1$ lifts for every selector from the base cyclotomic field. Proposition 5.12 proves that pure order- p selectors also have zero weight. The mixed Jacobi family escapes both cancellations, but Corollary 5.14 identifies the target with one inverse-Fourier coordinate of its first local phase. Proving that coordinate nonzero remains the theorem. Proposition 5.8 gives an equivalent characteristic-two boundary: failure means that every nonconstant binary section of $(1 + x)^{\mathcal{L}_{r,a}/p}$ vanishes at x_{a-1} , or, after descent, that the actual Conway minimal polynomial P_{α_r} divides every selected section polynomial $R_{e,c}$. Corollary 5.9 proves that norm and complement symmetry are automatic under this pattern. The remaining theorem is an explicit selected-polynomial nondivisibility, not a generic binomial-step statement. Proposition 5.19 and Remark 5.20 prove this distinction sharp: the simultaneous divisibilities occur in arbitrarily many full-degree irreducible factors while preserving the complete r -power ancestry, smooth degree, multiplicative orders, and full p -primary test. They also occur among scalings inside any fixed bottom Kummer coset. Only the actual Conway scaling $c = 1$ remains unevaluated.

C. *Cubic arm.* Prove C_k for every k : equivalently, for each $\ell^a \mid \Phi_{3^k}(2)$, prove

$$v_\ell(\text{ord } \gamma_k) = a.$$

The norm tower propagates all old factors, and Proposition 6.21 settles primary factors above its explicit threshold. The derivative regulator, secant saturation, and reciprocity calculations all recover the same one-dimensional Kummer class without evaluating it, while Proposition 6.10 proves that the coefficient recursion does not preserve primitivity even when it preserves irreducibility. Proposition 6.27 makes the remaining assertion purely combinatorial in the cyclic group algebra: the binary subset sums of $(2^{3^k} - 1)/\ell$ cannot have parity one at every residue except a single prescribed residue. This universal inequality, equivalently direct Kummer non-vanishing at the smaller current factors, remains open. Corollary 6.28 shows that both the full Frobenius product and the apparent ℓ -th-power exponent test are unconditional: failure is an extra proper-subproduct relation, so those identities alone cannot exclude it. Proposition 6.29 compresses that relation to one reciprocal unit and proves that mask complementation forces the prescribed exceptional residue rather than contradicting it. Proposition 6.30 gives the remaining exact factor-selection problem: failure is the occurrence of the distinguished minimal polynomial of η_k , together with its reciprocal translate, in a binary Wendt gcd. Every Wendt factor already has the required degree and a translation partner, so a closure must exclude this selected pair rather than all such factors. Proposition 6.31 factors the gcd through Singer coset intersections and gives their exact first two moments. In the range $\ell - 1 < \sqrt{q} + 1/\sqrt{q}$, every coset contains degree- n Wendt factors, proving that generic factor exclusion is impossible there. Proposition 6.32 descends the selected pair to the degree- 3^{k-1} minimal polynomial of $\eta_{k-1}^2 + \eta_{k-1} + 1$; deciding whether that trace fiber lies in the chosen multiplicative coset is the same Kummer evaluation. Corollary 6.33 identifies this quotient selector with γ_{k-1}^{-2} , gives its irreducible cubic recursion, and proves the exact order seam

$$\text{ord}(z_k) = \text{ord}(z_{k-1}) \text{ord}(\eta_k).$$

Thus the descent is a lossless reciprocal form of C_k , not an induction from C_{k-1} . Proposition 6.34 makes the last coordinate one extremal value $\Theta_\chi(\eta_{k-1}) = 3$. Other Singer trace fibres genuinely attain 3, so uniform character bounds cannot close the selected fibre.

D. *Exceptional arm.* Prove D'_k for every k , equivalently

$$\Psi_k \mid \text{ord}(M_k)$$

in Conjecture 8.4. Proposition 8.3 gives Capelli and resultant formulations, while Proposition 8.21 treats the large factors. The antiunit, translated-binomial, semiprimitive, Dirichlet, Dickson, and norm-blindness results show precisely why the remaining smaller factors are not consequences of the present generic identities. Proposition 8.12 adds an exact first-order boundary: failure is simultaneous π^2 -flatness of all mixed Jacobi sums of conductor $15h$. Ruling out that simultaneous flatness is still equivalent to evaluating the selected local phase. Proposition 8.13 gives the parallel characteristic-two boundary: failure is a complement-of-zero sub-mask parity pattern modulo $15h$, while its ℓ -th-power consequence is an unconditional idempotent identity. Proposition 8.14 splits the exponent into two Frobenius-related blocks and proves that the resulting quadratic norm equation, its Hilbert–90 condition, and its involution-fixed coefficients are all compatible with failure. The exact surviving coordinate is Proposition 8.15: the selected residue symbol must have nonzero relative trace. Proposition 8.17 then identifies

the phase with one explicit reciprocal sextic over $\mathbb{F}_{2^{3k-1}}$. Equivalently, failure is the selected divisibility

$$C^2Y^3 + gCY^2 + Y + g^2 \mid D_{(2^{2 \cdot 3^{k-1}} - 2^{3^{k-1}} + 1)/\ell}(Y).$$

The Dickson polynomial factors completely as $Y(Y+1)^2$ times squares of irreducible cubics. Thus degree, reciprocity, and factor shape admit many false positives. Proposition 8.18 proves more: every admissible cubic has constant coefficient equal to the square of its trace coefficient, exactly as G/C^2 does. Proposition 8.19 evaluates the two absolute traces

$$\mathrm{Tr}(g/C) = 1, \quad \mathrm{Tr}(C^{-2}) = k \pmod{2},$$

but the ambient factorization supplies no contrary coordinate identity. Remark 8.20 also shows that the cubic and exceptional current groups have coprime orders, so no rational-point homomorphism can transfer arm (C) to arm (D) . The remaining theorem is precisely exclusion of the recursively selected cubic.

Appendix A records finite computations, external factorization dependencies, and implementation provenance. Those data are valuable for falsification and audit, but none is used to promote a uniform conjecture to a theorem.

A Computational and implementation evidence

This appendix separates exact finite computation from theorem and from external data. A row is called *locally certified* only when the checkout contains enough exact arithmetic and primality information to verify it without trusting a probable-prime test or an external database. A *source-assisted* row has exact local field arithmetic but relies on an identified external primality or factorization record. A *consistent* row verifies only the currently known factors and does not control an unfactored cofactor. None of these finite labels asserts a uniform theorem in the level.

A.1 Source-pinned tower data

The implementation vendors the complete OEIS A380496 b-file for the 126 odd primes $3 \leq p \leq 709$ and diffs it in full against the source-pinned copy. These integer rows determine the implemented Kummer carries below the stated boundary; selected resulting ordinals receive independent value tests. In particular $m_{359} = 1$ is a source-pinned table value. Section 5 isolates the independent analytical trace whose unevaluated value would reprove that row. The next unsupported carry is α_{719} , and the implementation refuses to cross it.

The contributor-linked extended A380496 auxiliary table [18, 20] also records $m_{1093} = m_{3511} = 0$. These are the two known base-two Wieferich rows: $v_{1093}(2^{364} - 1) = v_{3511}(2^{1755} - 1) = 2$. The external calculator applies Lenstra’s exact exponent $(2^E - 1)/p$, so each zero asserts the full p^2 -primary order, not merely divisibility by p . The two rows lie outside the approved 126-term b-file, however, and the external repository stores no compact nonidentity remainder certificate for them. They are therefore source-assisted external computations, not locally certified rows or analytical instances of Proposition 4.1.

A.2 Independent small-row oracle

`experiments/ordinal_excess_probe.py` is a local term-algebra oracle independent of the Rust production tower. It implements the impartial term algebra used by the calculators, multiplicative-

order tests for small component fields, and fixed-base root tests for larger selected rows. Its audited sample is:

p	m	$\mathcal{Q}$	p -th root?
7	0	$\{3\}$	yes
7	1	$\{3\}$	no
19	1	$\{9\}$	yes
19	4	$\{9\}$	no
73	1	$\{9\}$	no
89	1	$\{11\}$	no
47	1	$\{23\}$	no
179	1	$\{89\}$	no

These are exact finite checks, not evidence for an all-level maximal-order theorem.

A.3 Cubic finite evidence

Proposition 6.38 proves C_1, C_2, C_3 without an order census. The maintained script `experiments/cyclotomic_3k_family.py` then performs exact factor reconstruction and power tests.

For $k = 4, 5$, every required factor is locally proved prime and the script certifies that γ_k contains the full primary part of $2^{3^k} - 1$; hence C_4 and C_5 are locally certified. At $k = 6$, the same exact field computations cover all listed factors, but the 42- and 90-digit factors are only probable primes under the local Miller–Rabin path. FactorDB records them as proven primes, but no certificates are vendored here. Thus C_6 is *source-assisted*, not locally certified.

At $k = 7, 8$, every currently known prime factor of $\Phi_{3^k}(2)$ passes the required order test, while unfactored composite cofactors remain. These levels are only consistent with C_k . The distinctions are summarized as follows:

levels	status	basis
$1 \leq k \leq 3$	theorem	prime-step argument
$k = 4, 5$	locally certified	exact factorization, primality, and order
$k = 6$	source-assisted	exact order; external primality for P42 and P90
$k = 7, 8$	consistent	known factors only; unfactored cofactors

A.4 Exceptional finite evidence

The maintained script `experiments/exception_column_m4.py` checks the corrected norm, the projected antiunit, and the current-factor power tests. The fully factored levels $k = 2, \dots, 5$ are locally certified and give $m_p = 4$ for every prime with

$$f(p) \in \{18, 54, 162, 486\}.$$

At $k = 6$, the field computations are exact but the 78-digit factor is only a local probable prime; FactorDB records it as proven. The complete $k = 6$ conclusion is therefore source-assisted unless a primality certificate is supplied. At $k = 7, 8$, all currently known prime factors pass, but the unfactored cofactors and their multiplicities remain unknown; these checks are merely consistent with D'_k .

The small anchor rows $m_{19} = m_{163} = m_{1459} = 4$ are reproduced rather than assumed. An independent Artin–Schreier compositum model also verifies $\sigma(4) = 5$, Proposition 8.1, and the direct power test on the smallest factor of each checked level.

A.5 Reproducibility boundary

The three maintained entry points are `experiments/ordinal_excess_probe.py`, `experiments/cyclotomic_3k_family.py`, and `experiments/exception_column_m4.py`. The additional programs under `experiments/excess/` are archived research probes and must be judged file by file under that directory’s status table before being used as evidence. The paper’s theorem statements depend only on the displayed proofs and cited standard results; the programs serve finite certification, falsification, and audit.

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